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HY Repro/Obgyn, by Dr Michael D Mehlman

I have revamped this PDF into chart/table format. I originally was considering splitting up the content into isolated Step 1 Repro vs 2CK/3 Obgyn PDFs, but after I finished the content, I decided to keep it as one PDF. What I've done, however, is clearly labeled if info is Steps 1 vs 2CK/3. If I don't explicitly label which Step information is, it's for both Steps.

After you finish the tables/charts, you will notice the PDF transitions into bullet point structure. All of this encompasses the original Repro/Obgyn PDF prior to the revamping. Once you reach the bullet points, you can use them as extra / rapid review, or you can move on to one of my other PDFs. So if you're freaking out about the length of this PDF, the core first-pass of the material will only take you out to around p.77.

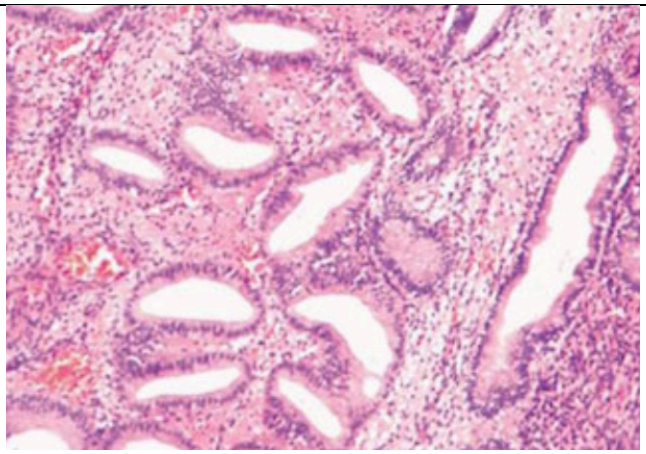
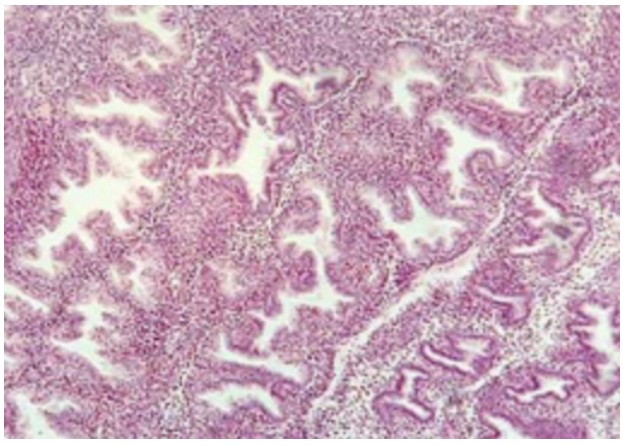
Repro-related hormone HY Points	
GnRH	- Gonadotropin-releasing hormone; produced by the hypothalamus. - Stimulates LH and FSH secretion from anterior pituitary. - "Gonadotropins" refer to the peptide hormones, LH and FSH. - Mechanism for amenorrhea in anorexia is ↓ GnRH pulsation. This is because leptin is required for adequate GnRH pulsation, and in anorexia, leptin levels are low. This is because leptin makes you feel full, and in anorexia, the patient is hungry, so leptin is low. ↓ GnRH pulsation causes LH and FSH to both be ↓, which leads to reduced ovarian hormone production and follicle stimulation. - In polycystic ovarian syndrome (PCOS), insulin resistance in high-BMI female causes abnormal GnRH pulsation, which causes ↑ LH/FSH ratio. I discuss this in extensive detail later in this PDF, but for the moment, just know that the starting point for the pathology in PCOS is insulin resistance causing abnormal GnRH pulsation. - Leuprolide, goserelin, and nafarelin are all GnRH receptor agonists. When administered continuously, they cause desensitization of the GnRH receptor at the anterior pituitary, leading to ↓ LH and FSH secretion (i.e., even though the drugs are pharmacologic agonists, they function clinically as antagonists). - GnRH secretion is inhibited by estrogen, androgen, and progesterone. This inhibition is called negative-feedback. - Exogenous anabolic steroids, for instance, ↓ GnRH secretion → ↓ LH + FSH → ↓ endogenous testicular production of androgens. - Menopause (which is physiologic), as well as any condition characterized by primary hypogonadism or insufficiency (i.e., Turner, Klinefelter, premature ovarian failure, etc., which are all pathologic), will cause ↑ GnRH secretion due ↓ negative-feedback. "Primary" means the organ itself (i.e., the ovary or testis) is the reason for the change in hormone secretion, whereas "central" means the hypothalamus or anterior pituitary is the reason. - Androgen-insensitivity syndrome causes ↑ GnRH due to inability of androgen to exert negative-feedback at androgen receptors at the hypothalamus.
LH	- Luteinizing hormone; produced by anterior pituitary. - Stimulates testicular Leydig cells in males, and ovarian theca interna cells in females, to make androgens. - LH is ↑ in PCOS, which is why these patients have hirsutism. In PCOS Qs for USMLE, select ↑ LH and ↓ FSH. This is due to "decreased/abnormal GnRH pulsation." - LH is ↓ in amenorrhea due to anorexia. In Qs on USMLE, select ↓ for both LH and FSH. This is due to "decreased/abnormal GnRH pulsation."
FSH	- Follicle-stimulating hormone; produced by anterior pituitary. - Stimulates testicular Sertoli cells in males, and ovarian granulosa cells in females, to make aromatase and inhibin B; in males also facilitates sperm maturation. - FSH is ↓ in PCOS for USMLE. The LH/FSH ratio is ↑. Some students will ask, "well aren't both LH and FSH ↑, just LH ↑ more than FSH?" Most literature sources say FSH is straight-up ↓ in PCOS. I've never seen an NBME Q assess an ↑ for it. Point is, if I were to sit the real deal again, I'd put a ↓ for FSH, and if it's wrong, everyone can go fuck themselves.
Aromatase	- Produced by Sertoli cells in males and granulosa cells in females in response to FSH. - Converts androgens into estrogens. - Testosterone → estradiol (potent estrogen). - Androstenedione → estrone (less potent estrogen). - DHEA-S → estriol (pregnancy-associated estrogen; weakest estrogen). - Anastrazole and exemestane are aromatase inhibitors. - There's an NBME Q that gives gynecomastia in a male where the answer is Sertoli-Leydig cell tumor (granulosa cell tumor not listed). SL tumors normally make androgens. The reason gynecomastia is possible is because of aromatization of androgens into estrogens. - Physiologic gynecomastia can occur in adolescent males, and virginal breast hypertrophy in adolescent females, due to ↑ androgen production → aromatization → ↑ estrogens. - Closure of growth plates occurs via estrogens, via aromatization from androgens.
hCG	- Human chorionic gonadotropin; produced by syncytiotrophoblastic cells of the placenta.

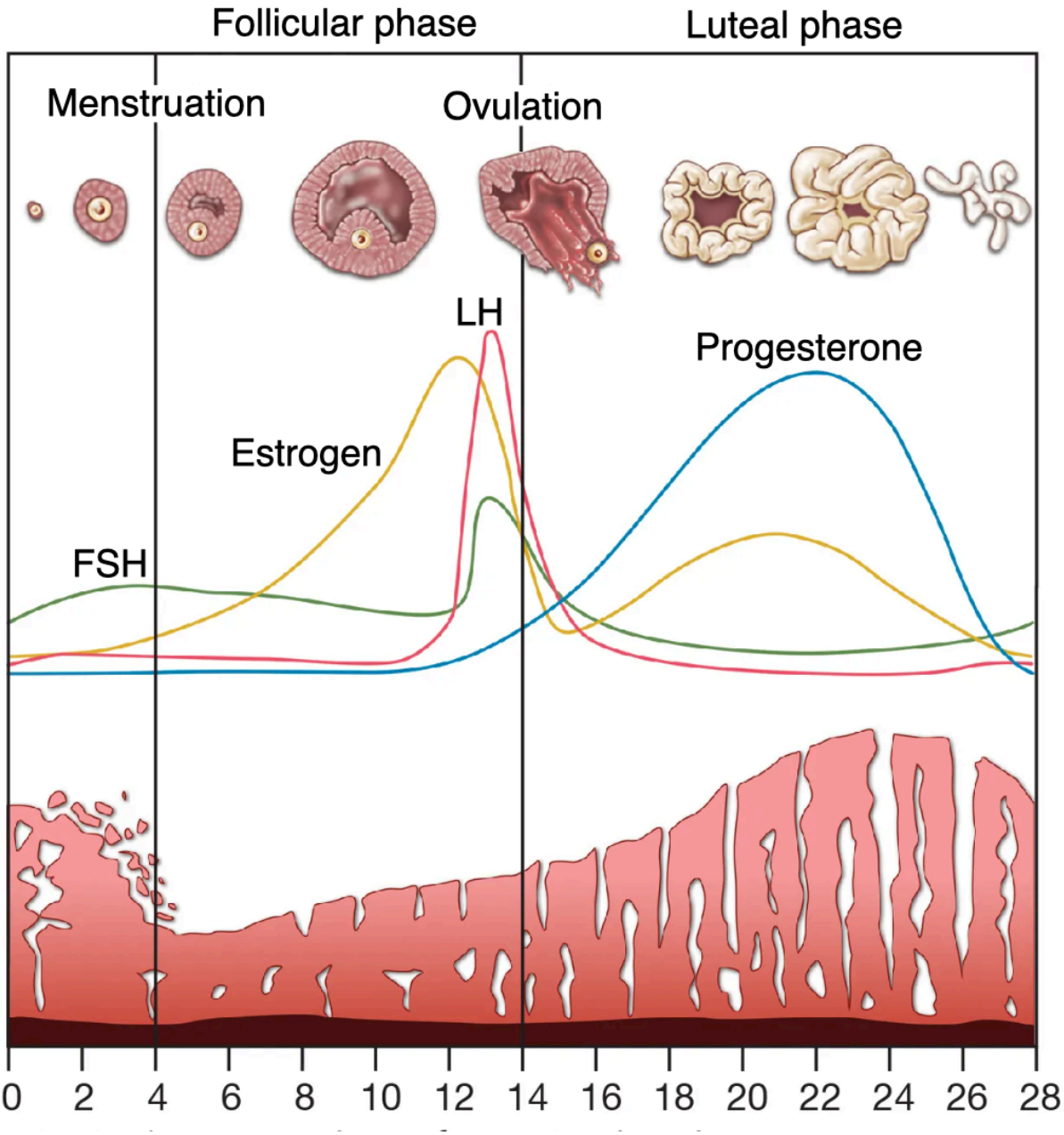
	- Maintains the corpus luteum (remnant of ruptured ovarian follicle) after fertilization has occurred. - Corpus luteum normally secretes progesterone, which maintains the endometrial lining during pregnancy. - Without fertilization, there is no placenta, hence no hCG present to maintain the corpus luteum → corpus luteum degrades → ↓ progesterone → endometrial lining cannot be maintained → sloughs → menstruation. - hCG peaks at 8-10 weeks of pregnancy. After this point, the placenta takes over production of progesterone. This obviates the need for the corpus luteum. Since we don't need the corpus luteum anymore to make progesterone, we don't need hCG anymore, which is why hCG falls after 8-10 weeks. - High hCG at 8-10 weeks is responsible for the nausea and vomiting in pregnancy. If the vomiting is severe enough where the female is ketogenic (i.e., ketones detected in the urine), we call this hyperemesis gravidarum. I will discuss this more later. - Beta-hCG simply refers to the subunit of hCG that is measured during a pregnancy test. This is because hCG shares the same alpha subunit with TSH, LH, and FSH, so we must measure for the beta-subunit to determine whether the hormone is present or not. - An NBME Q gives gynecomastia in a male patient taking hCG. They ask for the mechanism → answer = "testis producing estrogen." They can also have as answer "testicle production; estrogen." The reason this makes sense is because, since hCG shares the same alpha-subunit with LH and FSH (and TSH), giving hCG is as though we are giving LH and FSH at the same time. So LH effect → androgen production; FSH effect → aromatase production → therefore androgens are converted to estrogens in the testis → mechanism via which we could get gynecomastia. - hCG is ↑ in Down syndrome during second-trimester quad screen (↓ AFP, ↑ hCG, ↓ estriol, ↑ inhibin A). Just remember that the ones that have "Hs" are ↑ (i.e., hCG and inhibin A both have Hs, so those are the ones that are ↑). - USMLE wants you to know pregnancy can occur in Peds *before the first menses*. They can give classic Sx of pregnancy (i.e., morning nausea, bloating, breast tenderness, etc.) in 13-year-old who's never had a period → next best step = check serum β-hCG. - Always check β-hCG before commencing isotretinoin for acne, since it is teratogenic.
Estrogens	- Estradiol is produced mainly by the ovaries (and testes in males). - Estrone is produced mainly by adipose tissue. In post-menopausal women, adipose tissue becomes important source of estrogen, where low BMI ↑ risk of osteoporosis. Interestingly, the NBMEs assess BMIs 20s and lower in young females as also ↑ risk of osteoporosis, even though low BMI is <18.5. - Two different 2CK Pysch CMS Qs give BMI of 19 and 20 + say girl is at risk of what later in life → answer = osteoporosis. Sounds odd, but I'm letting you know the Qs exist. - Estriol is produced primarily by the placenta; ↓ in Down syndrome, as mentioned above. - Adrenal DHEA-S (an adrenal-specific androgen) can be aromatized into estradiol. - Estradiol stimulates the endometrial lining to grow during the follicular phase of the menstrual cycle. Its levels gradually increase during the follicular phase. Once a critical threshold is reached (called the estrogen threshold), its effects at the hypothalamus flip from negative-feedback to positive-feedback, causing the LH surge, which triggers the rupturing of the Graafian follicle. - Stimulates secondary sex characteristics in females – i.e., breast development and changes in body fat distribution. - Maintains bone density by ↑ OPG/RANK ratio. In post-menopausal women, ↓ estrogen → ↑ RANK in comparison to OPG → ↑ osteoclast activity → osteoporosis. - Important for healthy blood lipid profile (estrogen ↑ HDL and ↓ LDL). - Hormone-replacement therapy (HRT) is only approved for severe perimenopausal vasomotor symptoms (i.e., hot flashes, urge incontinence, atrophic vaginitis). It is not given to help preserve bone density or for positive role on mood and neurocognition, since ↑ absolute estrogen exposure in women ↑ risk of breast cancer, MI, and thromboembolic events (i.e., DVT, PE, stroke). The latter is because estrogen upregulates fibrinogen and factors V and VIII.

	- **Unopposed estrogen** is the mechanism for endometrial hyperplasia and adenocarcinoma. The three groups this refers to on NBME exams, as per my observation, are 1) PCOS, 2) HRT patients who stop taking the progesterone component, and 3) granulosa cell tumors. The reason is because estrogen stimulates the growth of endometrium; progesterone inhibits growth of endometrium; so the balance between these two hormones keeps the endometrial lining in check. USMLE might give a vignette of vaginal bleeding in a woman with unopposed estrogen, and the answer will just be "endometrial biopsy." - Estrogen-containing OCPs ↓ protein content of breast milk (asked on 2CK Obygn form).
Progesterone	- Produced by the corpus luteum (ruptured ovarian follicle) following ovulation and maintains the endometrial lining during the luteal phase of the menstrual cycle. As discussed earlier, if fertilization occurs, hCG will maintain the corpus luteum so that progesterone production continues + can maintain the pregnancy. - Also produced in small amounts in the adrenal glands. - Along with estrogen, facilitates proper mammary gland development. During pregnancy, progesterone helps maintain the structure of the mammary glands. - Progesterone inhibits the effects of prolactin at the alveolar cells in the breast (milk-producing cells), preventing lactation during pregnancy. Once the placenta is expelled at parturition and progesterone levels fall, prolactin can induce lactation. - Progesterone-only contraceptives are considered safe while breastfeeding, since the low amount of progesterone does not inhibit lactation, even though the high progesterone during pregnancy does. It is the effect of estrogen in combined OCPs that decreases protein content of breast milk. - Progesterone plays a role in thermoregulation, where ↑ secretion at ovulation ↑ body temperature, which is one way to predict timing of ovulation, albeit dubiously. There is an NBME question floating around where they show a graph of increased body temperature at day 14 of menstrual cycle, and answer is just progesterone. Not hard.
Prolactin	- Produced by anterior pituitary. - Stimulates lactation (milk production). - Levels rise during pregnancy, but lactation is inhibited by ↑ progesterone causing ↓ prolactin effects at its receptor on breast alveolar cells. As mentioned above, after expulsion of the placenta and progesterone falls, prolactin can induce lactation. - Dopamine inhibits prolactin. If the pituitary stalk is severed, prolactin ↑ due to lack of negative-feedback by dopamine. - Bromocriptine (D2 receptor agonist) is first-line Tx for prolactinoma. - D2 antagonists (i.e., anti-psychotics, metoclopramide) can cause hyperprolactinemia.
Oxytocin	- Produced by hypothalamus and stored in posterior pituitary. - Induces milk secretion (let-down) and uterine contractions. - Suckling reflex at breast causes ↑ oxytocin secretion, inducing milk let-down. - Known colloquially as the "love hormone," since its secretion facilitates maternal-neonate bonding. - Ferguson reflex → cervical distension by the fetus at parturition ↑ oxytocin production, causing ↑ uterine contractions. - Excessive administration during parturition can cause uterine hypertonus and tachysystole, ↑ risk of uterine rupture and fetal complications.
Inhibin	- Inhibin A and B are produced by Sertoli cells in males and granulosa cells in females. - Both exert negative feedback at the anterior pituitary to specifically ↓ FSH secretion. - Inhibin A is measured as part of the second trimester quad screen, not inhibin B, because it is more accurate in helping to predict fetal chromosomal abnormalities.
Androgens	- Produced by testicular Leydig cells in males and ovarian theca interna cells in females. - DHEA-S is an adrenal-specific androgen made in the zona reticularis of the adrenal cortex. The adrenal gland can also make androstenedione. These two androgens can then go peripherally to be converted into testosterone. - Testosterone is converted into dihydrotestosterone (DHT) via 5-alpha reductase. It is DHT that is the strongest androgen and has the greatest effect at tissues. This enzyme is inhibited by finasteride (used for BPH and androgenetic alopecia).

	- DHT is necessary for external male secondary sex characteristics, so if DHT is low, then the male will appear phenotypically female. - In 5-alpha reductase deficiency, the karyotypic male will appear female for the first 12 years of life, followed by "penis (phallus) at age 12," where the surge in testosterone from puberty can override the 5-alpha reductase deficiency where a threshold DHT level is achieved, leading to a 3-4-cm clitoral hood. The vignette will tell you a 12-year-old girl has grown 4 inches in past 4 months + has acne + hair on upper lip + clitoral hood growth. Karyotype is 46XY since this is a male. - Androgen-insensitivity syndrome is a karyotypic male (46XY) presenting as phenotypic female due to failure of testosterone and DHT to have effect. Both are ↑ in AIS due to ↓ negative-feedback at hypothalamus and anterior pituitary. So LH is also ↑. I talk about the sex disorders later in this PDF, but AIS will be a 15-year-old girl who's never had a menstrual period + vagina ends in blind pouch + has scanty/absent pubic/axillary hair. - Androgens build muscle, maintain spermatogenesis, enhance libido, and stimulate prostate growth. - High androgens (anabolic steroids) accelerate atherosclerotic plaque maturity (↑ LDL and TGAs, ↓ HDL) , increase plaque calcium scores (marker of plaque maturation), increase hematocrit (sometimes leading to hyperviscosity syndrome), suppress hypothalamic-gonadal axis (i.e., suppress endogenous androgen synthesis and spermatogenesis), and can cause liver damage.
ABP	- Androgen-binding protein. - Produced by Sertoli cells in response to testosterone (not FSH). - Binds androgen locally at the seminiferous tubules to facilitate spermatogenesis.
SHBG	- Sex hormone-binding globulin. - Binds to and carries androgen and estrogen around the blood. - Most androgen and estrogen is bound to SHBG; the free fraction is a minor % and is most biologically active. - Estrogens can ↑ SHBG. - Anabolic steroid users sometimes take other agents to ↓ SHBG to ↑ androgen effects.
hPL	- Human placental lactogen (aka human chorionic somatomammotropin). - Produced by syncytiotrophoblastic cells of the placenta (same as hCG). - Highest levels in 3rd trimester. - Has many effects, but one important purpose is it increases insulin resistance in order to ↑ serum glucose to maximize availability to the developing fetus. However, the biologic tradeoff is that this ↑ the risk of gestational diabetes for the mother.

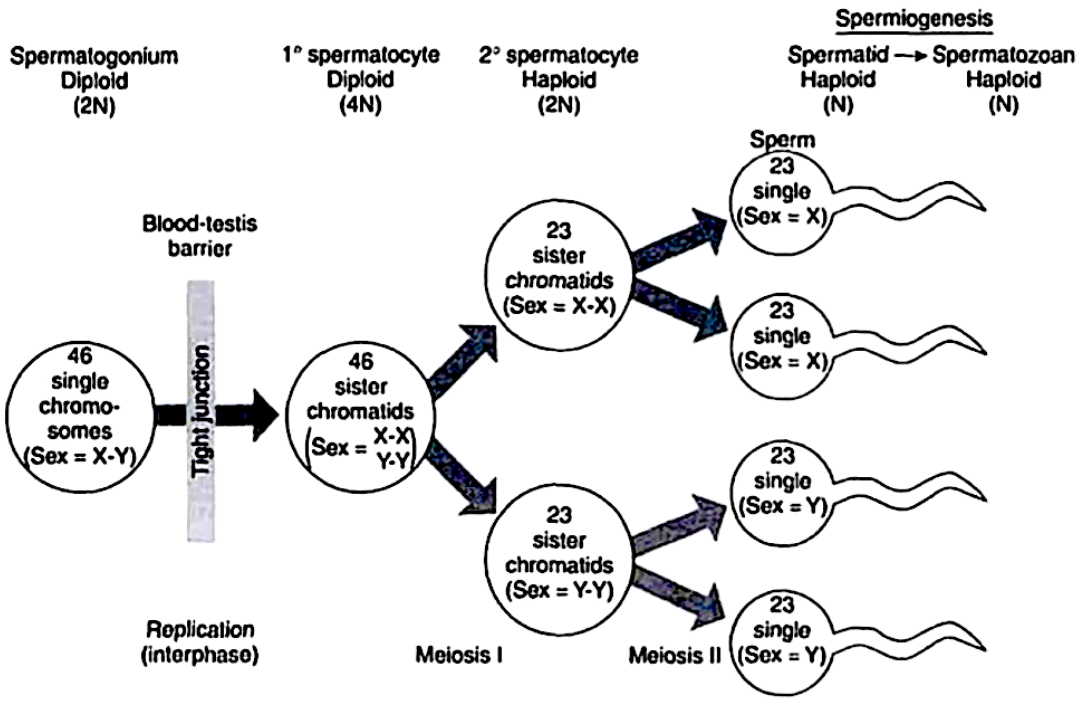
Menstrual cycle HY points	
Follicular phase	- Aka proliferative phase. - At the beginning of this phase, multiple ovarian follicles start to develop in the ovaries. These follicles contain an immature egg (oocyte) surrounded by granulosa cells. - As follicular phase progresses, one of the developing follicles becomes dominant and continues to grow while the others degenerate. - Granulosa cells within the dominant follicle produce increasing amounts of estrogen, specifically estradiol. The increasing estrogen levels cause thickening of the endometrial lining (proliferation) in preparation for implantation if conception occurs. - Rising estrogen reaches a critical level (i.e., estrogen threshold) where its negative-feedback effects on LH and FSH secretion flip to positive-feedback. This causes LH and FSH to rise sharply/acutely. - The ↑ in LH (i.e., LH surge) triggers ovulation, where the Graafian follicle ruptures and forms the corpus luteum (follicular remnant). - Follicular phase histo shows straight, tubular glands:

	
Luteal phase	- Aka secretory phase. - 14 days in length. If a female's menstrual cycle length is > or <28 days, it is the follicular phase that changes length. - The corpus luteum secretes progesterone, which maintains the endometrial lining in preparation for implantation. You need to know for USMLE that **progesterone** is the main hormone of the luteal phase. - The granulosa cells during follicular development transform into granulosa-lutein cells once the corpus luteum forms; it is these cells that secrete progesterone. - The theca-interna cells in the developing follicle transform into theca-lutein cells in the corpus luteum. These cells continue to secrete androgen, which are transformed into estrogens by aromatase from the granulosa-lutein cells. - The secondary, albeit less robust, increase in estrogen during the luteal phase occurs due to continued mild estrogen release from the granulosa-lutein cells. - If fertilization does not occur, the corpus luteum will degenerate, leading to a drop in progesterone levels, triggering a sloughing of the endometrium. - As discussed earlier, if fertilization occurs, hCG from the syncytiotrophoblast of the placenta maintains the corpus luteum, thereby maintaining progesterone secretion, and in turn prevents sloughing of the endometrium. - Secretory phase histo shows coiled and spiral-shaped glands: 
Menstruation	- First 3-7 days of menstrual cycle. - Menstruation is caused by the presence of progesterone followed by its withdrawal. - In other words, progesterone is high during the luteal phase. If no fertilization occurs and the corpus luteum degrades, the loss of progesterone causes apoptosis and sloughing of the endometrium. - Women who experience anovulation (e.g., PCOS), do not form a corpus luteum (since they're not ovulating), so there is no sequence of "progesterone present followed by its withdrawal"; therefore they have irregular menses and missed cycles.

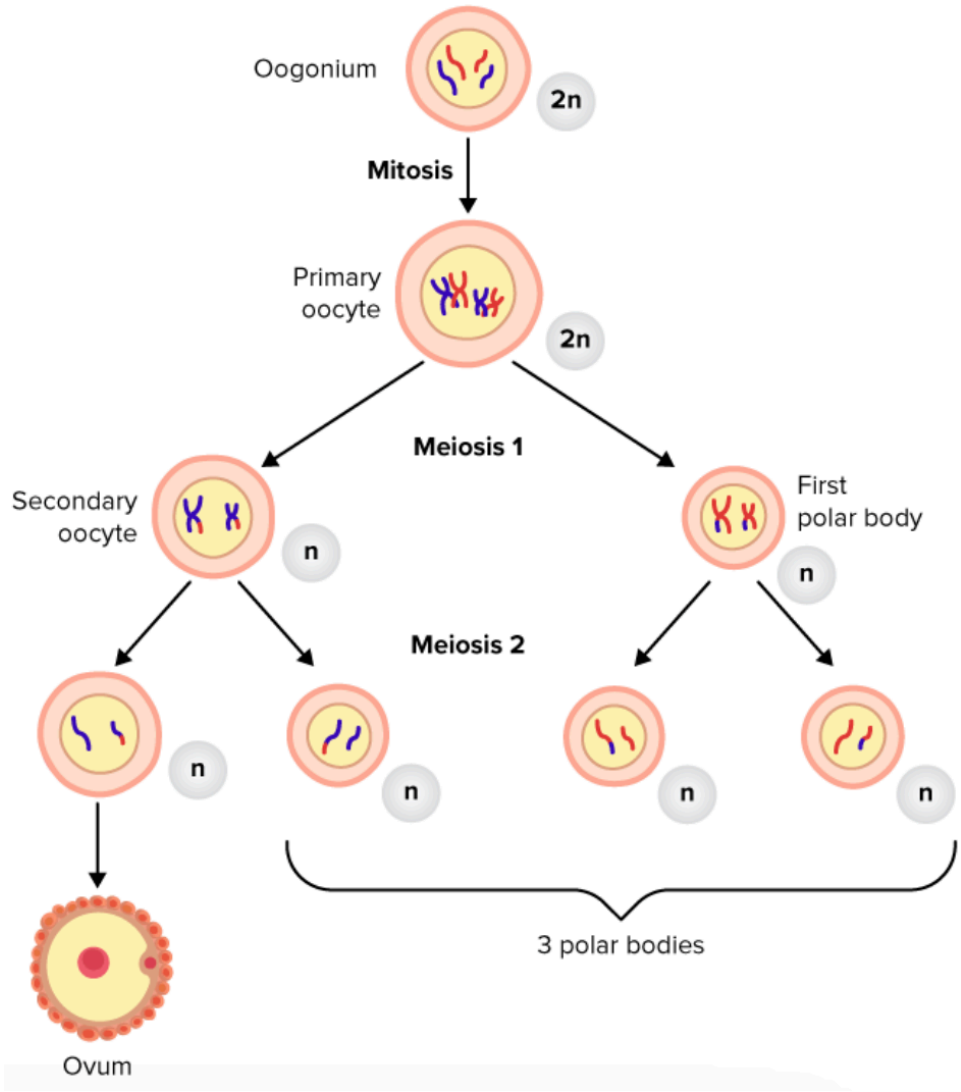


Spermatogenesis (Step 1 only)	
- Diploid (2n): A diploid cell has two complete sets of chromosomes, which refers to 46 chromosomes, with one set inherited from each parent (23 + 23); refers to all cells in the body except for sperm and ova. - Haploid (n): A haploid cell has only one complete set of chromosomes, which means only 23 chromosomes. Haploid cells are specific to sperm and ova. When a spermatazoan (haploid) fertilizes an ovum (haploid), the resulting zygote is diploid.	
Spermatogonia	- Diploid (2n). - Spermatogenesis begins at puberty when diploid germ cells (spermatogonia) undergo mitosis (not meiosis). Some spermatogonia remain as stem cells, while others differentiate into primary spermatocytes.
Primary spermatocytes	- Diploid (2n). - Primary spermatocytes are the result of the mitosis of spermatogonia. Each primary spermatocyte undergoes meiosis I, yielding two haploid secondary spermatocytes.

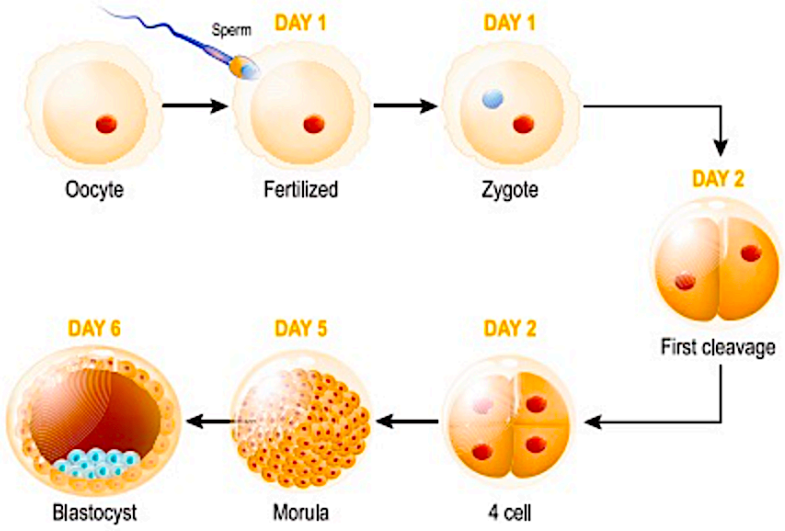
Secondary spermatocytes	- Haploid (n). - Secondary spermatocytes are haploid cells produced by meiosis I. Each secondary spermatocyte then undergoes meiosis II, producing four haploid spermatids.
Spermatids	- Haploid (n). - Spermatids are haploid cells produced by meiosis II. They then mature into sperm.
Spermatozoa	- Haploid (n) - Mature sperm; follows growth of flagellum (tail) by the spermatid.

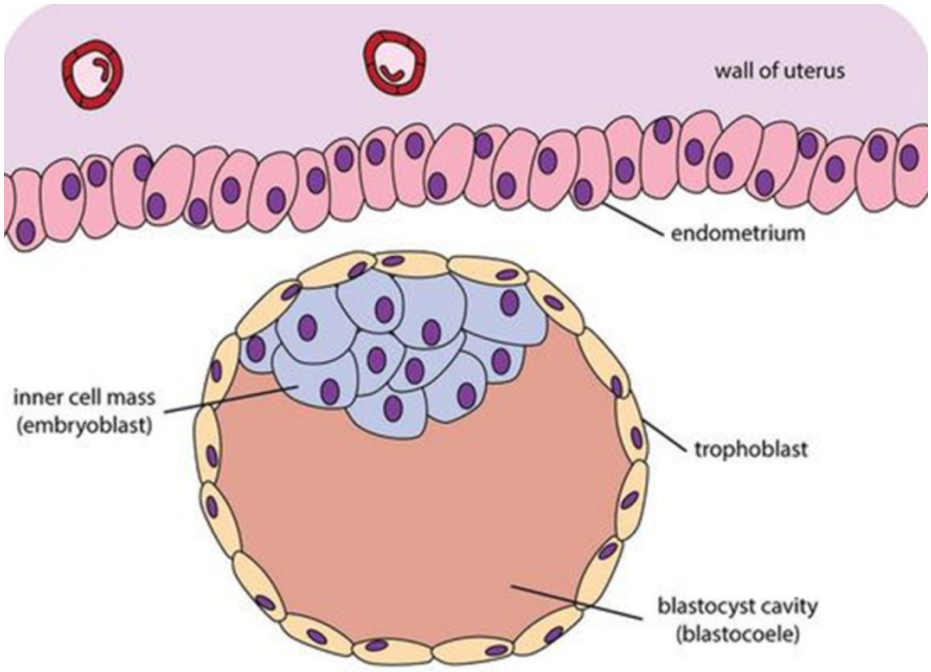


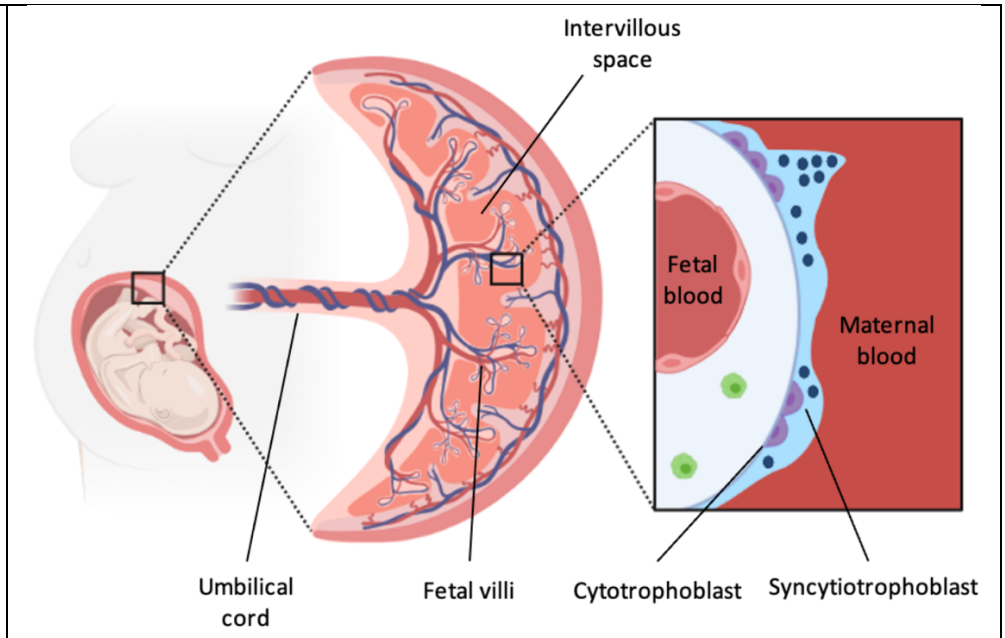
Oogenesis (Step 1 only)	
Oogonia	- Diploid (2n). - Oogenesis begins during fetal development when diploid germ cells (oogonia) undergo mitosis, forming primary oocytes.
Primary oocytes	- Diploid (2n). - **Primary oocytes are arrested in prophase I** of meiosis before birth. - At this stage, they are surrounded by granulosa cells; these structures are called primordial follicles.
Meiotic arrest	- Primary oocytes remain in prophase I until puberty. - Each month during the menstrual cycle, some of these primary oocytes are activated to continue development.
Meiosis I (Puberty)	- Upon activation, a primary oocyte completes meiosis I, resulting in the formation of a secondary oocyte and a smaller polar body. - **The secondary oocyte is arrested in metaphase II.**
Meiosis II (Fertilization)	- If fertilization occurs, the secondary oocyte completes meiosis II, yielding a mature ovum and additional polar body. - This second round of meiosis only occurs if a sperm fertilizes the secondary oocyte. - Fertilization results in the formation of a diploid (2n) zygote.



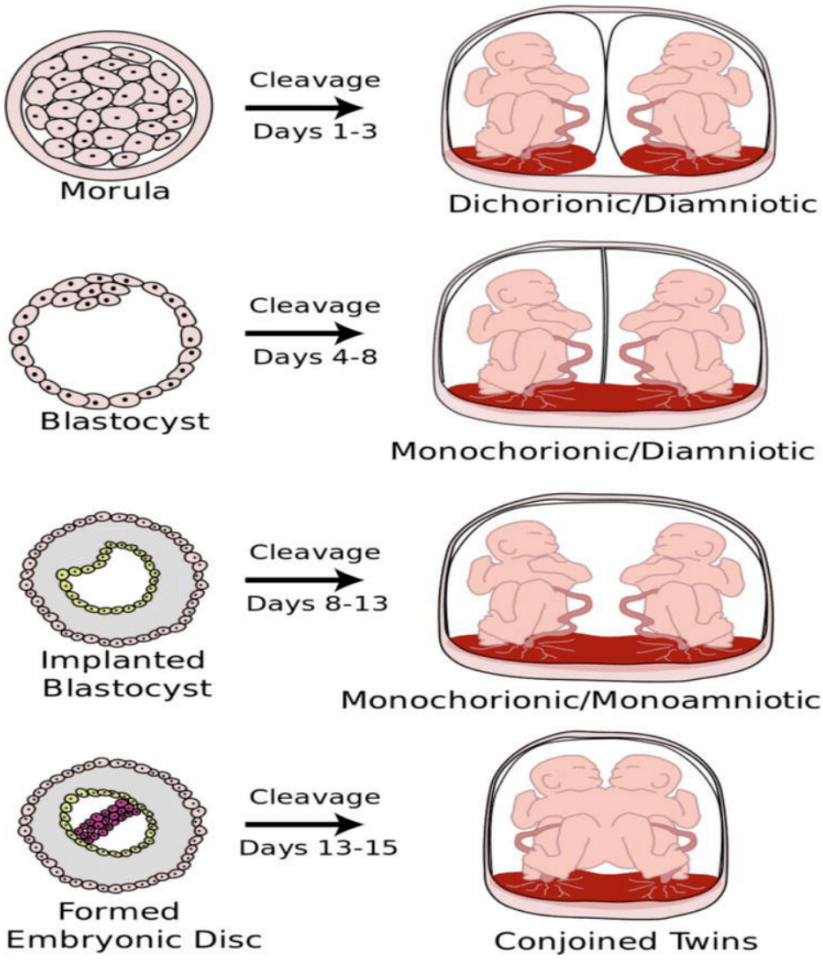
Fertilization to implantation (Step 1 only)

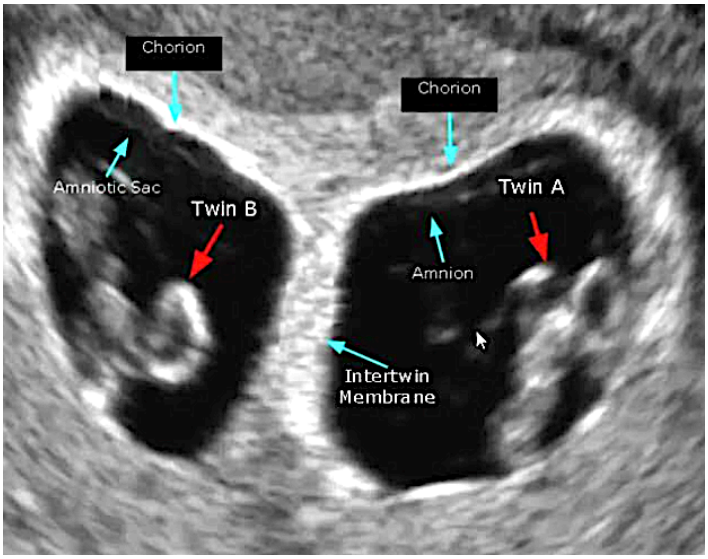
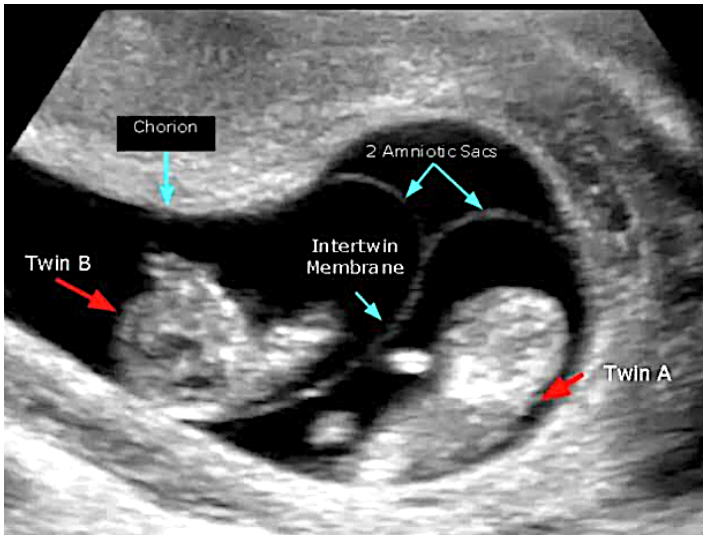


Fertilization	- Occurs when a haploid spermatogonia penetrates a haploid secondary oocyte, resulting in a diploid zygote. - Typically occurs in the ampulla of the fallopian tube.
Cleavage	- The zygote undergoes a few mitotic divisions, resulting in a cluster of cells called a morula. This is a solid ball of cells. - The morula continues to divide, forming a blastocyst.
Blastocyst	- The blastocyst is a hollow ball of cells and has three structures: 1) An outer layer of cells called the trophoblast. 2) The inner cell mass. 3) A central fluid-filled cavity called the blastocoel. 
Inner cell mass	- The inner cell mass becomes three important structures: 1) Embryonic disc, which gives rise to the three germ layers: ectoderm, mesoderm, endoderm. 2) Amniotic sac, which is the thin membrane surrounding the embryo. It produces amniotic fluid to cushion the embryo. 3) Yolk sac, which conducts early hematopoiesis and structural support. The human yolk sac does not contain a yolk, which is a source of nutrients, as with birds and reptiles.
Trophoblast	- The trophoblast becomes the fetal component of the placenta. - The trophoblast comprises two layers: an inner cytotrophoblast and outer syncytiotrophoblast.
Implantation	- The blastocyst enters the uterus after several days of transit in the Fallopian tube. - After trophoblastic cells invade the endometrium, the syncytiotrophoblast becomes more pronounced. It has four main functions: 1) Grows chorionic villi, which contain the fetal blood vessels, allowing for an interface between the maternal and fetal circulations; 2) Grows the chorion, which is the thick membrane surrounding the embryo; 2) Protects the fetus from the mother's immune system; 3) Produces hCG. - The embryo is now surrounded by an inner thin membrane (amnion), which was derived from the inner cell mass, and a thick outer layer (chorion), which was derived from the syncytiotrophoblast.
Decidua	- The endometrial tissue connected to the implanted embryo is called the decidua. - The decidua contains the maternal blood vessels and surrounds the fetal-derived chorionic villi; this interface enables nutrient and gas exchange.


Intervillous space
Fetal blood
Maternal blood
Umbilical cord
Fetal villi
Cytotrophoblast
Syncytiotrophoblast

Twinning

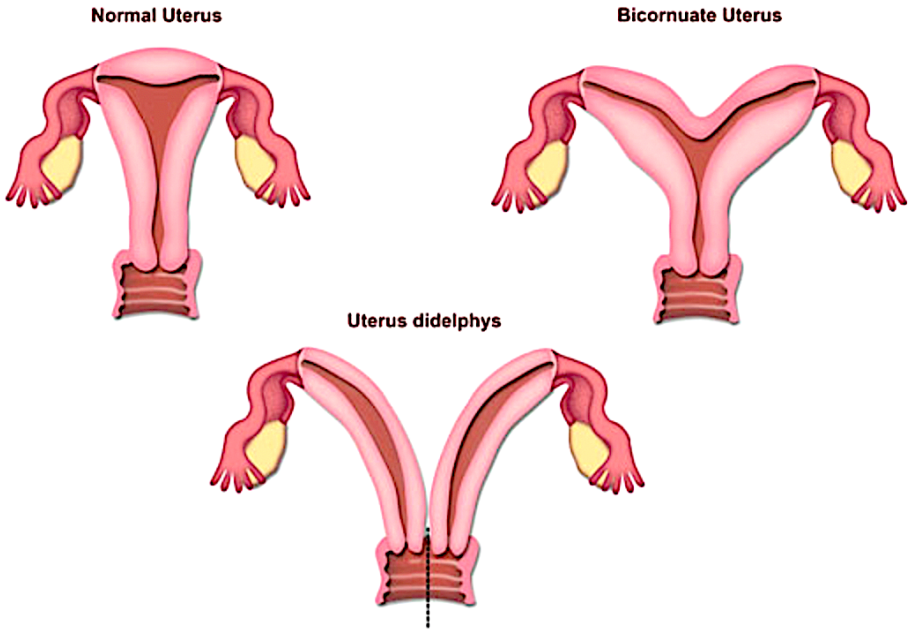

Morula
Cleavage
Days 1-3
Dichorionic/Diamniotic
Blastocyst
Cleavage
Days 4-8
Monochorionic/Diamniotic
Implanted Blastocyst
Cleavage
Days 8-13
Monochorionic/Monoamniotic
Formed Embryonic Disc
Cleavage
Days 13-15
Conjoined Twins

Dichorionic / diamniotic	- Refers to dizygotic (fraternal) twins, where there are two zygotes to start. - Refers to monozygotic (identical) twins, where the conceptus splits ~1-3 days post-fertilization. - # of chorions = # of placentas, so we have two placentas in Di-Di twins. - The NBME Q can say there is a "thick dividing membrane" between the fetuses. 
Monochorionic / diamniotic	- Refers to monozygotic twins, where the conceptus splits ~4-8 days post-fertilization. - There is no thick dividing membrane between the fetuses since they lie within the same chorion. 
Monochorionic / monoamniotic	- Refers to monozygotic twins, where the conceptus splits ~8-13 days post-fertilization. - High risk of twin-twin transfusion syndrome due to entanglement of fetal vessels.

Conjoined twins	- Refers to monozygotic twins, where the conceptus splits ~13 days or later.

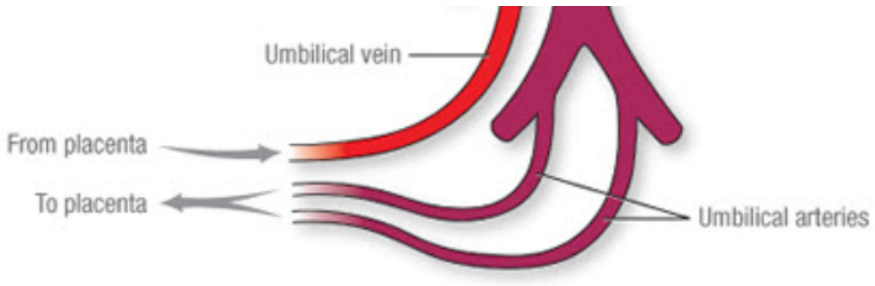
Embryologic development by week of development (Step 1 only)	
Week 1	- Fertilization + zygote begins to divide, forming the morula (solid ball of cells).
Week 2	- Morula divides into blastocyst (hollow ball of cells) + implants into endometrium. - Neural plate forms.
Week 3	- Primitive germ layers (ecto-, meso-, endoderm) appear. - Neural tube (early nervous system) begins to form. - **Fetus is most susceptible to teratogens at 3-8 weeks.** - Folic acid (vitamin B9) is most important in the first 3-4 weeks of pregnancy, when the neural tube begins to form. If B9 is deficient → neural tube defects (e.g., spina bifida).
Week 4	- Neural tube continues to develop. - Heart (4 chambers) and limbs (4 limbs) begin to form at week 4.
Week 5-onward	- Organogenesis. - Can go into elaborate detail by week, but USMLE doesn't give a fuck.

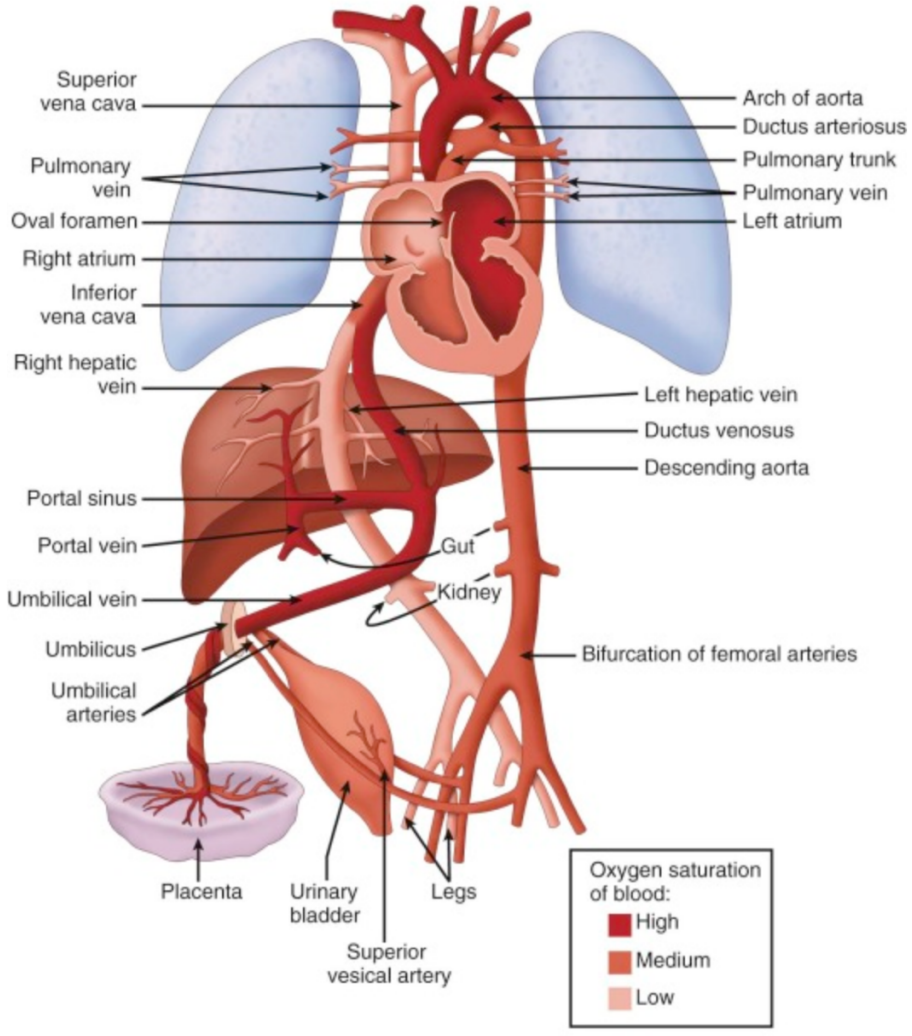
Germ layer points (Step 1 only)	
Ectoderm	- Derived from clefts (aka grooves). - CNS, PNS, skin, hair, nails, lens of eye. - Craniopharyngioma is derived from Rathke pouch, which is from the "roof of the primitive oral cavity," or "roof of primitive oropharynx"; surface ectoderm. - Neural crest is a specialized type of ectoderm that is derived from the neural plate.
Neural crest	- Melanocytes, PNS, adrenal medulla, various cardiac structures. - Pheochromocytoma, neuroblastoma, medulloblastoma are all neural crest-derived. - Neural crest is HY answer for Hirschsprung (failure of neural crest migration distally in the GIT) and fetal alcohol syndrome (cardiac defects).
Mesoderm	- Derived from arches. - Most bones, muscles, cartilage, blood vessels, and the heart.
Endoderm	- Derived from pouches. - GIT, lungs, thymus, thyroid, parathyroids. - I've seen NBME give tracheoesophageal fistula and then ask about the embryo of the esophageal atresia → answer = endoderm. - Thyroglossal duct cyst → NBME answer for embryo = "endoderm of foramen cecum." - 3rd pouch becomes the two inferior parathyroids + thymus. - 4th pouch becomes the two superior parathyroids. - The USMLE can ask the 3rd and 4th pouches as part of DiGeorge syndrome Qs. But they also like to assess that you know what they become independently – i.e., there is a missing superior parathyroid → answer = 4th pouch is fucked up. Or there is an adenoma of an inferior parathyroid → answer = 3rd pouch.

More *garbage* repro developmental stuff (Step 1 only)	
Müllerian ducts	- Aka paramesonephric ducts. - Embryonic structures that give rise to the female reproductive tract: - Fallopian Tubes, uterus, cervix, upper vagina. - USMLE will show you a hysterosalpingogram of uterine didelphys or bicornuate uterus, and then the answer is something like "failure of paramesonephric ducts to fuse." 
Wolffian ducts	- Aka mesonephric ducts. - Embryonic structures that give rise to several male reproductive structures: - Epididymis – coiled tube on the back of each testis where sperm are stored and mature.

	- Vas Deferens (aka ductus deferens) – tube that transports sperm from the epididymis to the urethra during ejaculation. - Seminal Vesicles – glands that produce a significant fraction of the seminal fluid. - Ejaculatory Ducts – tube that connects the vas deferens to the urethra. - Development is stimulated by testosterone.
MIF	- Müllerian-inhibitory factor (aka Müllerian-inhibitory hormone). - Produced by Sertoli cells starting around 8 weeks' gestation in males. - As the name denotes, inhibits the development of the Müllerian ducts, preventing the formation of female reproductive structures. This is essential for the differentiation of male reproductive structures and the suppression of female ones. - In females, the Müllerian ducts develop in response to the absence of MIF.
TDF	- Testis-determining factor (aka sex-determining region-Y [SRY] protein). - Coded for by *SRY* gene on Y chromosome. - Secreted by gonadal ridge (early/undifferentiated embryonic tissue) - Initiates development of male reproductive structures by promoting the differentiation of the embryonic gonadal tissue into testes.
Mini-summary:	- During early embryonic development, both males and females have two pairs of ducts: the Wolffian ducts and Müllerian ducts. - Production of TDF by *SRY* gene causes formation of testes. - MIF secreted by Sertoli cells causes regression of Müllerian (paramesonephric) ducts. - Testosterone secreted by Leydig cells promotes development of Wolffian ducts.

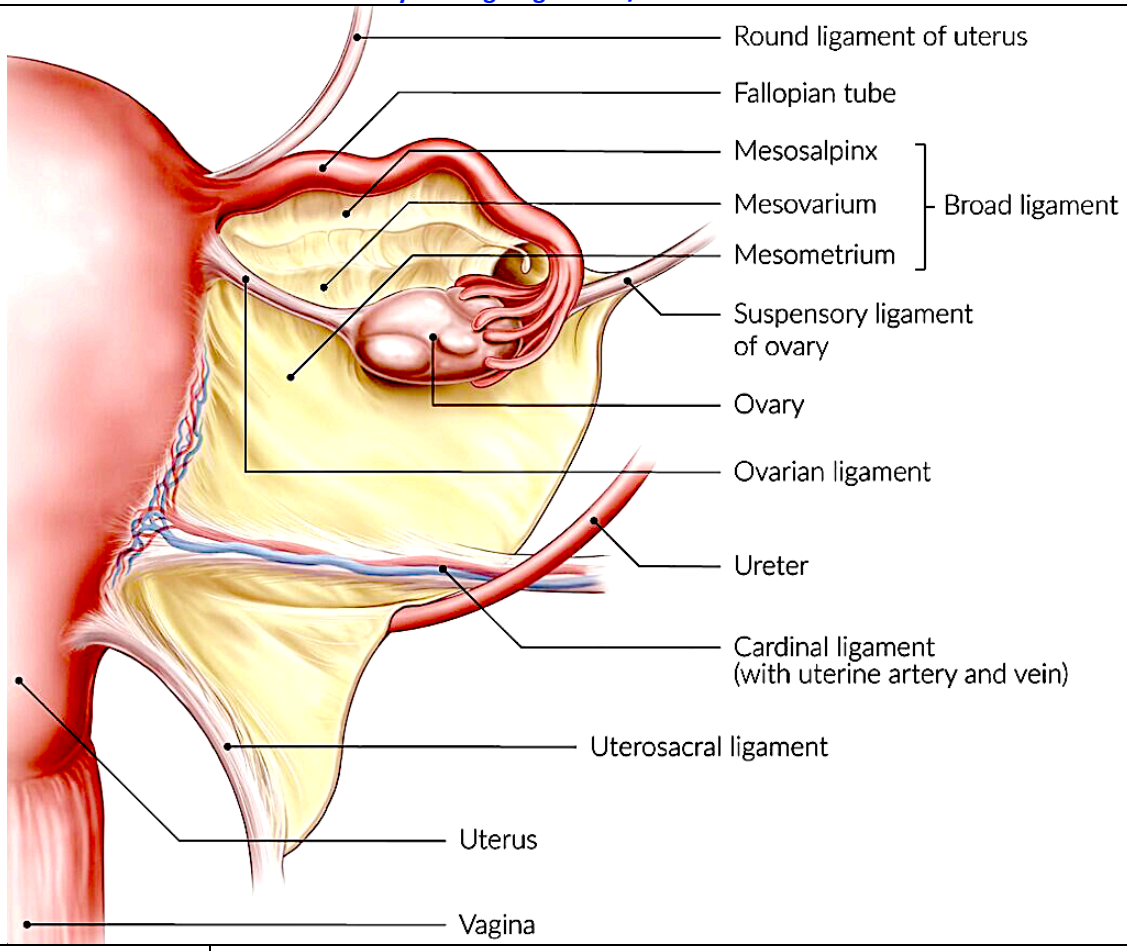
Male-female equivalent structures (Step 1 only)		
Derivative	**Males**	**Females**
Labioscrotal swelling	Scrotum	Labia majora
Urogenital folds	Ventral shaft of penis; penile urethra	Labia minora
Urogenital sinus	Prostate gland	Glands of Skene
Urogenital sinus	Cowper glands	Bartholin glands
Genital tubercle	Corpus cavernosum/spongiosum	Vestibular bulbs
Genital tubercle	Glans penis	Glans clitoris

Fetal circulation/umbilical cord anatomy	
Umbilical vein	- Carries oxygenated blood from the mother to the fetus. - Post-birth remnant is called ligamentum teres hepatis, or round ligament of liver. - Umbilical cord has one umbilical vein and two umbilical arteries. 
Umbilical arteries	- Carry deoxygenated blood from the fetus back to the mom. - Post-birth remnants are called medial umbilical ligaments. - Don't confuse the latter with the median umbilical ligament, which is the post-birth remnant of the fetal urachus, which is a canal-like structure that connects the fetal bladder to the umbilicus (belly button).
Wharton jelly	- Specialized connective tissue within the umbilical cord. - Protects the umbilical vein and two umbilical arteries within the umbilical cord.

	- Velamentous cord insertion is a rare condition in which the umbilical cord inserts into the fetal membranes rather than directly into the placenta. This can lead to exposure + risk of injury to the umbilical vessels where they are not protected by Wharton jelly.
Ductus arteriosus	- Fetal blood vessel that connects the pulmonary trunk to the proximal descending aortic arch. - Allows most oxygenated blood from the fetal RV to bypass the high-resistance, non-functional fetal lungs, allowing it to flow directly into the systemic circulation. - Normally closes within the first week post-birth. - Post-birth remnant is called the ligamentum arteriosum. - A patent ductus arteriosus (PDA) is when it does not close post-birth. However in the neonate, blood flows in the reverse direction, high pressure to low pressure, going from the descending aortic arch back to the pulmonary trunk. This causes a buzzy continuous, machinery-like murmur (aka pan-systolic-pan-diastolic, or to-and-fro). - PDA is classically seen in **congenital rubella syndrome.** - NBME can give easy vignette of PDA in neonate and then the answer is "extra-cardiac left-to-right shunt," where wrong answers are R→L and intra-cardiac combos. 
Ductus venosus	- Fetal blood vessel that connects the umbilical vein coming from the mother to the fetal IVC, allowing oxygenated blood from the placenta to bypass the high-resistance fetal liver and flow directly to the fetal RA. - Post-birth remnant is called ligamentum venosum.
Foramen ovale	- Opening in the fetal inter-atrial septum that allows for oxygenated blood in the RA to move directly into the LA, bypassing the high-resistance fetal lungs.

	- Post-birth, the remnant is called the fossa ovalis, which is a depression in the inter-atrial septum. If there is failure of closure, a patent foramen ovale results, which is a type of atrial septal defect (ASD) in the neonate (fixed splitting of S2).

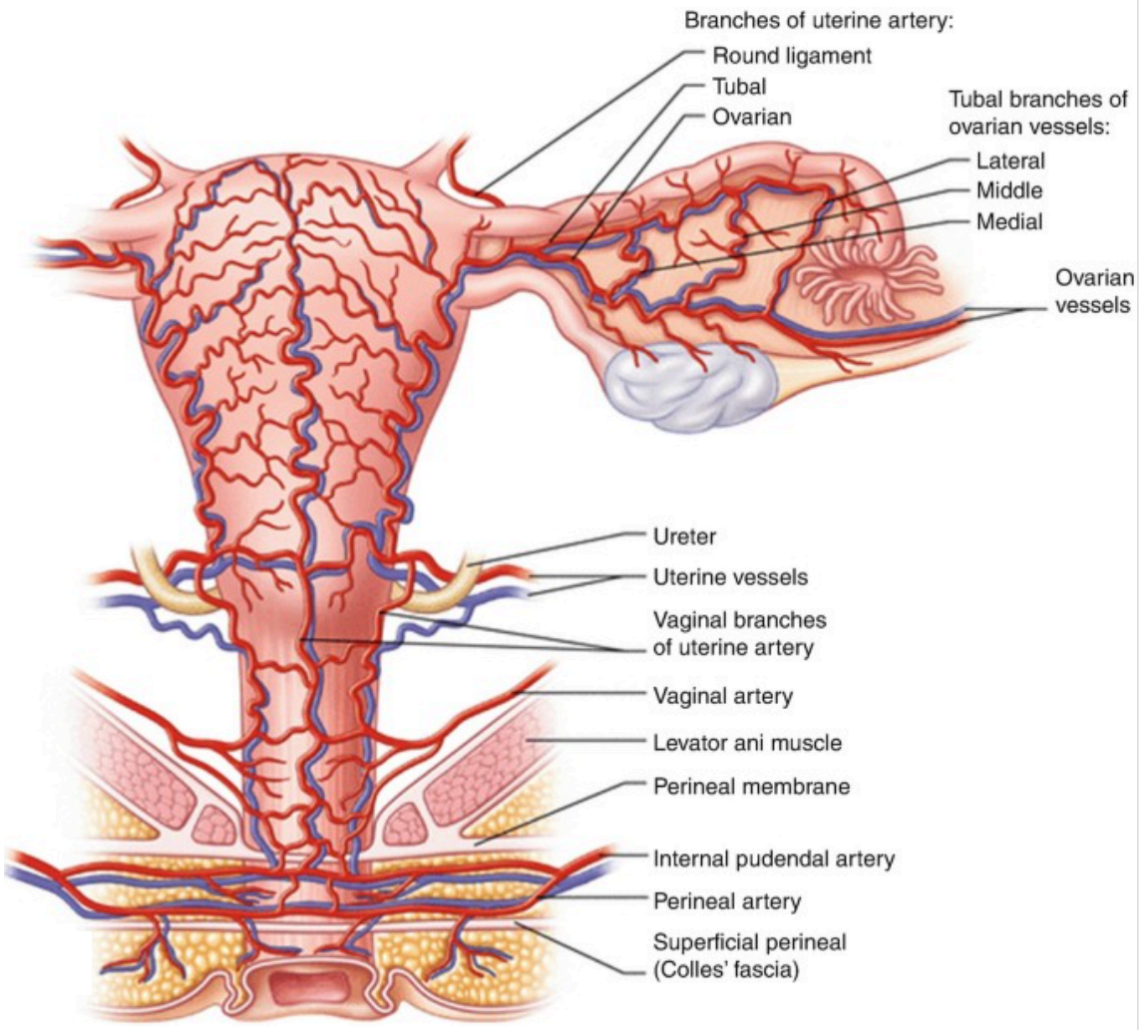
Gynecologic ligaments/structures



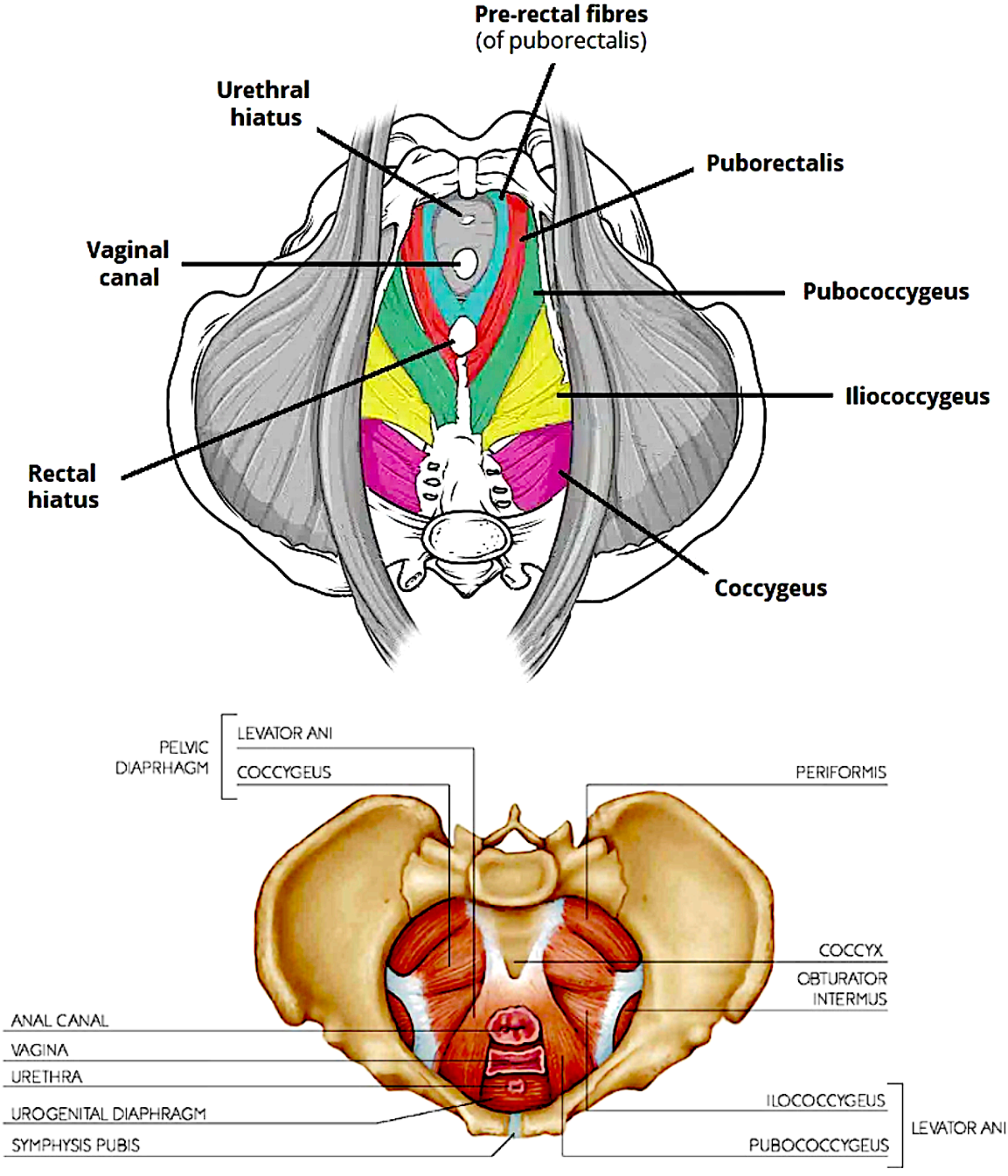
Broad ligament	- Connects the uterus, Fallopian (uterine) tubes, and ovaries to the lateral pelvic walls. - Contains uterine arteries and veins. - Consists of three layers of peritoneum: - The mesosalpinx is the upper fold that encloses the Fallopian tube. - The mesometrium supports the uterus and is the largest part of the broad ligament. - The mesovarium is the fold that surrounds the ovary.
Cardinal ligament	- Connects the cervix to the pelvic side wall. - Contains parts of uterine artery and vein.
Ovarian ligament	- Connects the ovary to the uterus.
Round ligament	- Connects the uterine horn (junction of uterus and Fallopian tube) to the labia majora.
Suspensory ligament	- Connects ovary to the pelvic side wall. - Contains the ovarian artery and vein.
Uterosacral ligament	- Connects the posterior cervix and uterus to the sacrum (hence the name).
Ureter relation:	- “Water under the bridge" is a mnemonic for remembering the relationship between the ureter and the uterine artery. - The uterine artery "bridges" over the ureter as it courses through the pelvis.

	- When performing gynecological surgeries, especially a hysterectomy, the ureter is prone to damage.
Parametrium	- Not a ligament, but instead is connective tissue that connects the broad ligament to the pelvic side wall. I'm mentioning this because there is an NBME Q where they say a fetus is observed on ultrasound in the parametrium of the uterus, and the answer is ectopic pregnancy. So just be aware it is a structure that exists and that ectopics can occur there.

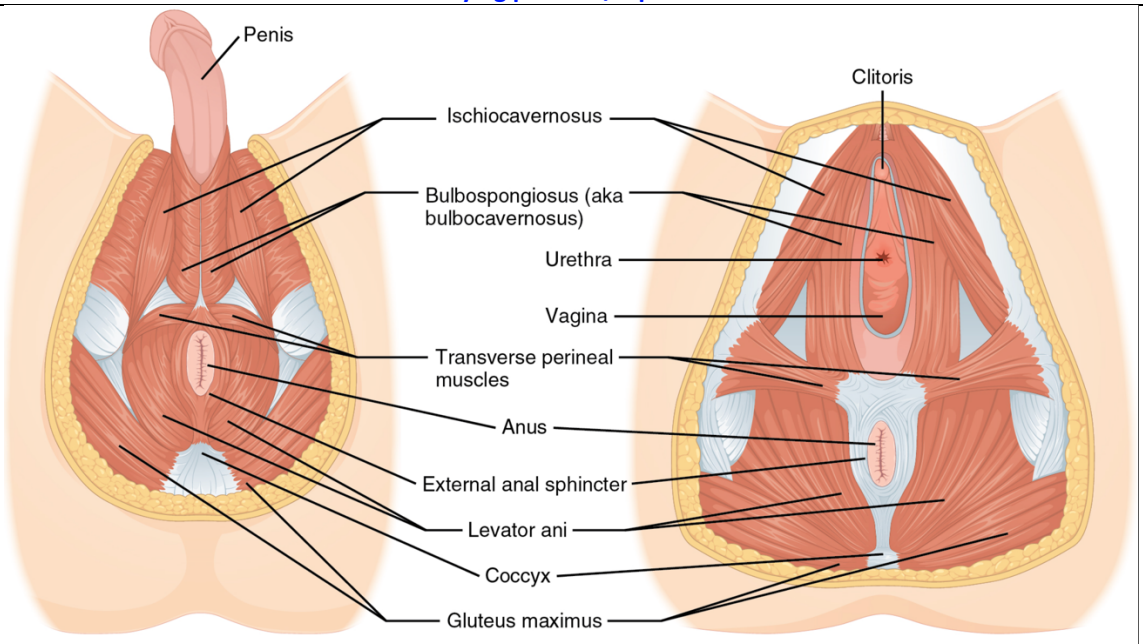
Gynecologic arterial/venous/lymphatic info



Arterial supply	- The uterus supplied by the uterine arteries, which branch off the internal iliacs. - The ovaries are supplied by the ovarian arteries, which branch directly off the abdominal aorta at L2.
Venous drainage	- The uterus is drained by the uterine veins, which feed into the internal iliac veins. - The ovaries are drained by the ovarian arteries, which feed directly into the IVC on the right ("right to IVC") and into the left renal vein on the left.
Lymphatics	- The uterus and cervix both drain to the external and internal iliac lymph nodes. - The ovaries drain to the para-aortic lymph nodes.

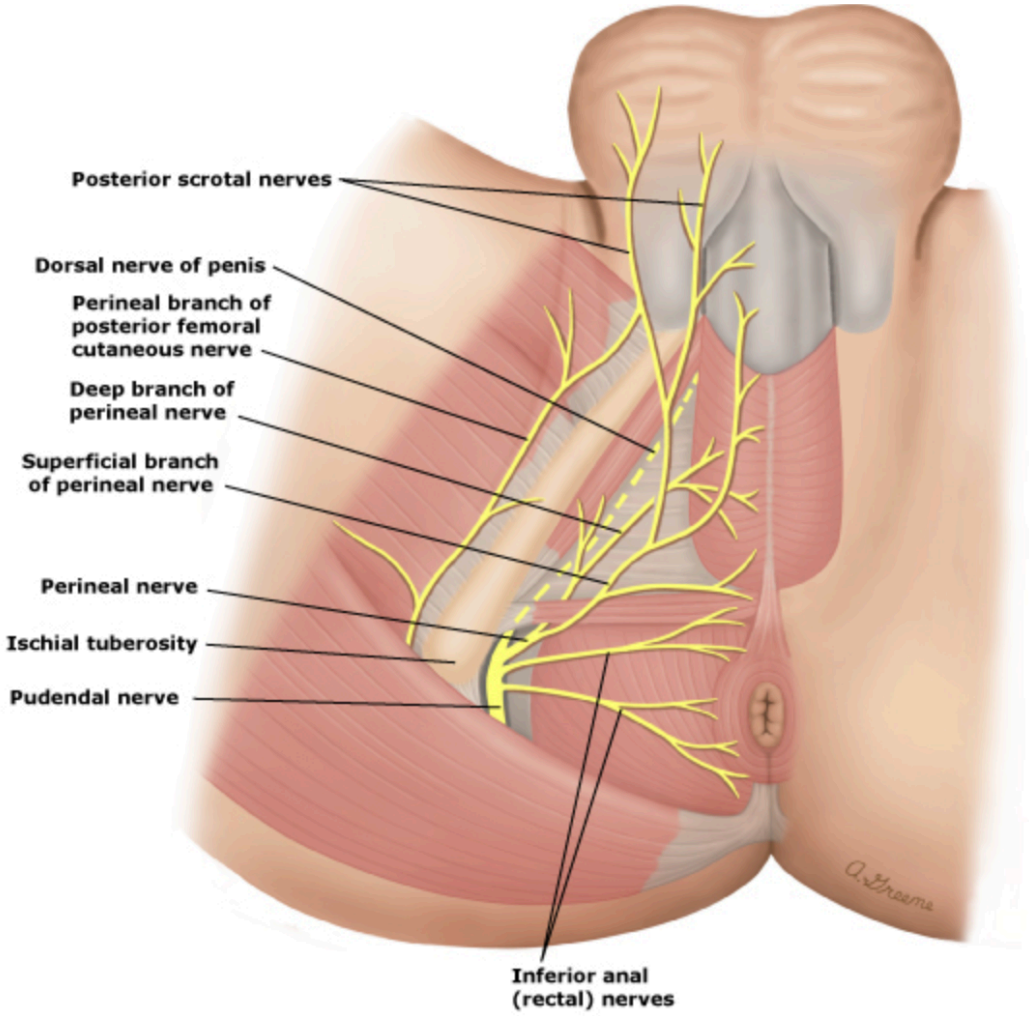
<table>
<tr><td colspan="2">Pelvic floor anatomy
 
</td></tr>
<tr><td>Kegel exercises</td><td>- Aka pelvic floor exercises.
- Used to strengthen the pelvic floor muscles in women who have stress incontinence.
- USMLE will ask which muscle is not strengthened by Kegel exercises. The student says, “Huh? That could be any muscle. The deltoid isn’t for instance.” Answer = internal anal sphincter. The big picture concept is: internal sphincters of the anus and bladder are under autonomic (sympathetic) control, not somatic (voluntary). So you say, “Even though I’m not an obstetrics expert, clearly you can’t voluntarily do exercises to strengthen a muscle that’s not under voluntary control.” The external anal and bladder sphincters, in contrast, are under voluntary control.</td></tr>
<tr><td>Episiotomy</td><td>- Procedure performed prior to Stage 2 of labor (i.e., delivery of the fetus) to cut the vagina posteriorly in the midline. The purpose is to prevent vaginal laceration during delivery of a macrosomic (large) fetus.
- USMLE will ask, “If an episiotomy is performed posteriorly in the midline and the doctor cuts too far, what will he/she cut into?” → answer = external anal sphincter.</td></tr>
</table>

Ultra-annoying perineal/repro nerve stuff



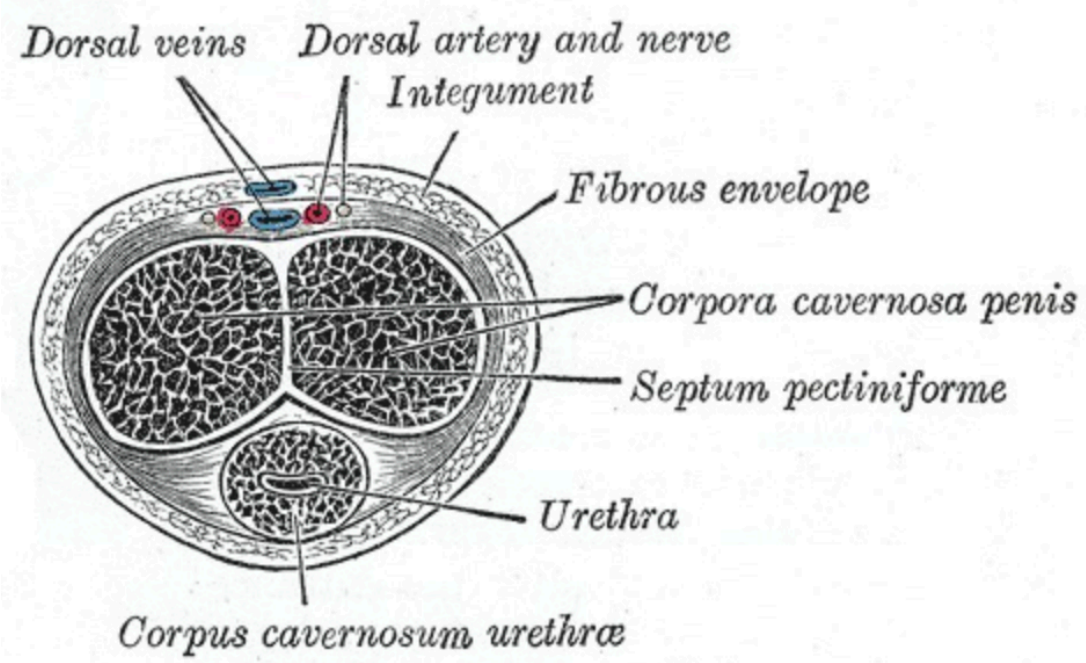
Male perineal muscles: inferior view

Female perineal muscles: inferior view

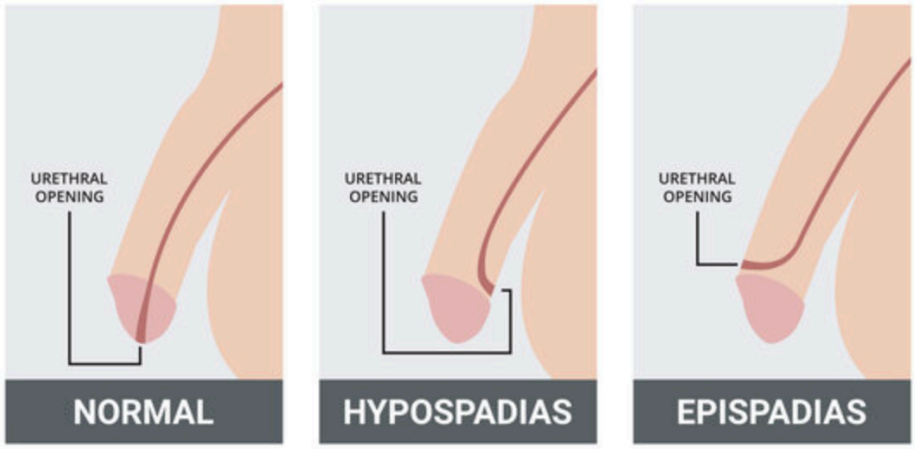


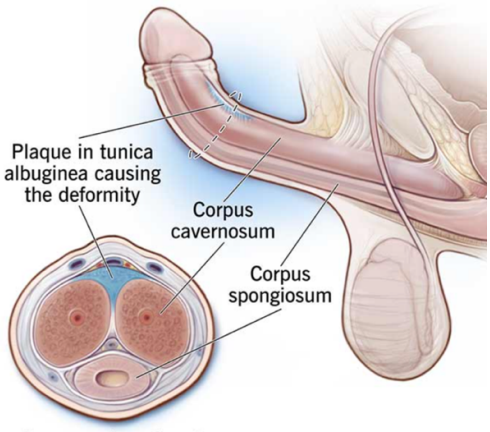
Erection	- Pelvic splanchnic nerves (parasympathetic) from S2-S4 ("2,3,4 keeps the penis off the floor.").
Emission	- Hypogastric nerves (sympathetic) from L1-L2.
Ejaculation	- Pudendal nerve (somatic; voluntary) from S2-S4.
Pudendal nerve	- 1) Main sensory nerve of perineum and external genitalia (i.e., to the penis, scrotum, clitoris, and labia). This sensory aspect facilitates erection due to stimulation, but it is not the pudendal nerve that mediates male/female erection in and of itself; pelvic splanchnic nerves do. - 2) Main motor nerve of pelvic floor muscles (i.e., external anal sphincter, levator ani, etc.) for Kegel exercises. - 3) Pudendal nerve block can be performed during parturition during episiotomy.

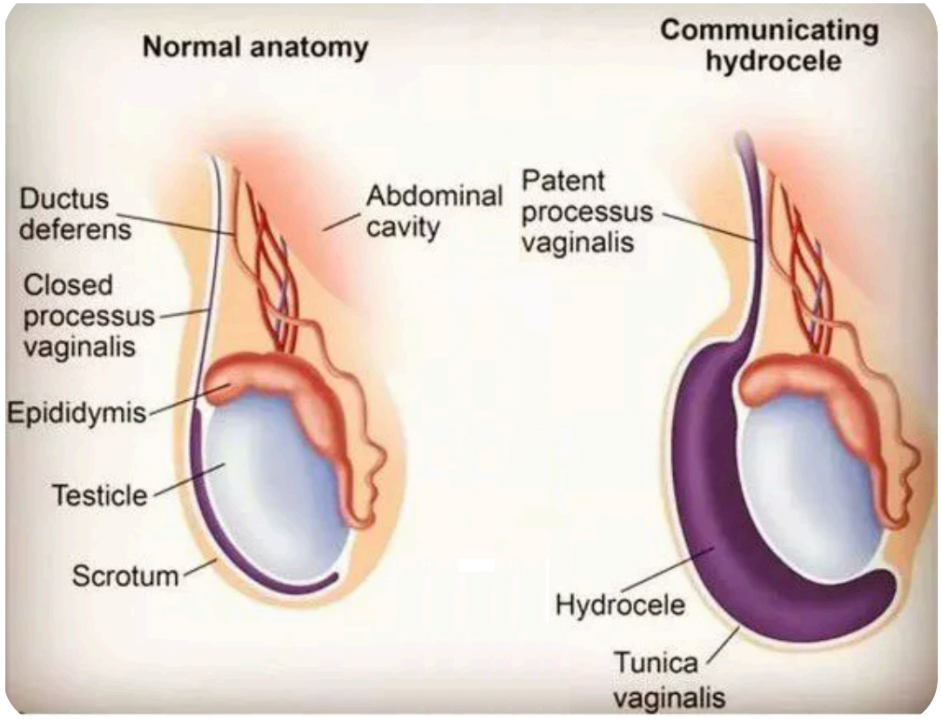
Penile anatomy + DDx

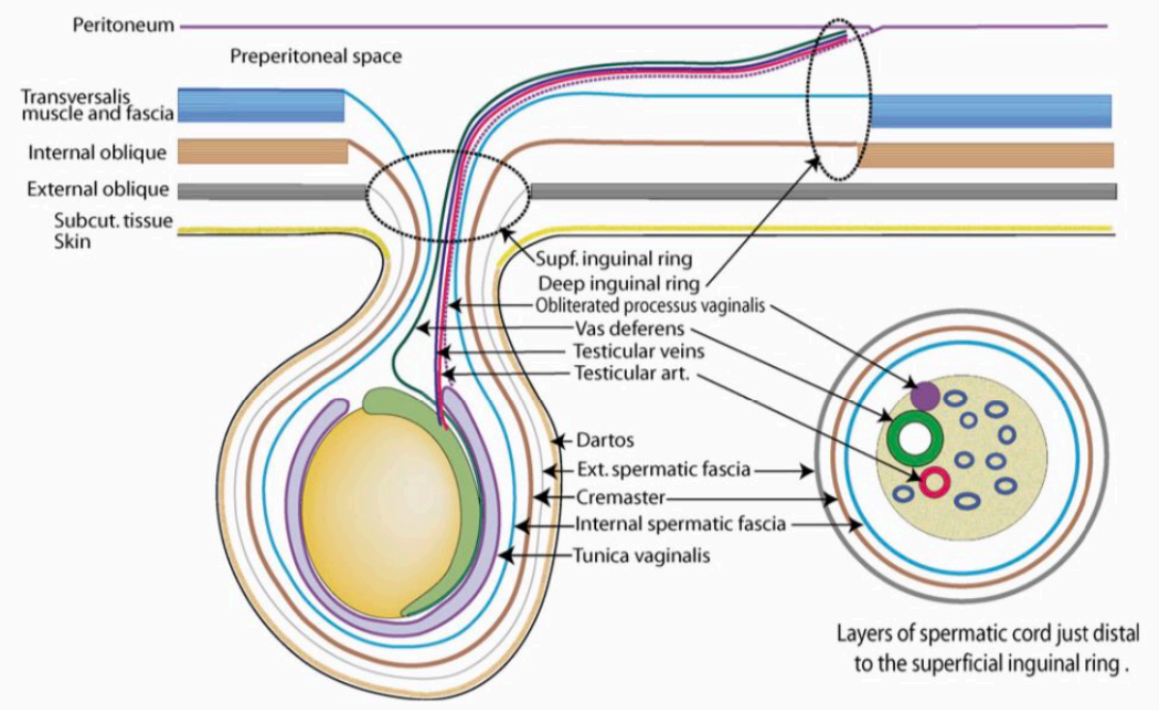
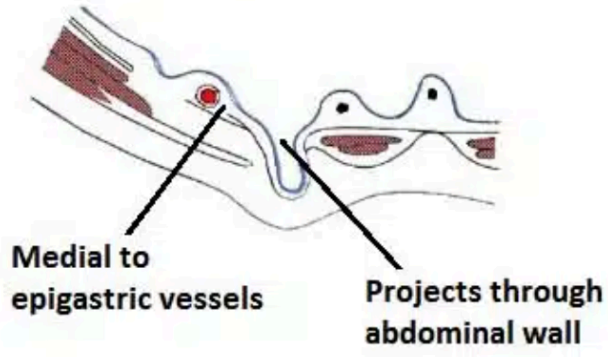
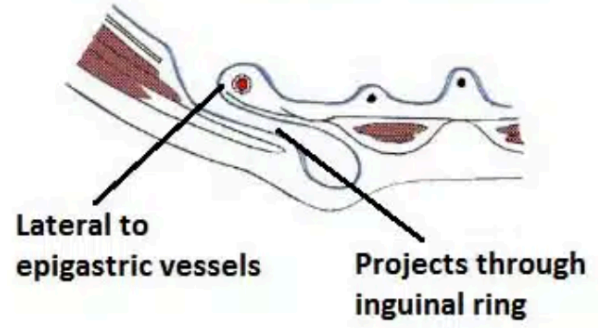


- NBME exam will give cross-section of penis and then ask you for location that sildenafil acts → answer = the erectile muscle. PDE-5 inhibitors act on smooth muscle cells within the small blood vessels inside the erectile tissue, not specifically on the dorsal artery of the penis.

Hypospadias	- Urethral meatus opens on underside of penis (ventral aspect). 

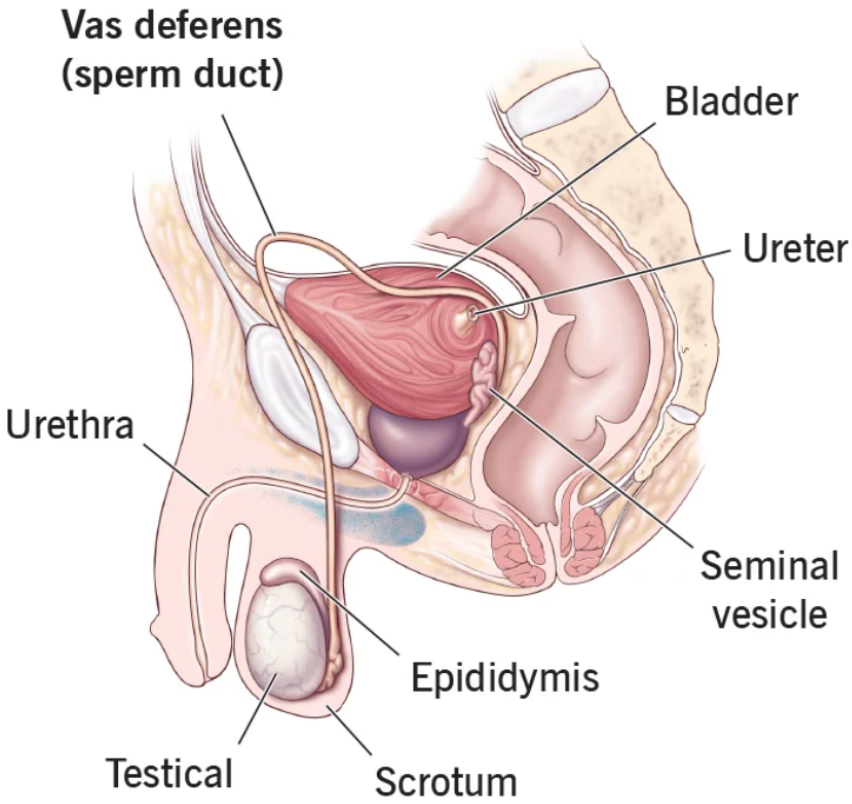
Epispadias	- Urethral meatus opens on top of penis (dorsal aspect).
Phimosis	- Cannot retract the prepuce (foreskin). - In other words, cannot pull it back to expose the glans.
Paraphimosis	- Cannot reduce the prepuce. - In other words, once it is retracted, cannot replace it back over the glans. - Just remember that paraphimosis is the bad one / emergency.
Cryptorchidism	- Undescended testis. - Increased risk of infertility (↑ temperature within abdomen → ↓ spermatogenesis). - Increased risk of testicular carcinoma. - USMLE wants no treatment prior to age 6 months; after 6 months, perform orchidopexy. - NBME Q gives adult with mass in the groin + no testicle palpable in the right hemiscrotum + they ask for what the groin mass is → answer = testicular carcinoma (increased risk because of cryptorchidism). Not hard, but I've seen students miss this.
Testicular torsion	- Acutely painful testis; medical emergency; can lead to ischemia and loss of testis. Can be caused by abnormality of the gubernaculum, which is a ligamentous cord that anchors the testis within the scrotum. This leads to "bell-clapper deformity," where improper anchoring = ↑ risk of rotation. - Negative Prehn sign, which is failure of relief of pain upon lifting of the scrotum. Apparently this is an unreliable test in real life, but NBME vignettes can still mention it. In contrast, a (+) Prehn (relief of pain upon lifting the scrotum) is more indicative of epididymitis. - Negative cremasteric reflex, which is failure of the scrotal skin to retract upon palpation of the medial thigh/scrotum. In contrast, a (+) cremasteric reflex (scrotal skin retracts) is more indicative of epididymitis. Normally, when the inner thigh / scrotum is stroked, the cremaster muscle contracts and pulls up on the ipsilateral testis. In torsion, the twisted spermatic cord can impair the nerve fibers responsible for triggering the reflex. - USMLE wants Doppler ultrasonography as next best step, followed by surgery.
Epididymitis	- Inflammation/infection of epididymis. - (+) Prehn sign and (+) cremasteric reflex. - Chlamydia and gonorrhea are most common causes in males <35. - Over 35, *E. coli* is most likely. - The USMLE will not play games where they give you a borderline case and force you to guess. They will give you either a 25- or 45-year-old male. This also goes for prostatitis (same organisms). There is an NBME Q of a 45-year-old where the answer is *E. coli*, and chlamydia is wrong.
Torsion of appendix testis	- Not the same as torsion of testis. - Asked on a 2CK Peds form. - Q will tell you there's a kid with a painful testis + the superior pole is blue + there's an intact cremasteric reflex ("regular torsion" it's not intact). - "Blue dot" sign = blue superior pole of testis. - Not my opinion if you think it's nitpicky or weird. Take it up with the NBME exam.
Peyronie disease	- Curvature deformity of penis caused by fibrous plaque within tunica albuginea.  - Doesn't have to be treated unless causing functional impairment.

Orchitis	- Inflammation/infection of the testis. - Mumps is HY cause in unvaccinated / immigrants (sometimes implies unvaccinated). - If not caused by mumps virus, then the bacterial causes are same as epididymitis and prostatitis for young vs old (i.e., STIs vs *E. coli*, respectively).
Priapism	- Erection lasting longer than 4 hours. - Can be caused by sickle cell or by PDE-5 inhibitors (i.e., tadalafil/sildenafil). - Injection of alpha-1 agonist (i.e., phenylephrine) into the erectile tissue is treatment → decreases blood flow due to blood vessel constriction.
Hydrocele	- Serous fluid collection within the scrotum between the layers of the tunica vaginalis surrounding the testis. - Mechanism is patent processus vaginalis. - The processus vaginalis is an embryonic outpouching of peritoneum that descends into the scrotum. It normally obliterates shortly after birth, but if it remains open, can cause hydrocele or indirect inguinal hernia.  - Transilluminates when a light is shone to it. - USMLE wants no treatment before the age of 1; most spontaneously close.
Spermatocele	- Obscure serous fluid collection within the epididymis containing sperm. - Occurs usually in adult men (rather than neonates as with hydrocele). - Transilluminates similar to hydrocele. But once again, it's an older adult, not neonate.
Varicocele	- Dilation/congestion of pampiniform venous plexus draining the scrotum. - Does not transilluminate (in contrast to hydro- and spermatoceles). - Can be described as buzzy "bag of worms." Sounds too easy, but an NBME Q literally uses this colloquialism. - Can also be described as a heavy, dragging scrotum lower on the left. - Occurs almost always on the left side due to the anatomy of venous drainage of the testis. - On the left, we have left testicular vein → left renal vein → IVC. - On the right, we have right testicular vein → IVC ("right to IVC"). - The left testicular vein drains at a 90-degree angle into the left renal vein, creating a proclivity for hydrostatic pressure backup ipsilaterally. - There is a 2CK NBME Q where bilateral varicocele is the answer. Highly obscure, but just be aware it's somehow possible/exists. - Can occur due to left renal vein thrombosis in setting of nephrotic syndrome (loss of antithrombin III in the urine → hypercoagulable state).

<table>
<tr><th colspan="2">Spermatic cord + hernias</th></tr>
<tr><td colspan="2"> 
</td></tr>
<tr><td>Direct inguinal hernia</td><td>- Abdominal contents herniate medial to inferior epigastric vessels within an area known as Hesselbach triangle, which is bordered by the inguinal ligament (inferior border), rectus abdominis muscle (medial border), and the inferior epigastric vessels (lateral border).
- The weakening of the abdominal wall can be idiopathic (aging), chronic straining, or previous surgical incisions.
 
</td></tr>
<tr><td>Indirect inguinal hernia</td><td>- Abdominal contents herniate lateral to inferior epigastric vessels due to a patent processes vaginalis.
- Follows the path of the inguinal canal into the scrotum.
 

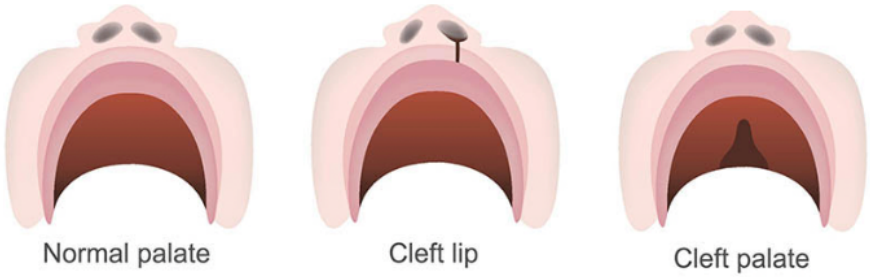
- I discuss all of the different hernia types in detail in the HY Anatomy/MSK PDF.</td></tr>
</table>

More male anatomy

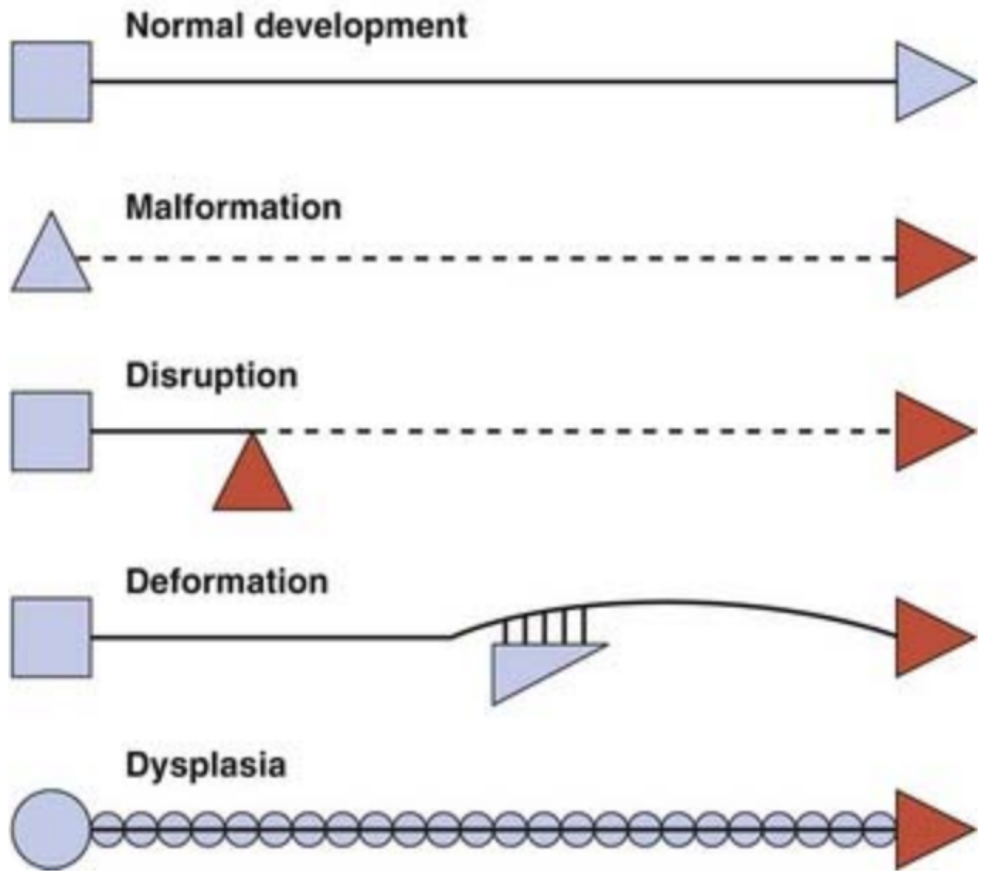


- Congenital bilateral absence of vas deferens (CBAVD) is seen in cystic fibrosis (absent sperm in sample).
- In primary ciliary dyskinesia (Kartagener), the vas deferens is present, but the sperm simply have impaired motility. In women, there can be ectopic pregnancy due to impaired Fallopian tube cilia function.
- In other words, recurrent pulmonary infections in male + absent sperm = CF; recurrent pulmonary infections in male + poorly motile sperm = Kartagener.
- I discuss CF and Kartagener in extensive detail in my HY Pulmonary PDF.

Cleft lip/palate



Cleft lip	- Failure of fusion of the maxillary and nasal prominences. - Polygenic; occurs 1 in 1,000-5,000 depending on population. - Cause on USMLE is “one-off developmental anomaly” (i.e., it is usually not part of a greater syndromic condition). This is same for pyloric stenosis and ASD/VSD. - If a couple has child with cleft lip or palate, the chance of having 2nd child with it is 3-5%. - If one of the parents has cleft lip or palate, chance of any child having it is 10-15%.
Cleft palate	- Failure of fusion of the lateral palatine processes (aka palatal shelves).

Birth defects	
	
Malformation	- Intrinsic/innate abnormality in the structure of an organ or body part due to genetic factors or early environmental influences like teratogens. - Spina bifida, cleft lip/palate, congenital heart defects, phocomelia (severe flipper-like malformation of limbs; caused by thalidomide).
Deformation	- An initially normal organ or body part becomes abnormally shaped later in development, usually due to mechanical forces, such as uterine compression or oligohydramnios. - Club foot (talipes equinovarus), congenital hip dysplasia, plagiocephaly (abnormal head shape).
Disruption	- An initially normal organ or body part is interrupted by a destructive event, leading to the breakdown of a body structure. - Amniotic band syndrome → fibrous bands from the amniotic membrane classically entangle a fetal limb or fingers, leading to amputation.

HY incontinences for USMLE	
Stress	- The answer for loss of urine with ↑ intra-abdominal pressure from laughing, sneezing, coughing. - Stereotypical risk factor is grand multiparity (i.e., Hx of many childbirths) leading to weakened pelvic floor muscles. - I'd say only ~50% of Qs will mention Hx of pregnancy. The other ~50% are idiopathic. - Buzzy vignette descriptors and answer choices are: "downward mobility of the vesicourethral junction," "urethral hypermobility," and "urethral atrophy with loss of urethrovesical angle." - Treatment is pelvic floor (Kegel) exercises. These notably strengthen **levator ani, pubococcygeus,** and the **external urethral sphincter.** - USMLE is known to ask which muscle is *not* strengthened by Kegel exercises, which sounds obscure, since any muscle could theoretically be the answer ("Well the deltoid

	isn't strengthened."). But a favorite answer here is **internal urethral sphincter.** The way you know this is the answer is because internal sphincters are under sympathetic (i.e., involuntary; autonomic) control, which means it's impossible to strengthen it via a voluntary (i.e., somatic) exercise. USMLE doesn't expect you to be an obstetrician. The bigger picture concept is simply knowing internal sphincter control is involuntary. It is external sphincter control that is voluntary (somatic). - Do **not** give medications for stress incontinence on USMLE. - If Kegel exercises fail, patients can get a mid-urethral sling (LY; asked once).
Urge	- The answer on USMLE for patient who has an "urge" (NBME will literally say that word and I've seen students get the Q wrong) to void 6-12+ times daily **unrelated** to sneezing, coughing, laughing, etc. (otherwise stress incontinence). - Ultra-HY for **multiple sclerosis.** I've had students ask whether MS is urge or overflow. It shows up repeatedly on the NBMEs as urge; I've never seen it associated with overflow. I'd say ~1/3 of urge incontinence vignettes on NBME forms are MS. - Other vignettes will be peri-menopausal women, or idiopathic in old women. - Mechanism is "detrusor hyperactivity," or "detrusor instability." - Vignette can mention woman has urge to void when stepping out of her car, or when sticking her key in the car/front door of her house. Sounds weird, but these are important Qs to ask when attempting to diagnose urge incontinence. - UTIs can present similarly to urge incontinence. Some students have asked, "Well isn't that because UTIs are a cause of urge incontinence?" Not really. It just happens to be that UTIs can sometimes cause transient urinary urgency. For example, if they give a Q where they say patient had Hx of urinary catheter + now has dysuria and urinary urgency, answer = "urinary tract infection" on NBME; "detrusor hyperactivity" is wrong answer. - Treatment is **oxybutynin** (muscarinic receptor antagonist); this ↓ activity of the detrusor muscle of the bladder. - Some students get hysterical about mirabegron (β3-agonist), but I've never seen NBME forms assess this.
Overflow	- Will be due to either BPH or diabetes on USMLE. - Will have ↑ **post-void volume.** Normal is < ~50 mL. On USMLE for overflow, they'll give you 300-400 mL as post-void volume. - As I talked about earlier for BPH, they will give old dude + high creatinine (post-renal azotemia). Next best step is "insertion of catheter" to relieve the obstruction. If they don't have this listed, "measurement of post-void volume" can be an answer. We then treat the BPH with finasteride (5α-reductase inhibitor) or an α1-blocker (tamsulosin, terazosin). - For diabetes, the mechanism is neuropathy to the bladder causing "neurogenic bladder," or "hypotonic bladder," or "hypocontractile bladder/detrusor muscle." - For neurogenic bladder causing overflow incontinence + ↑ post-void volume, remember that USMLE is first obsessed with "measure post-void volume" and "insertion of catheter" if they are listed. They will not force you to choose between the two. But they like these answers prior to giving medications. - Give **bethanechol** (muscarinic receptor agonist) for hypotonic bladder due to diabetes; this stimulates the detrusor muscle. - Making sure you don't confuse oxybutynin and bethanechol is pass-level for USMLE. - Spinal trauma, spinal metastases, and cauda equina syndrome can also sometimes cause urinary retention with overflow incontinence. The NBME Q might give you patient with history of cancer who now has urinary retention + positive Babinski sign, and the answer is just "epidural spinal metastases," as an example.

Prolapses (2CK/3 only)	
Uterine prolapse	- Descent of the uterus into the vagina due to weakening of the pelvic floor muscles. - Can present as sensation of heaviness/fullness in the pelvis (i.e., "sitting on a ball"), or as a visible or palpable vaginal bulge.
Cervical incompetence	- Premature dilation and shortening of the cervix during pregnancy, without contractions, usually in second trimester. - Can lead to premature rupture of membranes and preterm birth. - Risk increased with history of excision procedures such as coning or LEEP. - USMLE Q will tell you one of two things: 1) woman is pregnant and that "fetal parts are palpable in the vagina," or 2) "funneling" is observed, which refers to the amniotic sac protruding into the cervical canal; it means that the internal os is dilating prematurely. - For example, 2CK Obgyn CMS Q gives massive nonsense paragraph stem with woman at 14 weeks' gestation + they mention at bottom of stem "funneled lower uterine segment" → answer instantly = cervical incompetence.
Urethral prolapse	- The urethra protrudes outward through the external urethral meatus. - Most common in prepubertal girls and postmenopausal women.

The "celes" (2CK/3 only)	
Urethrocele	- Herniation of urethra into the vagina. - **Anterior inferior vaginal wall.** - Can present with stress incontinence, recurrent UTIs, or a palpable bulge in the anterior inferior vaginal wall. - There is one 2CK Obgyn Q where they say the vaginal wall appears well-supported (i.e., doesn't sound like a urethrocele), but then they say a cotton-tipped applicator placed into the urethra arcs at a 45-degree angle when the woman coughs → answer = urethrocele. Call it weird all you want, but it's on the CMS form.
Cystocele	- Herniation of bladder into the vagina. - **Anterior superior vaginal wall.** - Similar to urethrocele, can present with stress incontinence, recurrent UTIs, or a palpable bulge on the vaginal wall (anterior superior).
Rectocele	- Herniation of the rectum into the vagina. - **Posterior inferior vaginal wall.** - Can present as difficulty with bowel motions.
Enterocele	- Herniation of small bowel into the vagina. - **Posterior superior vaginal wall.** - NBME Q will tell you a woman feels "movement" within her vagina, which refers to the peristalsis of the bowel. - I've seen NBME write "superior" wall as "high on the vaginal wall."
Cystovaginal fistula	- Including this here because I've seen it as a distractor in quite a few "cele" Qs. - Abnormal communication between the bladder and the vagina, leading to continuous, involuntary leakage of urine into the vagina. - Can be caused by prior obstetric trauma (e.g., prolonged labor), surgery, or radiation therapy due to cancer (e.g., cervical). - I've seen it once as a correct answer for woman who had history of radiation treatment for cervical cancer. NBME will say there is "pooling of urine" in the vagina.

Hydatidiform moles
- Nonviable pregnancies that result in abnormal proliferation of trophoblastic (placental) cells. - Considered a form gestational trophoblastic disease (GTD), which is an umbrella term that encompasses moles and choriocarcinoma. But you should know it for USMLE. In other words, if you see GTD as an answer choice, you should think, "that refers to either a mole or choriocarcinoma."

- Moles present on USMLE as a pregnant woman whose uterus is much larger than expected for gestational age + has β-hCG levels that are super elevated. - The NBME Q can tell you the female is 16 weeks pregnant, where the uterine fundal height is measured at the umbilicus (normally where it is at 20 weeks), and her β-hCG level is in the hundreds of thousands. - The Q can also give you a presentation that looks like preeclampsia, where the female has hypertension and proteinuria, but she's under, not over, 20 weeks' gestation. I discuss preeclampsia and eclampsia later in this PDF, but you should think, "sounds like preeclampsia but <20 weeks = mole till proven otherwise."	
Complete mole	- Empty egg fertilized by a single sperm. The sperm then duplicates its DNA. The resultant karyotype is usually 46XX. The karyotype 46YY is not compatible with formation of hydatidiform mole, apparently. - Results in "snowstorm" appearance on ultrasound, or as "bunches of grapes" on gross pathology. There are no fetal parts present. - Both complete and partial moles have high risk of progression to invasive mole + choriocarcinoma, which is overt cancer. But if you're forced to choose which carries higher risk of progression, the answer is complete. - USMLE wants surgical evacuation, either via D&C or suction curettage, as the Tx.
Partial mole	- Aka incomplete mole. - Normal egg fertilized by two sperm. Resultant karyotype is 69XXX, 69XXY, or 69XYY. - Fetal parts are present on ultrasound / gross path specimen.
Invasive mole	- Form of GTD where molar tissue penetrates into the myometrium. - Carries the potential for metastasis and progression to choriocarcinoma. - Gross path will show presence of chorionic villi (in contrast to choriocarcinoma).
Choriocarcinoma	- Cancer of syncytiotrophoblastic (i.e., placental) tissue. - Usually develops from hydatidiform moles, but not always. - Produces super-elevated β-hCG levels, same as moles. - Loves to metastasize to lungs and brain. USMLE will give a woman who has ultra elevated β-hCG with pulmonary nodules or stroke-like presentation → answer just = choriocarcinoma. - Described as "bloody mess" on gross path; no chorionic villi present (in contrast to invasive mole).

Ovarian neoplasms	
Serous cystadenoma	- Has Fallopian tube-like epithelium. - Can occur bilaterally. - Benign.
Serous cystadenocarcinoma	- Can have psammoma bodies. - 20-30% bilateral at time of diagnosis. - Malignant (i.e., has metastatic potential). I discuss tumor nomenclature in detail in the HY Path PDF.
Mucinous cystadenoma	- Has intestine-like epithelium. - Loculated or locular (meaning, has tiny cavities, like honeycomb). - Benign.
Mucinous cystadenocarcinoma	- Same as mucinous cystadenoma, but malignant. - Can cause pseudomyxoma peritonei, which is accumulation of mucin-producing tumor cells and gelatinous ascites in the peritoneal cavity.
Dysgerminoma	- Seminoma equivalent in females. - Large uniform cells with clear cytoplasm (same as seminoma). - Highly sensitive to chemo- and radiotherapy (same as seminoma). - Can occur idiopathically or in Turner syndrome. - LDH is a tumor marker.
Sertoli-Leydig cell tumor	- Produces androgens. - Can cause virilization in females. - Rarely can also produce estrogen. There is NBME Q where male has gynecomastia + they ask for which tumor the patient has → answer = Sertoli-

	Leydig cell tumor, where granulosa cell tumor isn't listed. Implication is, either the SL cell tumor produces estrogen in this case, or there is considerable aromatization of androgens into estrogens causing gynecomastia.
Granulosa cell tumor	- Produces estrogen. - Can cause endometrial hyperplasia and ↑ risk of endometrial cancer due to unopposed estrogen. - NBME Q gives female with ovarian neoplasm + ↑ thickness of endometrial stripe + vaginal bleeding → answer for ovarian lesion = granulosa cell tumor.
Yolk sac tumor	- Aka endodermal sinus tumor. - The answer on USMLE for ovarian tumor in pediatrics (usually up to age 3). - Secretes AFP as tumor marker. - Schiller-Duval bodies on histo, which resemble glomeruli.
Embryonal carcinoma	- Aggressive and rapidly growing. - Undifferentiated/anaplastic cells on histo. - Secretes both AFP and hCG together as tumor markers. - Usually occurs in adult men, but can be seen as ovarian tumor also. The key for USMLE is that it's the answer if AFP and hCG are both high, whereas if only AFP is high, then the answer is yolk sac tumor. You can also remember yolk sac tumor is kids, whereas embryonal cancer is usually adults.
Brenner tumor	- Benign ovarian tumor that has "bladder-like" epithelium. In other words, histo shows urothelium (i.e., transitional cells). - Can appear tan/yellow grossly.
Struma ovarii	- Benign ovarian tumor that secretes thyroid hormone.
Krukenberg tumor	- Gastric metastases to the ovaries bilaterally. - Has mucin-containing signet-ring cells on biopsy. - Spread to ovaries from stomach is usually hematogenous.

Endometrial cancer
- Highest yield point on USMLE is that it is caused by "unopposed estrogen," usually due to anovulation/PCOS. - Estrogen normally stimulates growth of endometrium; progesterone prevents overgrowth. - In anovulation/PCOS, no corpus luteum is formed, since the latter is the follicular remnant. - The corpus luteum normally secretes progesterone to maintain the endometrial lining if fertilization occurs. If no corpus luteum is formed, there's no progesterone produced to balance estrogen. This leads to endometrial hyperplasia, which increases the risk of endometrial adenocarcinoma. - USMLE can give an overweight female who's perimenopausal who has breakthrough bleeding (i.e., mid-cycle bleeding), or a post-menopausal woman with any vaginal bleeding, and they want endometrial biopsy as the answer. The fact that the patient has high BMI insinuates that anovulation/PCOS may have been a part of her past, where unopposed estrogen is a part of her history. - Can occur secondarily as a result of granulosa cell tumor of the ovary (as discussed above). This is a rare cause of unopposed estrogen. - Can occur in women who take hormone-replacement therapy (HRT) who stop taking the progesterone component (mentioned in one NBME Q). - Can be associated with hereditary non-polyposis colorectal cancer (HNPCC; Lynch syndrome).

Fibroid points
- Fibroids are aka leiomyomata uteri and are most common benign tumor in women. - They are smooth muscle tumors of the myometrium. - Stain positive for desmin (muscle marker); can appear grossly as white tumors with whorled structure. - Often found incidentally on ultrasound or autopsy. - If fibroids are picked up incidentally on ultrasound, the answer is observe / follow-up. The highest yield point for USMLE is that these are not managed almost always.

- There is negligible risk of progression into leiomyosarcoma (malignant version that doesn't exist on USMLE). In other words, do not choose answers like myomectomy. - If a woman has bleeding due to fibroids, OCPs and NSAIDs can be attempted in theory, although I have not seen USMLE assess these. Other treatments like leuprolide are also nonexistent on USMLE. - **(2CK/3 only):**	
Submucosal	- Closer to endometrial side of myometrium; can bleed.
Subserosal	- Closer to outside of uterus; doesn't bleed per vaginum; can cause globular uterus.
Intramural	- Features common to both submucosal and subserosal.
Pedunculated	- Grow as stalk-like structures. An obscure NBME Q for 2CK mentions a beefy red mass protruding from the cervix → answer = pedunculated submucosal leiomyoma; cervical cancer is wrong answer; "beefy red" tends to mean endometrial in origin; cervical SCC will usually be an ulcerated exophytic mass.
Parasitic	- Detach from the uterus and become attached to adjacent structures such as the bladder.

Vulvovaginal lesions	
Vulvovaginal carcinoma	- Cancer of the vagina and/or vulva. - Almost always squamous cell carcinoma due to HPV 16 or 18. - Can occur rarely due to lichen sclerosus.
Condylomata acuminata	- Aka warts; caused by HPV 6/11. - Present as painless, skin-colored or slightly hyperpigmented cauliflower-like papular lesions. - I discuss all of the STDs in more detail later.
Lichen sclerosus	- Presents as whitish-grey, rough, irritated or scratchy patch on the vulva or perineum in usually woman >50 years. - Thought to be caused by a mix of chronic irritation and autoimmunity; not HPV-related. - Characterized by atrophy, hyperkeratosis, and a band of lymphocytes in the dermis. It is not neoplasia, dysplasia, or metaplasia. - USMLE wants biopsy as next best step to rule out SCC (on an old Step 1 Free 120 even though NBS Qs are classically 2CK). If SCC is negative and the diagnosis is LS, USMLE wants topical steroid as treatment.
Sarcoma botryoides	- Rare as fuck. Rhabdomyosarcoma of the vagina. - Seen in pediatrics as bunches of grapes protruding from the vagina.
Clear cell vaginal carcinoma	- Obscure cancer that occurs in women 30s-50s due to diethylstilbestrol (DES) exposure *in their mothers* while they were pregnant 30-50 years ago.
Bartholin gland cyst	- Shows up as a tender/painful bump at a 4- or 8-o'clock position on the vulva. - If the cyst becomes infected (i.e., warm and red), we call it Bartholin gland abscess. - USMLE wants "polymicrobial" as the most likely organism. - For cysts, sitz bath + warm compresses are first Tx. - For overt abscesses, drainage is the answer.
Vaginal foreign body	- Vignette will be a child with foul-smelling discharge from the vagina. - There will be no signs of physical trauma or lacerations (means not abuse).
Sexual abuse	- Can occur in children or elderly (I've seen both on NBME). - Vignette likes to mention lacerations. - There may or may not be discharge.

Cervical cancer + Pap smear screening (2CK/3 only)
- Squamous cell carcinoma; occurs at the transitional zone between the stratified squamous ectocervix and columnar endocervix. - Caused by HPV 16/18. There are other strains students get pedantic about, but USMLE doesn't give a fuck. - Can cause vaginal bleeding + uni- or bilateral hydroureter / -nephrosis. - Pap smears should be started at age 21. If the vignette gives you a teenager who's sexually active, the answer is you test for STDs but do not do a Pap smear. - Perform Pap smear age 21-65. - From age 21-30, they are done every 3 years without HPV co-testing. - From age 30-65, the female has the option of continuing Paps every 3 years, or doing Pap-HPV co-testing every 5 years. - Pap smears are stopped at age 65. - If the female has a history of a high-grade squamous intra-epithelial lesion (HSIL) or cervical intra-epithelial neoplasia (CIN), Pap smears are continued after age 65. - Patients who have had hysterectomy due to cervical cancer should have continued Pap smears of the vaginal cuff. Patients with hysterectomy due to endometrial cancer do not. - Patients with HIV require two Pap smears in the first year following diagnosis (6 months apart), followed by annual Pap smears thereafter. The vignette will tell you HIV patient had normal Pap smear last year, and student is like, "Cool, she doesn't need one for another 2 years." No. This is important. You need to know HIV patients need once/year Paps. - Low-grade squamous intra-epithelial lesions (LSIL) and atypical squamous cells of undetermined significance (ASC-US) are managed with either repeat Pap smear or HPV co-testing, where (+) vs (-) HPV test determines whether colposcopy + biopsy is performed. **USMLE will not** assess the algorithmic specifics. If you memorize the algorithm based on the female's age, you're wasting your time for USMLE purposes. - HSIL on Pap smear is managed with immediate colposcopy + biopsy. - CIN I is what we call LSIL that is confirmed on colposcopy + biopsy. - CIN II/III is what we call HSIL that is confirmed on colposcopy + biopsy. - CIN I has high rate of spontaneous regression. Repeat Paps are done in a year. - CIN II/III requires excisional or ablational treatment due to higher risk of progression to invasive cervical cancer. Such treatments include LEEP, coning, cryotherapy, or laser ablation. - Cone biopsy is also an answer on NBME for next best step in patient who has colposcopy performed for HSIL on Pap + "the entire squamocolumnar junction cannot be visualized." In other words, do a cone biopsy if colposcopy is insufficient. - CIN III is **not** excised/ablated during pregnancy. HPV lesions (including warts) are known to get temporarily worse during pregnancy due to relative immunosuppression (biologic attempt to minimize attack against fetal antigens). In the post-partum period, CIN III lesions are evaluated for signs of regression over the course of weeks; if they do not regress, they are excised same as non-pregnant women with CIN II or III.

Cervical conditions confused with cancer (2CK/3 only)	
Cervical trauma	- Will present as vaginal bleeding in young woman usually within 12 hours of intercourse. - Doesn't sound hard/dramatic, but 2CK Qs can make vignettes somewhat nebulously where you second guess, i.e., they might say there's blood on the underwear. The key is that part of the Hx will mention post-coital bleeding.
Cervical polyp	- Benign, non-cancerous, pedunculated tumor that grows typically on the ectocervix. - Idiopathic. - Can cause metrorrhagia (mid-cycle bleeding). - Will not be described as ulcerated. If they say ulcerated, think cervical SCC.

Breast neoplasia + HY points	
Ductal carcinoma *in situ* (DCIS)	- Most common malignant (i.e., has metastatic potential) breast cancer. - *In situ* means confined superficial to basement membrane. - "Microcalcifications" on mammography = DCIS till proven otherwise. - 2CK NBME Q gives breast lesion with microcalcifications + answer is just "needle-guided open biopsy," where FNA is wrong. - Paget disease of breast (looks like a nipple with eczema in patient over 50) is often the cutaneous extension of an underlying DCIS.
Invasive ductal	- Invasive form of ductal carcinoma. - Cells on biopsy have stellate morphology.
Lobular carcinoma *in situ* (LCIS)	- Same as with DCIS, has malignant potential but is still confined superficial to basement membrane. - Sometimes bilateral. In other words, if a Q ever mentions bilateral breast cancer, diagnosis is lobular till proven otherwise. But you have to use your head. If the Q says stuff like "microcalcifications" or "stellate morphology," for instance, you know you're dealing with ductal carcinoma instead. - Linear rows of cells ("Indian file") on histo. This is supposedly due to ↓ E-cadherin expression, where the cells don't clump together the same way they do as with ductal carcinomas. I haven't seen USMLE give a fuck about this detail, but I'm mentioning it because if you ever talk to an attending about breast cancers, they become like hysterical over this detail.
Invasive lobular	- Invasive form of lobular carcinoma. - 20-30% bilateral.
Intraductal papilloma	- Unilateral bloody/rusty nipple discharge = intraductal papilloma till proven otherwise.
Inflammatory carcinoma	- Can appear as a red/inflamed breast, as though there's an infection. - Rubor (redness), dolor (pain), tumor (swelling), calor (heat). - Peau d'orange is an orange peel-appearing texture of the breast due to tethering of the skin by Coopers ligaments in the setting of blockage of lymphatic drainage by tumor cells.
Comedocarcinoma	- Cancer with cheese-like consistency. Nonexistent on USMLE. - Don't think I've ever seen it assessed.
Medullary carcinoma	- Another nonsense / garbage diagnosis on USMLE. - Mentioning comedo- and medullary carcinomas because if I don't, then at some point I'll get some whiny/mousy DMs from students about why I didn't mention them.
Fibroadenoma	- Most common breast tumor overall; benign. - Presents as rubbery, mobile, non-tender breast lump in woman 20s-40s. - Will appear as solid breast mass without calcification on ultrasound. If calcifications, think ductal carcinoma instead. - FNA is next best step to confirm Dx. If confirmed fibroadenoma, they do not need to be excised.
Cystosarcoma phyllodes	- Fast-growing breast lesion that appears "leaf-like" on light microscopy. - Can be benign or malignant based on histo.
BRCA points	- *BRCA1/2* are tumor suppressor genes but are inherited in an autosomal dominant fashion with incomplete penetrance (i.e., sometimes can skip a generation). A pedigree requiring you know this pattern is asked on NBME. - NBME also wants you to know that, with *BRCA* mutations, the molecular process that's fucked up is "recombinational double-stranded DNA repair." - 2CK Q has "bilateral mastectomy and oophorectomy" as the next best step in women who have confirmed *BRCA* mutations. Might sound overkill as NBS, but it's answer on an NBME.
Screening/Mx points **(2CK/3 only)**	- For USMLE, mammography is done every two years between ages 50-74. - Since committees / cancer task forces have varying recommendations, the USMLE doesn't play trivia as far as "oh em gee do we start at 45 vs 50, etc."

	They won't be borderline about it. They will just give you a patient who's, e.g., 56 who needs a mammogram since she hasn't had one for two years. - If first degree family member (i.e., parent or sibling) has Hx of breast cancer, then start mammography at age 40, not 50. - Any breast screening done for a women under 30, do not do mammography on USMLE. Ultrasound is typically done. This is because younger women have denser breast tissue, making mammography less specific (i.e., more false-positives) and hence less reliable of a diagnostic modality. - Estrogen and progesterone receptor (ER/PR) positivity is correlated with better prognosis for breast cancers. This is partially because SERMs and aromatase inhibitors can be used as Tx. - HER2/neu positivity is correlated with worse prognosis. Trastuzumab (Herceptin) can be used to target HER2/neu. - Triple-negative ER/PR/HER2/neu is associated with bad prognosis due to ↑ aggressiveness, ↑ rates of recurrence, and ↑ risk of metastasis to lungs/brain. - 2CK NBME wants "bilateral mastectomy + oophorectomy" in patients who have confirmed *BRCA* mutation, as mentioned above. - Breast cancer risk is ↑ as a result of HRT, even when progesterone is given as part of it. Breast cancer risk is ↑ as a result of ↑ absolute estrogen exposure the female has in her life. This has nothing to do with unopposed estrogen, which is the major risk factor for endometrial cancer. Insofar as the female is taking the progesterone component of the HRT appropriately, there is no increased risk of endometrial cancer. - Combined OCPs have been suggested in some literature as ↑ the risk of breast cancer, but there is no significance. They won't assess this on USMLE either way. But they do assess ↑ breast cancer risk as a result of HRT. - As I talked about before, the reason the only approved indication of HRT is severe vasomotor Sx (i.e., and not to maintain bone density, etc.) is because HRT ↑ the risk of breast cancer, thromboembolic events, MI, and stroke. This is because estrogen upregulates fibrinogen and factors V and VIII.

Other breast conditions	
Fibrocystic change	- Benign cystic alteration in breast tissue in women 20s-50s. - Shows up three ways on NBMEs: 1) Highly buzzy/classic presentation is a woman 20s-30s who has bilateral, waxing/waning breast tenderness over the course of her menstrual cycle; 2) Unilateral tenderness or sharp pain in one breast (students are often surprised that it's not always bilateral). 3) Painless, unilateral breast cyst in woman in her 40s that drains brown/greenish fluid → answer = fibrocystic change straight-up on the NBME. Call it weird all you want. Take it up with NBME, not me. - You do not need to biopsy or do ultrasound. It is a clinical diagnosis and Tx is symptomatic only (i.e., warm showers, evening primrose oil). - Biopsy of fibrocystic change wil occasionally show some buzzy findings in Qs: "sclerosing adenosis," apocrine metaplasia," and/or "blue dome cysts." These occasionally are mentioned in Qs.
Simple breast cyst	- "Simple" means hypoechoic, or black on ultrasound. It reflects a fluid-filled cyst. - "Complex" means at least in some part hyperechoic, or white on ultrasound. It reflects a solid component to the cyst. - Simple cysts are observed and do not require FNA drainage unless they cause discomfort to the patient (e.g., large cysts). There is no specific size that requires drainage on USMLE.

	- There is an NBME Q where they say a woman was started on HRT three months ago and now has a simple cyst → answer = "biopsy of the lesion." So you should be aware that sudden growth in size can be indication for biopsy. - All complex cysts require FNA biopsy, similar to fibroadenoma.
Mastitis	- Usually caused by *S. aureus.* - Presents two ways on USMLE: 1) Classically a red, cracked, fissured nipple in breastfeeding woman; 2) High-yield point is that mastitis need not affect the nipple and can present as a red, warm, tender, ***non***-fluctuant mass of the breast. - When we talk about infection of the breast, non-fluctuant = mastitis; fluctuant = abscess. This is asked insidiously on a 2CK form and students get it wrong all the time thinking it's abscess – i.e., "They say it's a warm, red, tender mass of the breast though!" Yeah, but they say non-fluctuant, not fluctuant. Not my issue if you get emotional over dumb shit. - USMLE wants oral dicloxacillin (not doxycycline) + continue breastfeeding through the affected breast as treatment. There is no risk of harm to the neonate. Methicillin-class beta-lactams (e.g., dicloxacillin) and first-generation cephalosporins (e.g., cephalexin) are equivalent on USMLE for all intents and purposes. They both treat *S. aureus* skin infections.
Abscess	- Warm, red, tender *fluctuant* mass of the breast. - Drain as next best step.
Galactocele	- Aka milk retention cyst due to blockage of lactiferous ducts; usually seen in recently breastfeeding women. - Often confused with breast abscess. - Can present as painless or tender fluctuant mass of the breast that is peri-areolar. - USMLE will say the patient is afebrile and that the lesion is not warm or red. - Tx is warm compresses followed by drainage.
Ductal ectasia	- As the name implies, this is dilation of lactiferous ducts. - Diagnosis of exclusion on USMLE (i.e., we eliminate to get there). - Presents 2/3 Qs as an inverted nipple. - Presents 1/3 Qs as some random breast lesion where they say biopsy shows "dilation of the ducts," which is what the name means.
Virginal breast hypertrophy	- Virginal breast hypertrophy is benign asymmetric growth of the breasts that can occur in young women. USMLE will try to trick you into thinking it's sinister by saying a first-degree relative had breast cancer. - They will say usually 8-16-year-old girl has large breast lump under the nipple on one side. Answer is just follow-up / observe.
Gynecomastia	- Physiologic gynecomastia can occur in teenage boys on USMLE, where there can be ↑ breast tissue, either tender or not, occurring uni- or bilaterally. Just observe. - Can occur as a result of ↑ estrogen in liver failure, or due to certain drugs like cimetidine, spironolactone, ketoconazole, or even marijuana. - Caused by estrogen effect on breast tissue. It is not caused by prolactin. - As I talked about earlier, an NBME Q gives gynecomastia in a male patient taking hCG. They ask for the mechanism → answer = "testis producing estrogen." The reason this makes sense is because, since hCG shares the same alpha-subunit with LH and FSH (and TSH), giving hCG is as though we are giving LH and FSH at the same time. So LH effect → androgen production; FSH effect → aromatase production → therefore androgens are converted to estrogens in the testes → mechanism via which we could get gynecomastia.
Necrosis	- Non-enzymatic fat necrosis of breast due to trauma; can calcify. - Don't confuse with enzymatic fat necrosis, which is acute pancreatitis. - Can occur due to overt trauma (e.g., getting hit by softball) or sometimes by tight-fitting sports bras.

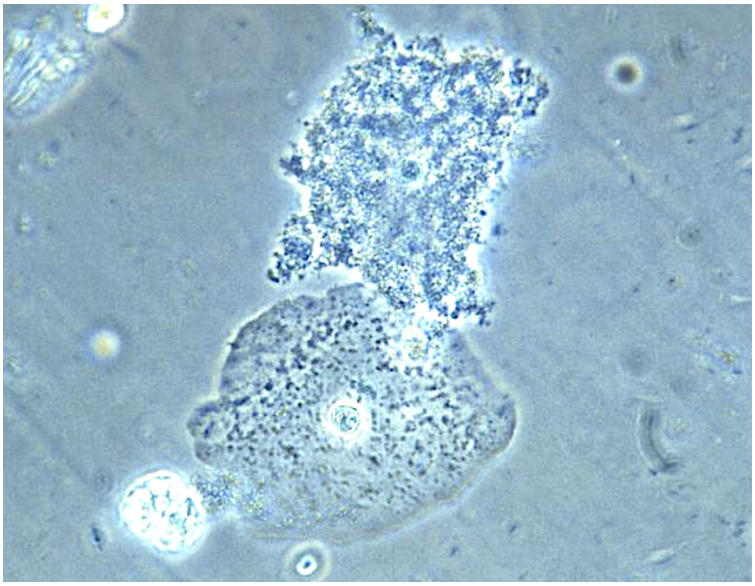
Other ovarian conditions	
Ruptured ovarian cyst	- Presents as sudden, sharp pain in an adnexa (adnexa = combo of Fallopian tube + ovary). - The Q will often say there is 10-15 mL of serosanguineous fluid in pouch of Douglas on culdocentesis (means aspiration of pouch of Douglas). - The Q will say there is no mass / will not mention mass. - Treated symptomatically (i.e., NSAID). Do not do laparoscopy.
Adnexal/ovarian torsion	- 3/4 Qs will give intermittent adnexal pain that becomes constant over hours to weeks (I've seen NBME give both time frames). - 1/4 Qs will just say few-hour Hx of constant pain without the intermittent detail first. - Presents as "large" adnexal mass (i.e., 8 x 12-cm). - There will be no mention of fluid in pouch of Douglas. - Treatment is laparoscopic detorsion. - Tricky detail is that the biggest risk factor is a pre-existing anatomic abnormality (e.g., cyst). So if they say Hx of cyst + now has large adnexal mass and pain, that is torsion, not ruptured cyst.
Theca-lutein cyst	- Arises from theca cells of the ovary; usually bilateral. - Frequently associated with conditions where hCG is very high, such as moles, multiple gestation pregnancies, or fertility drug use. - Can occasionally bleed, where the answer is "hemorrhagic theca-lutein cyst" as a diagnosis of exclusion (i.e., we eliminate to get there), where endometrioid cyst can be eliminated because the Q won't mention anything about dysmenorrhea or signs of endometriosis.
Endometrioid cyst	- Aka endometrioma, or "chocolate cyst." - Cyst filled with endometrial tissue that bleeds. - Occurs in endometriosis (endometrial tissue growing outside the myometrium, usually on the ovary, as with this case; discussed more later).
Dermoid cyst	- Aka mature cystic teratoma. - Contains tissue from multiple germ layers – i.e., hair, skin, teeth, etc. - Can calcify.

Testicular cancer	
Seminoma	- Most common testicular cancer; occurs teenage years and older. - Large, uniform cells with watery cytoplasm. - Chemo- and radiosensitive. - Placental ALP can be a tumor marker.
Yolk sac tumor	- Aka endodermal sinus tumor. - Most common in kids under 3. - AFP is tumor marker.
Embryonal carcinoma	- Immature/anaplastic cells. - Combined AFP and hCG are tumor markers. - Usually adult men.
Sertoli-Leydig cell tumor	- Secrete androgens. - Can cause gynecomastia (shows up on NBME). This is because ↑ androgens can be aromatized into estrogens.
Granulosa cell tumor	- Secretes estrogen. - Can also cause gynecomastia. If NBME gives you Q with Sertoli-Leydig cell tumor causing gynecomastia, they won't simultaneously list granulosa cell tumor as an answer.
Choriocarcinoma	- Composed of syncytiotrophoblastic placental cells. - Can occur in males, yes. Usually 20s-30s. - ↑↑ hCG as tumor marker.

Prostate cancer vs BPH points	
Prostate cancer	- Adenocarcinoma (i.e., glandular); most common cancer in men. - Stimulated by DHT. - Causes osteo**blastic** metastases (highest yield point on USMLE). - Can present as random guy over 50 with no PMHx who just suddenly has neurologic symptoms in the legs and two blastic lesions in his spine picked up on MRI. - If neuro Sx are present due to spinal mets, give IV methylprednisolone. - If no neuro Sx are present and they just say vertebral blastic lesions, answer will be radiotherapy. I've seen distractor answers in this case such as "morphine + corticosteroids," where the student says, "I thought we give steroids though." Yeah, if the patient has neuro Sx, which he doesn't here, plus jumping to morphine before trying other pain management first is wrong. - Orchiectomy will ↓ androgen production → leads to atrophy and shrinkage of prostatic cancer cells. - Flutamide blocks androgen receptor. It is given prior to leuprolide, which is a GnRH receptor agonist that, when given continuously, causes desensitization at GnRH receptor. Since leuprolide can ↑ LH and FSH prior to desensitization, the flutamide is given first. - Prostate-specific antigen (PSA) will often be elevated. Free-PSA (i.e., not bound to protein) is low compared to bound-PSA in prostate cancer. - Gleason scoring is used to grade prostate cancer. - The parasympathetic nerves (for erection) are at greatest risk of injury during radical prostatectomy (asked somewhere on NBMEs).
BPH	- Benign prostatic hyperplasia. - All old dudes have enlarged prostates due to lifetime exposure to DHT, which causes prostatic enlargement. - Presents as dribbling, hesitancy commencing urination, or interrupted stream in a male over 60 (but usually 70s+). - "Old dude + high creatinine = BPH till proven otherwise." - Discontinue anti-cholinergic meds because they cause urinary retention. This concept shows up a lot on NBMEs. Three classes of drugs in particular: 1) First-gen H1 blockers (i.e., diphenhydramine, chlorpheniramine); 2) TCAs (i.e., amitriptyline). 3) Anti-psychotics. - You'll get a Q where an older male has high Cr and then the answer is just "discontinue anticholinergic meds," or "discontinue the diphenhydramine." - Can be described as "nodular" on palpation. - Prostate-specific antigen (PSA) will often be elevated. Free-PSA is high compared to bound-PSA in BPH. - Tx is alpha-1 blocker, such as tamsulosin/terazosin; or the 5-alpha-reductase inhibitor, finasteride. - Transurethral resection of prostate (TURP) for refractory cases.

HY STDs	
Human papilloma virus	- HPV 6/11 cause condylomata acuminata (warts). This is not limited to the genitalia and can cause laryngeal papillomatosis in neonates (warts of the vocal cords), which is asked on NBME. - HPV 16/18 cause squamous cell carcinoma of genitalia/anus; risk of overt SCC is ↑ in immunocompromised (i.e., HIV in MSM) and heavy smoking.
Chlamydia trachomatis	- The "regular" STD are *Chlamydia* D-K strains. - Causes mucopurulent discharge.

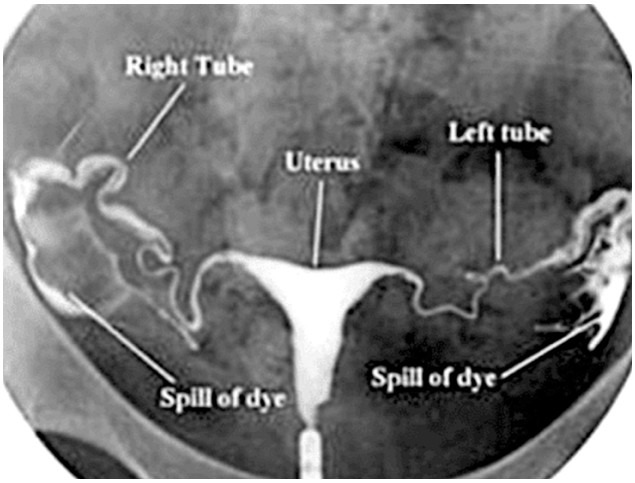
	- Can advance to pelvic inflammatory disease (PID) in females, which is when infection causes inflammation and scarring of Fallopian tubes, leading to ↑ risk of ectopic pregnancy. - Obligate intracellular, so cannot be grown. Discharge will show WBCs under light microscopy with **no organisms.** If the vignette tells you no organisms grow, this is pass-level for *Chlamydia*. - Treat with stat/one-off oral dose of **azithromycin;** can cause GI disturbance. - Can also be treated with one-week of BID (i.e., twice/day) **doxycycline;** cannot be taken with dairy or divalent cations (impaired absorption); can also cause photosensitivity; considered to be slightly more efficacious than azithromycin but much more annoying and arduous to take. - USMLE won't play trivia as to which drug is first-line; there will only be one correct answer. For example, doxy is not given in pregnancy, so if you're forced to choose between the two, in this case you know it's azithromycin (or even sometimes erythromycin in pregnancy). - Doxy isn't given in pregnancy because it can cause teeth discoloration in the eventual neonate. - *Chlamydia* can cause reactive arthritis (triad of urethritis, arthritis, and eye-itis – i.e., any inflammation of the eye, e.g., conjunctivitis, anterior uveitis, etc.). - To prevent ophthalmia neonatorum (neonatal conjunctivitis), treat the female while she is pregnant; to Tx the actual condition in neonates, give oral erythromycin. Conjunctivitis in a neonate can lead to *Chlamydia* pneumonia. - *Chlamydia* A-C are not STDs and cause trachoma (cause of blindness in Africa). - *Chlamydia* L1-3 cause lymphogranuloma venereum (anal strictures).
Neisseria gonorrhea	- Causes mucopurulent discharge, same as *Chlamydia.* - Gram-negative diplococci on light microscopy. - Same as with *Chlamydia*, can advance to PID, with ↑ risk of ectopic pregnancy. - Doesn't cause reactive arthritis; it causes gonococcal arthritis, which will present one of two ways on NBME: 1) monoarthritis of the knee; or 2) triad of mono- or polyarthritis + **cutaneous papules/pustules** + tenosynovitis (inflammation of tendon sheaths; stems like to give deQuervain tenosynovitis). - Treatment is IM ceftriaxone. - Always cotreat for *Chlamydia.* In other words, if the gram-(-) diplococci are seen under LM, there's no way to know if *Chlamydia* is also there or not since the latter shows no organisms, so if a patient has *Gonorrhea*, the proper Tx is IM ceftriaxone, PLUS either oral azithromycin or doxycycline. - If patient develops PID despite having been treated early with ceftriaxone for *Gonorrhea*, the answer for why this happened can be "Hx of improper antibiotic treatment," where the patient was supposed to be cotreated for *Chlamydia* with azithromycin but was only given the ceftriaxone for *Gonorrhea.* - If patient presents with PID who's septic (i.e., high fever, tachy, high WBCs), USMLE wants "admit to hospital + IV antibiotics," not the outpatient combo of IM + oral. - 2CK Obgyn form assesses that if an asymptomatic patient comes in after a partner tested positive for *Gonorrhea* or *Chlamydia*, the answer is give treatment without waiting for test results. - Erythromycin ointment is used on neonates as prophylaxis for conjunctivitis; if neonate already has it, give IM cefotaxime (preferred 3rd-gen cephalosporin in peds if listed).
Trichomonas vaginalis	- Causes trichomoniasis. - Flagellated protozoan seen on wet mount. - Presents as yellow-green discharge. - Can cause "strawberry cervix," or punctate hemorrhages on the cervix. If they don't say this, they can sometimes say yellow-green discharge + a vaginal canal that is erythematous. - Treat with metronidazole for patient **and partner** (high rate of reinfection).

Gardnerella vaginalis	- Causes bacterial vaginosis. - Gram-negative rod that causes a thin grey/watery discharge. - Positive whiff test (KOH prep causes fish-like odor). - Clue cells exceedingly HY (squamous epithelial cells studded with bacteria). They want you to know this image for USMLE: 
Candida spp.	- Causes candidiasis. - Buzzy thick, white, cottage cheese-like discharge in ~2/3 of questions. - The other ~1/3 of Qs will mention an itchy/erythematous vaginal canal without any overt discharge (in contrast to trichomoniasis which can present with erythema of the vagina but has characteristic yellow-green discharge). - Treat with topical nystatin or oral fluconazole. - Step 1 NBME says oral Tx is given + wants MOA for drug inhibition → answer = "P-450-mediated demethylation reaction," where fluconazole inhibits 14α-demethylase in the conversion of lanosterol to ergosterol.
Treponema pallidum	- Causes Syphilis. - Spirochete (spiral-shaped bacterium) visible under dark-field microscopy. - Primary syphilis = painless chancre (painless ulcer) on genitalia. - Secondary syphilis = 6 weeks to 6 months after appearance and disappearance of the initial chancre, patient can get body rash that **includes palms + soles,** and condylomata lata (painless genital plaques). - Tertiary syphilis = years later, patient can get gummas (appear as painless chancres but are on other areas of the body such as the face/nose), arthritis, and ascending aortitis (tree-barking of vasa vasorum). - Neurosyphilis **can occur at any stage;** it is not sequential where we have 1 → 2 → 3 → neurosyphilis. There is a 2CK Neuro Q that gives neurosyphilis in an 18-year-old. - Neurosyphilis presents as tabes dorsalis (obliteration of dorsal columns, with loss of vibration/proprioception + a positive Romberg sign, where patient falls over when standing with eyes closed), Argyll-Robertson pupil (i.e., "prostitute pupil"; accommodates but doesn't react), and "stroke without hypertension" (i.e., sometimes findings akin to stroke but in a younger patient). - Diagnosis of primary syphilis is made via visualizing the spirochetes from a chancre scraping under dark-field microscopy. - Diagnosis of secondary, tertiary, and neurosyphilis can be done with serology, where a VDRL and/or RPR is done first (sensitive but not specific); an FTA is done as confirmatory (specific but not sensitive).

	- VDRL test mixes the patient's serum with cardiolipin antigen. If antibodies against *T. pallidum* are present, the test demonstrates clumping/flocculation. The RPR enhances this reaction by using charcoal particles. There is an NBME Q floating around where they say something about a patient whose test results demonstrate clumping with charcoal particles, and the answer is SLE. - Patients with SLE who have anti-phospholipid syndrome can get false-positive VDRL tests because this syndrome is often caused by antibodies against cardiolipin (in SLE, we simply call these antibodies "lupus anticoagulant"). - FTA mixes a patient's serum with fluorescent *Treponema* antibodies. If binding occurs, this confirms the diagnosis of syphilis. - USMLE will show you 24-year-old male with rash on his back + KOH prep is negative + ask what's most likely to diagnose → answer = FTA. - Treatment for all syphilis types is penicillin. - If patient has Hx of anaphylaxis to beta-lactams but is pregnant or has tertiary or neurosyphilis, the answer is desensitize + give penicillin. This is because penicillin is the most efficacious and needs to be given in severe cases. - If patient has Hx of mere rash to beta-lactams, but not anaphylaxis, then the beta-lactam can be given anyway.
HSV 1/2	- Herpes simplex virus 1/2. - Causes painful vesicular lesions that recur at varying intervals (usually months). - Primary infection is most severe, often with fever, regional lymphadenopathy, burning/stinging/itching pain (herpetic neuralgia), and many vesicles. - Recurrences are often less severe and preceded by herpetic neuralgia. - HSV1/2 can also cause encephalitis (confusion + blood in CSF due to temporal lobe hemorrhage) and herpetic whitlow (vesicle[s] on the finger). - Viral culture can be negative in stem (not 100% sensitive). - Treat with acyclovir (or valacyclovir).
Haemophilus ducreyi	- Causes chancroid, which is a painful ulcer. - Gram-negative rod. - "-Oid" means looks like but ain't. So it looks like a syphilitic chancre, but it's not. - The syphilitic chancre is painless; the *H. ducreyi* lesion is painful. - Often a wrong/distractor answer for HSV Qs, where students get trigger-happy and erroneously choose the weird answer (*H. ducreyi*). - Chancroid will be the answer if they tell you there's a single painful genital lesion in someone who went abroad, classically backpacking in Africa or South America. - If they tell you there's a single, small painful lesion, *but that it's a recurrence*, this is HSV, not *H. ducreyi*. The latter is bacterial and doesn't cause recurrences the way HSV does; HSV can rarely appear as a single vesicle. - There is a 2CK NBME Q where answer is actually *H. ducreyi*, but I once again caution that this is usually a wrong answer, so be careful. But I have seen it correct as a one-off. - USMLE won't assess treatment, but either azithromycin or ceftriaxone is considered first-line.

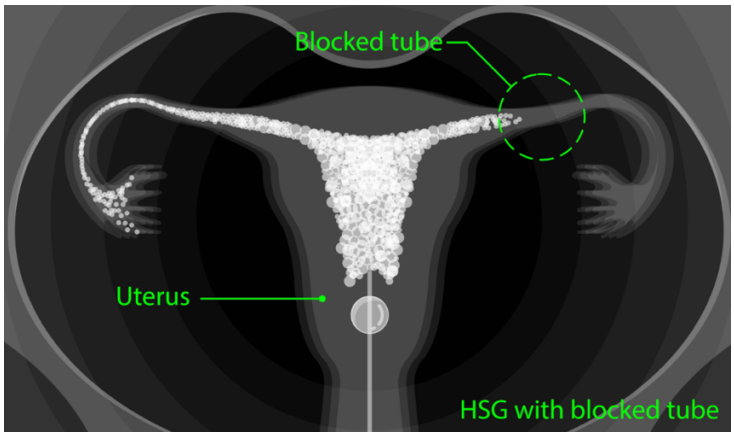
Pelvic inflammatory disease (PID)
- As discussed earlier, is caused by either *Chlamydia* or *Gonorrhea*. - When the infection ascends the uterus and Fallopian tubes, it can lead to inflammation and scarring, thereby increasing the risk for ectopic pregnancy (i.e., loss of Fallopian tube cilia leads to premature implantation usually in the ampulla of the Fallopian tube). - Adnexal and **cervical motion tenderness** are buzzy. But particularly the latter. If they say cervical motion tenderness, you know right away they're talking about PID. - USMLE wants you to know that the Fallopian tubes are normally open on both ends, where the release of the ovum from the ruptured Graafian follicle is "caught" by the fimbriae of the Fallopian tube, prior to its

transit to the uterus. Because the Fallopian tubes are open on both ends, when a hysterosalpingogram is performed, spillage of dye bilaterally into the peritoneal cavity *is normal.*



Normal hysterosalpingogram

- The USMLE will tell you a hysterosalpingogram is performed and spillage is observed on both sides, and I've seen students choose answers like "rupture of Fallopian tubes." No. This is normal anatomy.
- If there is Hx of PID and the Fallopian tubes are scarred, one or both may be sealed, where the hysterosalpingogram shows failure of spillage.



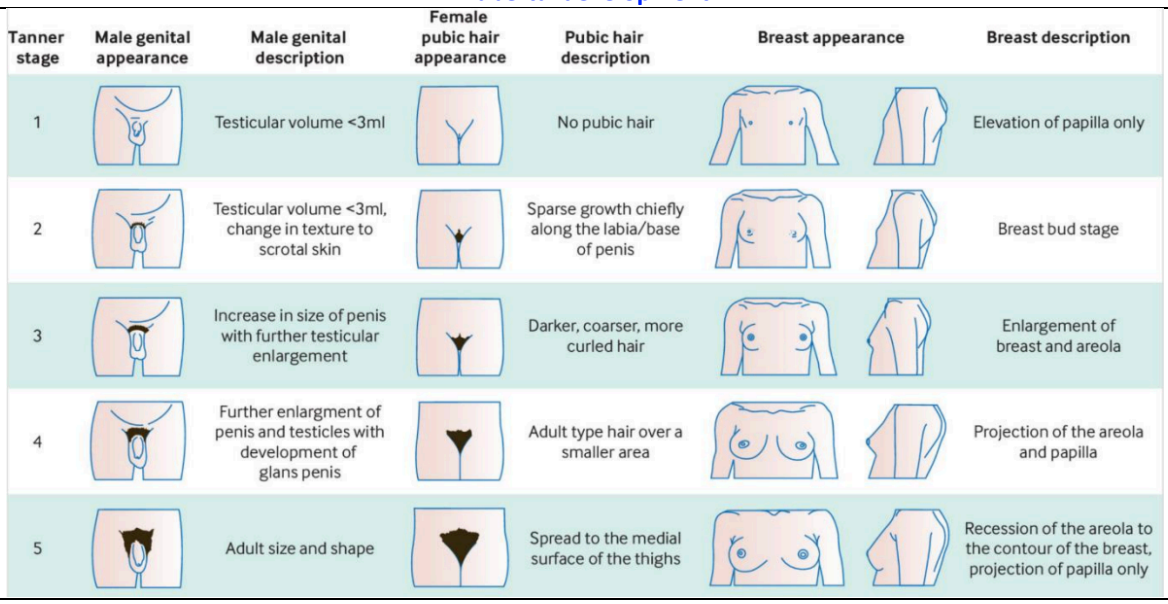
- If the Q tells you a girl has PID and is treated with antibiotics but has persistent fever and adnexal pain, next best step = ultrasound to look for **tubo-ovarian abscess,** which is a potential sequela of PID.
- Extension of PID to the liver capsule is called Fitz-Hugh-Curtis syndrome.

TORCHeS infections	
- Refers to congenital infections in the neonate due to infection in the mother while pregnant.	
Toxoplasmosis	- Presents as HY triad in neonate of 1) hydrocephalus, 2) chorioretinitis, and 3) intracranial calcifications. - They don't have to say the mom sat in a litter box with her cat during pregnancy. - Can be acquired from pork consumption.
Other (Parvovirus B19, VZV)	- Parvo B19 can cause aplastic anemia *in utero* or in the neonate, with increased risk in sickle cell. - Aplastic anemia = all 3 hematologic cell lines are down (i.e., RBCs, WBCs, and platelets). - Can sometimes cause a pure RBC aplasia (i.e., only RBCs down).

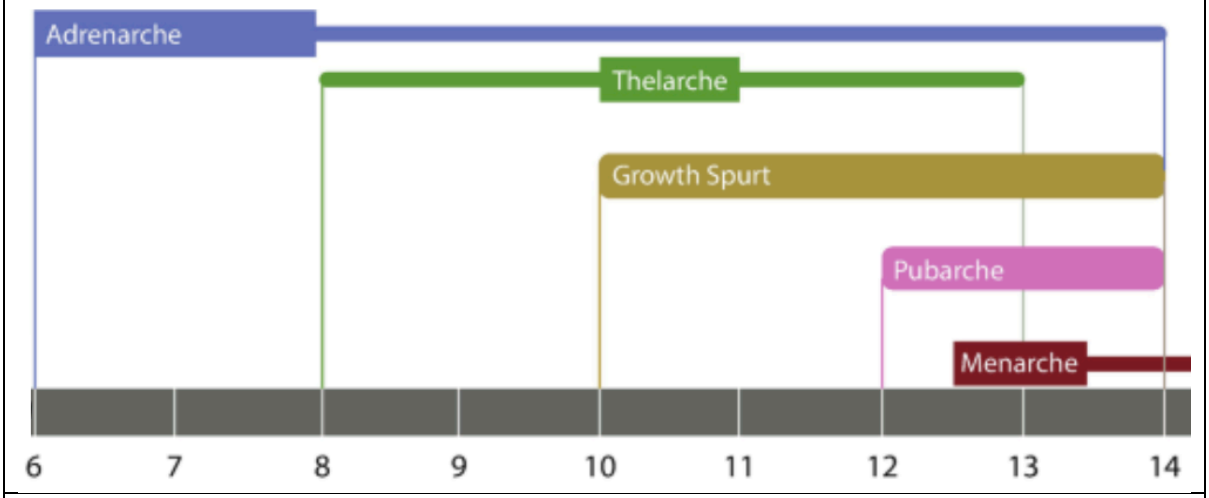
	- The term "fetal hydrops" refers to heart failure *in utero,* and can suggest severe anemia, such as with maternal Parvo infection. In other words, the stem can say the mom had a flu-like illness or cold, followed by hydrops in the fetus → answer = Parvo. - Q can ask how to confirm Parvo infection in neonate if suspected → answer = check Parvo IgM titers in neonate. - Varicella (VZV) during pregnancy can cause microcephaly and limb hypoplasia. - If mother contracts VZV for the first time between 5 days prior to parturition until 2 days after, Varicella immunoglobulin must be given to the neonate. - Kids with congenital VZV syndrome (as well as those who are immunocompromised), are at increased risk of pediatric shingles (just know this Dx exists / "is a thing"). - If neonate is exposed to child with active varicella, if the mother of the neonate previously had chickenpox or was vaccinated, you do **not** need to give VZV IVIG to the neonate. This is because the neonate has passive immunity from the transplacental IgG. - Should be noted that pregnant women who contract VZV can get **pneumonia** from it (sounds weird, but it's a factoid you could be aware of).
Rubella	- Presents as patent ductus arteriosus (PDA) in a neonate. Exceedingly HY / pass-level. - Cataracts and deafness also possible. - Q doesn't need to say mom had rash while pregnant. Sometimes pregnant women get only arthritis. This is not unique to rubella, but I'm just mentioning it because the stem can say she had arthritis but no rash, and student is like "Well it couldn't have been rubella in the mom then." No. Adults sometimes get only arthritis if they contract rubella or measles. - MMR vaccine is live-attenuated and is contraindicated during pregnancy due to theoretical risk to the fetus. If a woman inadvertently receives the vaccine while pregnant or within the month prior to pregnancy, it is not an indication for abortion, but risks to the fetus are increased and proper monitoring is important.
CMV	- Congenital cytomegalovirus is a diagnosis of exclusion on NBME Qs, as per my observation (i.e., we eliminate to get there). - Congenital deafness, hepatomegaly, "blueberry muffin rash," and intracranial calcifications are HY, but these are non-specific and seen in other conditions too. - The USMLE Q might say, "Kid is born with intracranial calcifications, hepatomegaly, and deafness." And the student says, "Aren't calcifications toxo?" Yes, but toxo is the strict triad as mentioned above. The student might also say, "Well can't deafness be rubella or syphilis." Yes, but for rubella they always mention PDA, and for syphilis they always mention unique findings like tooth or shin abnormalities (discussed below). So we eliminate to get there and are left with CMV. It's a bit circuitous, but it's what I've observed they do with CMV Qs. - CMV causes owl-eye appearance of cells due to intranuclear inclusions. - Tx = ganciclovir. MOA is DNA polymerase inhibitor. Mechanism of resistance is alterations to viral thymidine kinase.
Herpes/HIV	- Vaginal HSV1/2 infection in mother can lead to vertical transmission, increasing risk for encephalitis in neonate. - If a pregnant woman experiences prodromal symptoms (i.e., tingling, burning, etc.), even if no visible lesions are present, C-section is still recommended. - If a pregnant woman has predictable intervals of vesicular episodes, acyclovir is often given within 4-6 weeks of parturition to decrease risk of peripartum episode. - HIV in pregnancy is HY. Most important point is that highly active anti-retroviral therapy (HAART) is started immediately in any HIV patient regardless of CD4 count and that it is three-drug therapy. - Efavirenz (an NNRTI) is avoided in pregnancy. - TMP/SMX is avoided in first-trimester, or if woman is immediately trying to conceive, even if CD4 count is under 200. It is okay to use in 2nd and 3rd trimesters. - Intrapartum zidovudine is given to HIV (+) mothers + C-section is performed. - Zidovudine is given to the neonate within 6-12 hours of birth + given for 6 weeks.

	- HIV is present in breastmilk. It is generally recommended to avoid breastfeeding if mother is HIV (+). If the mother's viral load is undetectable and she is on continued HAART, transmission to neonate might not occur, but it is still advised against.
Syphilis	- Highest yield point is that it can cause tooth abnormalities (mulberry molars/incisors). - Can cause "saber shins" (bone abnormalities), saddle nose, deafness, and cataracts.

Pubertal development



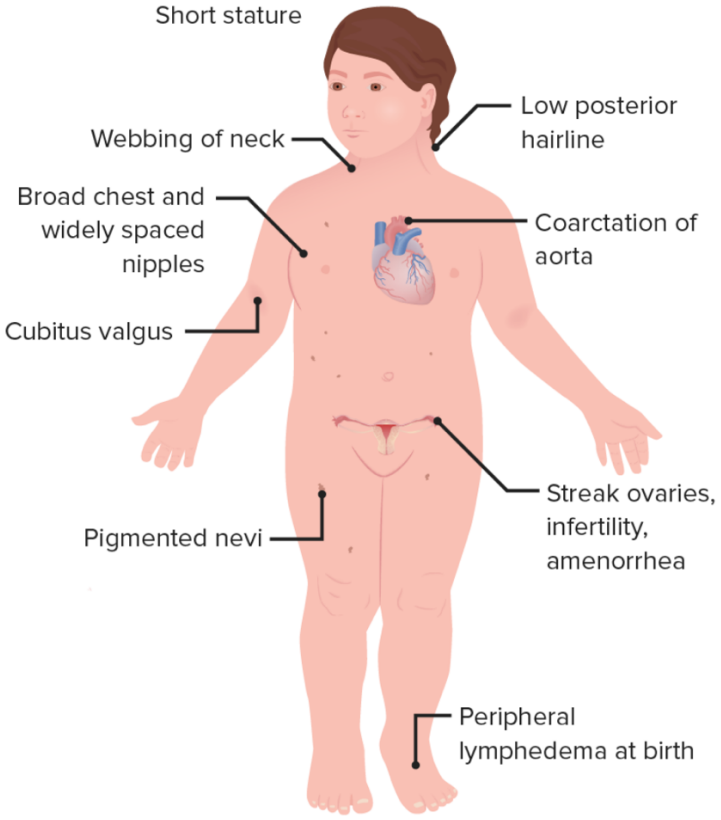
- USMLE wants you to know that at Tanner stage 3, "menarche is imminent." That is an answer straight-up on one of the 2CK Obgyn forms, but there are also various Repro Qs where mom will bring in her daughter who's 13/14, who's never had a menstrual period, and they'll say she's either Tanner stage 2 or 3, and the answer is just "schedule follow-up in 6 months."
- Turner syndrome can present with Tanner stage 1-2 breasts and 4-5 pubic/axillary hair. The latter are due to the effects of adrenal, rather than ovarian, androgens. In other words, don't be confused if you get a Turner syndrome Q and they say Tanner stage 4 pubic hair. But "shield chest" in Turner means "Tanner stage 1-2 breasts."
- In androgen insensitivity syndrome (46XY karyotype, but female phenotype), breast development is normal (i.e., Tanner 4-5), but pubic/axillary hair are Tanner stage 1 (i.e., scant/absent).
- Bone age refers to degree of maturation of a child's bones via radiographic examination.
- USMLE wants you to know that bone age < chronologic age means the short child will catch up (i.e., the growth chart is merely right-shfited). This is called constitutional short stature. The vignette will usually say both parents are average height. For example, a 14-yr-old boy is shortest in his class but has a bone age of 12, meaning his skeleton is aged 12, so he'll catch up later.
- For constitutional short stature, sometimes instead of mentioning bone age, they'll mention Tanner stage. For example, a 14-yr-old boy is shortest in his class + pubic hair is Tanner stage 2. The implication is he's still very early in development and will probably catch up.
- Bone age = chronologic age means genuine short stature. I've seen this in Turner syndrome Qs, where the female is usually < 5 feet.



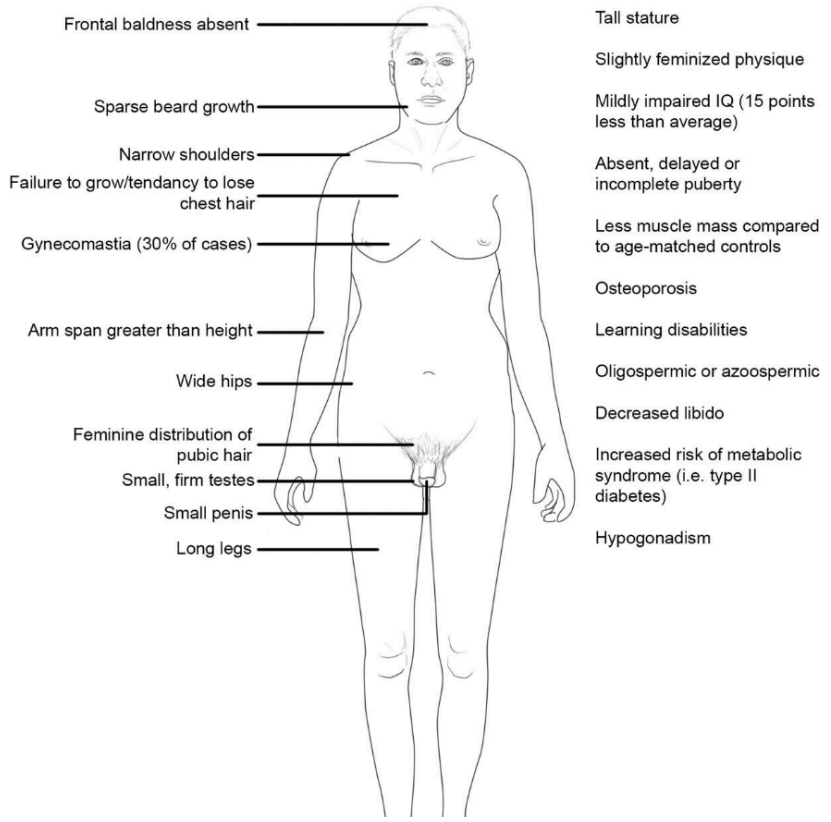
- A mnemonic to remember the above sequence is "**ATP M**an." Dumb. Don't know what to tell you.
- **A**drenarche: ↑ production of adrenal androgens, leading to the development of pubic and axillary hair.
- **T**helarche: Onset of breast development.
- **P**ubarche: Pubic hair growth
- **M**enarche: Onset of menstruation.
- USMLE asks straight-up what is most likely to occur immediately before puberty in female → answer = breast bud development.

Sex disorders

Turner syndrome



- 45XO karyotype; female phenotype.
- Since there is only one X chromosome, the Q can say there is no Barr body visible on electron microscopy.
- Infertile due to streak ovaries. In Turner, this is colloquially referred to as "menopause before menarche."

	- Can still have children with IVF using donor egg + exogenous hormones (asked sometimes on behavioral/psych Qs). - LH and FSH both ↑ due to primary hypogonadism (i.e., ↓ negative-feedback at hypothalamus and anterior pituitary due to ↓ ovarian hormones and inhibin). - Short stature (usually < 5 feet), Tanner stage 1 breast development ("shield chest"), cystic hygroma (webbed neck due to lymphatic insufficiency; asked on NBME); scattered nevi (confuses students for things like NF1, but I don't know what to say; you need to know scattered nevi are seen in Turner); normal pubic and axillary hair (Tanner 4-5). - ↑ risk of dysgerminoma developing from the streak ovaries; prophylactic oophorectomy is recommended, especially if there is Y-chromosome material present in rarer mosaic forms of Turner. - If Q tells you there's a fraction of a female's cells that are 46XX and another fraction that's 45XO (i.e., somatic mosaic Turner), the answer for the mechanism is "post-fertilization mitotic error." - Coarctation of the aorta + bicuspid aortic valve (aortic stenosis) HY.
Klinefelter syndrome	 - 47XXY karyotype; male phenotype. - Tall, eunuchoid body shape; small testes; gynecomastia; reduced IQ. - Q might say there is presence of electron-dense mass in the cell on electron microscopy (i.e., a Barr body). - Testes are present but are small; there is dysgenesis of the seminiferous tubules, leading to ↑ LH and FSH due to ↓ negative-feedback from ↓ androgens and inhibin. - Estrogen/androgen ratio is ↑ due to dysgenesis of Leydig cells (which make androgens). Absolute estrogen is also ↑ due to ↑ adipose tissue (therefore ↑ aromatase). - ↑ risk of breast cancer; risk of testicular cancer is only very subtly ↑, so prophylactic orchiectomy is not indicated.
Androgen insensitivity syndrome (AIS)	- 46XY karyotype; female phenotype. - Defect in androgen receptor. Since DHT is needed for external male development, the patient appears female.

	- Testosterone and DHT are normal or elevated due to ↓ negative-feedback at the hypothalamus and anterior pituitary → ↑ LH production → ↑ testicular androgen production. - The testes in AIS are usually located within the abdomen or inguinal canal. - Even though there is resistance to androgens, testes still develop because the the *SRY* gene on the Y chromosome induces differentiation of the primitive gonads into testes. This differentiation isn't dependent on androgens. - The testes produce Müllerian-inhibitor factor, so there is no uterus or Fallopian tubes in AIS. - Even though the gonads also produce androgen, since we have resistance to it, the karyotypic male appears female (i.e., phenotypically female). - Breast development is Tanner 4-5 but pubic/axillary hair is Tanner 1. - USMLE vignette will say: 15-year-old girl is brought to the physician by her mother for not yet having a menstrual period + vagina ends in blind pouch + pubic/axillary hair is scant/absent → answer = AIS. - Next best step = ultrasound first if listed, then karyotype. - ↑ risk of testicular malignancy (dysgerminoma), so prophylactic orchiectomy is indicated after puberty (maximizes estrogen effects for phenotypic female development), unless evidence of malignancy is seen prior.
Müllerian agenesis	- Aka Mayer-Rokitansky-Küster-Hauser syndrome. USMLE will never assess this name, but students have asked about this before. - 46XX karyotype. - Presents identical to AIS on USMLE except they'll say pubic/axillary hair is normal/coarse. - In other words, 15-year-old girl + never had a menstrual period + vagina ends in blind pouch + pubic/axillary hair is coarse → answer = Müllerian agenesis. - Ovaries are present, since no Y chromosome is present (i.e., no *SRY* gene to convert primitive gonads into testes, so ovaries as the default ensue). - But since the Müllerian (paramesonephric) ducts don't develop, the Fallopian tubes, uterus, cervix, and upper vagina don't develop. - Next best step = ultrasound first if listed, then karyotype.
MIF deficiency	- Müllerian-inhibitory factor deficiency. - 46XY karyotype; phenotypic male with testes (i.e., internal and external structures are male), but failure of regression of the Müllerian ducts means a uterus, Fallopian tubes, cervix, and upper vagina are present. - The Müllerian structures are usually located in the pelvis between the bladder and rectum (similar to location for uterus in a female), or in the inguinal canal. They are often discovered incidentally during orchidopexy for cryptorchidism. - In other words, the Q can say there are undescended testes, where upon during surgical reduction into the scrotum, a primitive uterus and Fallopian tubes are discovered.
5α-reductase deficiency	- 46XY karyotype; phenotypically female, or neonate with ambiguous genitalia. - 5α-reductase is enzyme that converts testosterone → DHT. If there is a deficiency and DHT is ↓, male external structures don't develop, so phenotype is female. - Presents as "phallus (penis) at age 12." This is buzzy. They will say a 12-year-old girl has grown 4 inches in the past 4 months + has acne + hair on upper lip + clitoral hood is 3-4 cm → answer = 5α-reductase deficiency. - At puberty, the ↑↑ in testosterone can partially override the enzyme deficiency, leading to threshold production of DHT where male secondary sex characteristics can develop.
SRY gene translocation	- 46XX karyotype; phenotypic male. - The *SRY* gene from the Y chromosome is translocated usually onto one of the X chromosomes, resulting in formation of testes instead of ovaries. - The presence of testes leads to testosterone and DHT → develops male external structures; MIF production → no Müllerian structures develop.

	- Testes are usually small, and there can be penile abnormalities like hypo- or epispadias. - The USMLE will tell you a neonate appears male but is confirmed 46XX karyotype → answer = *SRY* gene translocation. - Rarely, this condition can be seen in 46XY karyotype, where the *SRY* gene is lost, rather than gained. Testes are absent, and instead there are streak ovaries + Müllerian structures.
Congenital adrenal hyperplasia (CAH)	- I discuss this in detail in the HY Arrows PDF. If you hate these conditions, you can go there for practice with the Qs I've written. - Ambiguous genitalia due to deficiency of 21-, 11-, or 17-hydroxylase in the adrenal glands. - If the Q says 17-OH substrates are high (i.e., 17-OH-pregnenolone or 17-OH-progesterone) and/or elevated DHEA-S or androstenedione, you know right away 17-hydroxylase deficiency is wrong, and the answer must be either 21- or 11-deficiency. - Then, if BP is low/normal or K^+ is high, the answer is 21-deficiency. - If BP is high or K^+ is normal/low, the answer is 11-deficiency. - 21-deficiency will basically present like Addison disease, where you get high K^+, low Na^+, low bicarb, and low pH. They can also say glucose is low due to low glucocorticoids. - Since the Step 1 has gone to pass/fail, you **do not** need to memorize the adrenal substrate pathways. Waste of time. Just know the simple rules I've written above + go to my HY Arrows PDF for practice doing the Qs. - Answer for mechanism of CAH on an Obgyn CMS form for confirmed 46XX neonatal girl who has fused labia is "ACTH hypersecretion." In other words, the ↑ in adrenal androgens she has (i.e., DHEA-S in particular) can result from 21- or 11-hydroxylase deficiencies, where cortisol is ↓, so ACTH goes ↑ to compensate.
Swyer syndrome	- 46XY; phenotypically female. - Caused by failure of the primitive gonads to develop into testes, but they don't develop into ovaries either. They develop into streak gonads. - ↓ androgen production causes external female phenotype. Also leads to scanty/absent pubic/axillary hair. - Absence of MIF (since no testes) means Müllerian structures develop. - Presents similar to AIS – i.e., phenotypic female with scanty/absent pubic/axillary hair, but there are Müllerian structures (i.e., uterus, Fallopian tubes, cervix, and upper vagina) and ↓ androgens, whereas in AIS, Müllerian structures are absent and androgens are normal or ↑ (tissues are just insensitive to them).

Imperforate hymen (2CK/3 only)	
- The hymen is a thin membrane that partially covers the vaginal opening in females. It normally has a small hole that allows the passage of menstrual blood out of the vagina. The size and shape of the hymenal opening can vary between females. - Imperforate hymen is when the hymen completely covers the vaginal opening, which can lead to hematocolpos and hematometra.	
Hematocolpos	- Blood accumulation within the vaginal canal due to imperforate hymen. - USMLE Q will give a teenage girl who's never had a menstrual period. Physical exam shows a "blueish bulge" behind the hymen. The Q might say the girl has had monthly pain (i.e., she has in fact had menses, but blood is blocked from leaving the vagina due to the imperforate hymen). - Answer is either hematocolpos or "cruciate incision of the hymen."
Hematometra	- Blood backing up all the way to the uterus causing uterine distention.

	- Q will give what sounds like imperforate hymen + hematocolpos but you'll notice the blood pressure is low → distention of the uterus can trigger a vagal response leading to low BP.

HY Trisomy disorders	
Trisomy 21	- Down syndrome; caused by having 3 chromosome 21s, instead of the normal 2. - 95% due to extra chromosome 21 as a result of meiotic nondisjunction in advanced maternal age (1 in 350 at age 35; 1 in 100 at age 40; 1 in 10 at age 50). - 4% due to Robertsonian translocation – i.e., obscure process where long arm of chromosome 21 attaches to another chromosome (usually 14) and is passed to conceptus along with two normal chromosome 21s; the result is the conceptus inherits 3 chromosome 21s. - 1% is mosaic Down, which is due to post-fertilization mitotic (not meiotic) error. In this case, not all cells of the conceptus contain the trisomy. - Most common genetic cause of mental retardation; second most common genetic cause is Fragile X. Most common cause of mental retardation overall is fetal alcohol syndrome. - Associated with flattened facies; slanted palpebral fissures; epicanthal folds; single palmar crease (this finding non-specific but still often mentioned in Down); large gap between the first and second toes; Hirschsprung disease; duodenal atresia (double bubble sign on AXR); endocardial cushion defects (AVSD, VSD, or ASD); pulmonary artery malformations; acute lymphoblastic leukemia; congenital hypothyroidism; Eustachian tube atresia; Brushfield spots (iris lesions). - First trimester tri-screen (8-10 weeks): ↑ nuchal translucency, ↑ hCG, ↓ **PAPP-A**). Students often remember something is abnormal about PAPP-A, but they forget the direction, so remember that it's ↓, not ↑. - Although not technically part of the tri-screen, a hypoplastic nasal bone is also an important finding in Downs (reflects flattened facies). - Second-trimester quad screen (↓ AFP, ↑ hCG, ↓ estriol, ↑ inhibin A). As discussed earlier, just remember that the ones that have "Hs" are ↑ (i.e., hCG and inhibin A both have Hs, so those are the ones that are ↑).
Trisomy 18	- Edward syndrome. - Low-yield for USMLE. But you have to know it exists. - Low-set ears; prominent occiput; rocker-bottom feet; clenched hands / overlapping fingers; omphalocele or gastroschisis. - Some students get emotional about the Edward and Patau syndrome quad screens, but USMLE doesn't give a fuck. But in theory the arrows are all ↓. - Usually fatal *in utero* or shortly after birth.
Trisomy 13	- Patau syndrome. Rare as fuck. - Holoprosencephaly (cylopia) / failure of halves of brain to separate; polysyndactyly. - Quad screen arrows all ↓, except inhibin normal. USMLE doesn't care though. - Usually fatal *in utero* or shortly after birth.

Trisomy diagnostic modalities (2CK/3 only)	
Amniocentesis	- Performed at 15-20 weeks. - Thin needle is inserted through the abdominal wall, under ultrasound guidance, into the amniotic sac surrounding the fetus. A small amount of amniotic fluid, which contains fetal cells, is drawn out. The collected cells are analyzed for chromosomal abnormalities. - Risk of iatrogenic abortion is up to ~1/1,000, depending on location and expertise. - >99% accurate for detecting trisomies.
CVS	- Chorionic villus sampling. - Performed at 10-13 weeks.

	- Ultrasound-guided needle is inserted transcervically or -abdominally to remove a sample of cells from the chorionic villi of the placenta, which have the same genetic material as the fetus, allowing for chromosomal analysis. - Risk of iatrogenic abortion is higher than amniocentesis, at ~1/100-200. - >99% accurate for detecting trisomies. The reason some patients choose CVS in spite of its higher iatrogenic abortion risk and amniocentesis' comparable accuracy is because it allows for diagnosis of trisomies earlier in pregnancy, thereby ↑ time for parental decision making.
cfDNA	- Cell-free DNA; aka non-invasive prenatal testing (NIPT). - Performed after 10 weeks. - Involves taking a blood sample from the pregnant woman which contains fragments of DNA from both her and the fetus. Laboratory techniques then analyze the fetal DNA fragments for chromosomal abnormalities. - >99% accurate, but ↑ false-positive rate compared to amniocentesis and CVS. As a result, cfDNA is considered to be a screening, rather than diagnostic test. Positive results are followed up with amniocentesis or CVS.
Cordocentesis	- Performed after 18 weeks. - Transabdominal ultrasound-guided needle insertion into umbilical cord to retrieve fetal blood. - Classic indication is to diagnose fetal hematologic abnormalities such as anemia. - Not typically done for chromosomal investigations, but allows for rapid karyotyping if a quick chromosomal analysis is required, especially if amniocentesis or CVS results are inconclusive or if there's a need for faster results than these tests can provide.

Abnormal AFP measurements	
Dating error	- Most common cause of abnormal AFP measurement is "dating error." - The USMLE Q will tell you that a pregnant woman has AFP measured 2.6 times the upper limit of normal for her gestational age; next best step = "ultrasound"; "re-measure AFP" is wrong answer. This is because the AFP value measured by the lab isn't wrong; it's merely that the estimated gestational age of the pregnancy is wrong. An ultrasound will allow for a re-measuring of the crown-rump length (CRL), which is a typical method for ascertaining approximate gestational age.
Too high	- Apart from dating error, classic cause is neural tube defects (i.e., spina bifida). - Other causes of ↑ AFP for USMLE are anencephaly, multiple gestation pregnancies, and omphalocele / gastroschisis.
Too low	- As discussed before, AFP is "down in Down syndrome." - It's also low in Edward and Patau but USMLE doesn't care.

First antenatal visit (2CK/3 only)

- Usually at 8-10 weeks' gestation.
- Comprehensive medical history is taken.
- Complete blood count; blood type; Rh factor (+) or (-); check for anemia.
  - If Rh(-), give RhoGAM at 28 weeks and again at delivery.
  - Also give RhoGAM if any interventions (e.g., amniocentesis), or if there's complications like spontaneous abortion or abruptio placentae.
  - If 2nd pregnancy onward, if Rh (-) woman is found to have titers against Rh, do not give RhoGAM during the pregnancy, since it's too late.
- Check for STDs (especially *Chlamydia* and *Gonorrhea*).
  - Pap smears are not routinely done at first-trimester antenatal visit, but STD checks are.
- Rubella, syphilis, Hep B, and HIV.
  - Standard screening performed for these.
- Urinalysis looking for WBCs/bacteria, proteinuria, and kidney function.

- Always treat asymptomatic bacteriuria in pregnancy due to ↑ risk of pyelonephritis (progesterone slows ureteral peristalsis, and, in third trimester, uterine compression causes backup).
- Blood pressure.
  - "Chronic hypertension" = HTN prior to the pregnancy or diagnosed before 20 weeks' gestation; usually persists postpartum since BP present in early pregnancy, or prior to it, often reflects underlying issue unrelated to the pregnancy.
  - "Gestational hypertension" = new-onset HTN that develops after 20 weeks' gestation; typically resolves postpartum.
  - Chronic hypertension can pre-exist or persist beyond pregnancy, while gestational hypertension is specific to the pregnancy period. Both conditions require close monitoring due to potential risks to the mother and fetus.
- Glucose only if high-risk.
  - Discussed in more detail later.
- Ultrasound to confirm pregnancy, establish gestational age, and detect multiple pregnancies.
  - Crown-rump length to measure gestational age.
- Flu shot is safe in first trimester (if fall or winter).
  - Give killed intramuscular; safe for everyone annually age 6 months and older.
  - Intra-nasal live-attenuated only for non-pregnant, immunocompetent persons age 2-45.

Vaccines in pregnancy (2CK/3 only)

- Recommended during pregnancy:
  - Influenza (as discussed above).
  - Tdap (tetanus, diphtheria, pertussis: given in third trimester).
  - Hepatitis B (only if maternal surface antibody negative, meaning she is not immune).
- Contraindicated during pregnancy:
  - Live-attenuated vaccines, including MMR, varicella, intra-nasal influenza, and BCG (TB).
  - HPV (insufficient evidence on safety during pregnancy, so currently not given).

Iron studies in pregnancy (2CK/3 only)

- USMLE loves assessing thalassemia vs iron deficiency anemia (IDA) in pregnancy.
- Hemoglobin is normally 12-17.5 g/dL in menstruating women. In pregnancy, this can fall naturally due to dilutional effects down to around 10 g/dL.
- What USMLE wants you to know is that thalassemia is the diagnosis if the patient has a microcytic anemia *despite* iron supplementation. In thalassemias, Hb and MCV are both low *despite* normal serum iron and ferritin.
- You can't rely on target cells as a crutch for thalassemia. You need to know the iron studies.
- If a pregnant female has low Hb and MCV, the next best step is "check serum iron and ferritin" if it's listed. This is because, if they come back low, we know the diagnosis is IDA; if they come back normal, we know the diagnosis is likely thalassemia.
- If the serum and iron come back normal in the setting of low Hb and MCV, the next best step is hemoglobin electrophoresis. It will be normal in the setting of 1 or 2 alpha mutations; it will show ↑ HbA2 and HbF in beta-thalassemia minor.
- 3 or 4 alpha mutations, or beta-thalassemia major are irrelevant diagnoses because these apply to sick children, not adults. 1 or 2 alpha mutations, or beta-minor, will be adults who've lived out life either asymptomatic or with mild fatigue, or were previously misdiagnosed as having IDA.
- If the pregnant woman has low Hb and MCV and they tell you she's been on pregnancy multivitamin for at least a month (which means she's been taking iron), and "check serum iron and ferritin" is not listed as a next best step, the answer is just hemoglobin electrophoresis.
- "Alpha-/beta-globin DNA testing" is a wrong answer for thalassemia diagnosis. USMLE wants hemoglobin electrophoresis. DNA testing might be done as part of further work up.
- I talk about this stuff in more detail in my HY Heme/Onc PDF.

Hemolytic disease of newborn types	
Rhesus (Rh) type	- ~85% of people have Rh antigen on their RBCs – i.e., Rh(+). Women who are in the ~15% that are Rh(-) are at risk of developing IgG antibodies against Rh antigen if the fetus is Rh(+) and the circulations mix inadvertently. - When the mother develops antibodies against Rh (i.e., has become sensitized to it), we call this "Rh isoimmunization." NBME will throw this word around a bit, so you need to know it means "she's developed IgG antibodies against Rh." - Mechanism: Rh(-) mom in first pregnancy is exposed to fetal blood that is Rh(+); mom makes Abs against Rh; subsequent pregnancy results in IgG against Rh crossing placenta and targeting fetal Rh(+) RBCs, leading to hemolysis in the fetus. - As mentioned earlier, Rh status is checked at the first antenatal screening at 8-10 weeks. If the mother is Rh(-), then give RhoGAM at 28 weeks as well as at parturition. It must also be given if there are any interventions (e.g., amniocentesis), or if there's complications like spontaneous abortion or abruptio placentae. - If 2nd pregnancy onward, if Rh (-) woman is found to have titers against Rh, do not give RhoGAM during the pregnancy, since it's too late. - Kleihauer-Betke (KB) test is done in setting of hemorrhage or suspected mixing of maternal and fetal circulations in Rh(-) women. The mother's blood is drawn, and the test works by exploiting the resistance of HbF to acid, allowing fetal RBCs to be distinguished from maternal cells on a blood smear. Depending on the fraction of fetal RBCs present, this determines the dose of RhoGAM necessary.
ABO type **(2CK/3 only)**	- Most anti-A and -B Abs people have to opposing blood types are IgM, but people with O blood can have fractionally greater IgG; if mother with O blood has higher % of anti-A and -B Abs that are IgG, fetus can be symptomatic. - USMLE will give you O+ mom usually in first pregnancy and A or B fetus (can occur in Rh- women in second pregnancies, but the USMLE wants to assess you specifically know the ABO type of HDN, so they'll give O+ mom in first pregnancy).
Kell, Duffy, Kidd type **(2CK/3 only)**	- Minor antigens beyond Rh (i.e., Kell, Duffy, Kidd) can sometimes cause HDN. - The process for isoimmunization is same as Rh. - What will happen is, an A+/B+/AB+ mother will have a second pregnancy where fetal hemolysis is occurring, but based on her blood type, you know it's not ABO or Rh type HDN. - The Q will say her blood titer for, e.g., Kell is high. The next best step is "check Kell status of the father." - In other words, since the mother has titers against Kell, you know she's Kell negative. So if the father's RBCs are Kell(+), the fetus is likely Kell(+), and that would explain the HDN.

Diabetes in pregnancy (2CK/3 only)
- Diabetes screening for pregnancy is done in all women at 24-28 weeks' gestation. - First-trimester glucose screening is not done unless female is considered high risk (i.e., obesity, FHx of type II diabetes, or previous pregnancy with fetal macrosomia). - Screening is done with non-fasting 50-g glucose challenge. If after 1 hour, glucose is >130 mg/dL, a 100-g oral glucose tolerance test (OGTT(is performed. - With the OGTT, gestational diabetes mellitus is diagnosed if 2 or more of the following are seen: - Fasting glucose (prior to the test in the morning): >95 mg/dL; - 1 hour after: >180 mg/dL; - 2 hours after: >155 mg/dL; - 3 hours after; >140 mg/dL. - If a woman has GDM, she must be followed up postpartum: - Do a 75-g OGTT 6-12 weeks postpartum. - Do routine diabetes screening every 1 year thereafter (asked on NBME). This refers to standard fasting glucose and HbA1c.

- Insulin is gold standard for glycemic control during pregnancy.
- Oral hypoglycemics (e.g., metformin, glyburide) can be used during pregnancy insulin is preferred. During parturition, if a patient is on one of these agents, she is switched over to insulin because this allows for tighter glucose monitoring and control.
- **Main intrapartum fetal complication of GDM is shoulder dystocia,** where the anterior fetal shoulder gets caught behind the maternal pubic symphysis. This is because GDM often causes macrosomia (big baby; >4,000g).
- **Main maternal complication is vaginal tear/laceration** due to the shoulder dystocia. In the setting of vaginal delivery, McRobert's maneuver is implemented (i.e., flexion of maternal hips + application of suprapubic pressure) + an episiotomy is often performed, followed by suturing postpartum. This prevents uncontrolled laceration into the peroneal body / external anal sphincter, leading to fecal incontinence.
- GDM can also cause polyhydramnios due to fetal polyuria (↑ glucose in the fetus pulls water with it through the fetal kidneys, similar to children and adults with DM).
- Maternal insulin does not cross the placenta. Only glucose does. The fetus will produce its own insulin in response to hyperglycemia, leading to fetal hyperinsulinemia. Following parturition, when the transplacental glucose is removed, the fetal insulin levels remain high. This can cause hypoglycemia in the neonate, presenting with rigors or even seizure. There is a difficult Peds Q on a 2CK form where they ask why a neonate whose mom had GDM has seizure, and the answer is "decreased gluconeogenesis" (because insulin is high). Makes sense, but difficult Q to draw the conclusion.
- One or more of the pentad of hypoglycemia, hyperbilirubinemia, polycythemia, hypocalcemia, and hypomagnesemia can be seen in neonates of diabetic moms.
- If the Q gives you neonate with long QT on ECG, choose hypocalcemia over hypomagnesemia (even though the latter can cause the former).

Oligo- vs polyhydramnios	
Polyhydramnios	- Excessive amniotic fluid. - Important maternal causes: - **GDM.** As discussed before, ↑ glucose in the fetus pulls water with it through the fetal kidneys, causing ↑ urinary volume. - **Multiple gestation pregnancies.** This is due to the combined ↑ in urinary output in the setting of multiple fetuses. - Important fetal causes: - **Fetal hydrops** (severe edema in the fetus usually as a result of heart failure). In the setting of Parvo B19 infection, fetal anemia, or HDN, cardiac output ↑ to compensate, leading to ↑ renal perfusion and ↑ urine production. In the setting of fetal heart failure where cardiac output is ↓, ↑ venous pressure will cause edema, where the kidneys attempt to rid the body of excess fluid by ↑ urinary output.
Oligohydramnios	- Insufficient amniotic fluid. - Important maternal causes: - **Uteroplacental insufficiency.** Reduced blood flow to the fetus as a result of smoking, preeclampsia, and SLE (discussed in below tables) can result in compromised renal function and urinary output. This is likely related to impaired nutrient delivery in the setting of minimally sufficient oxygenation. - Important fetal causes: - **Posterior urethral valves.** Most common genitourinary abnormality in neonatal males. "Posterior" refers to the prostatic urethra developing thin membranes, or "valves," that prevent urinary outflow. This can lead to obstructive uropathy with vesicoureteral reflux and hydronephrosis. Severity can vary; some Qs give severe oligohydramnios; other Qs are male neonate hours old who hasn't urinated, or male neonate weeks old with urinary retention. - **Renal agenesis (Potter sequence).** Failure of kidney development → ↓ urinary output → ↓ amniotic fluid → ↓ swallowing of fluid by fetus →

	impaired lung development → pulmonary hypoplasia. The ↓ amniotic fluid also leads to flattened facies and rocker-bottom feet.

Hypertensive disorders of pregnancy	
Chronic hypertension	- "Chronic hypertension" = HTN prior to the pregnancy or diagnosed before 20 weeks' gestation; usually persists postpartum since BP present in early pregnancy, or prior to it, often reflects underlying issue unrelated to the pregnancy. - Can pre-exist or persist beyond pregnancy, while gestational hypertension is specific to the pregnancy period. Both conditions require close monitoring due to potential risks to the mother and fetus.
Gestational hypertension	- "Gestational hypertension" = new-onset HTN that develops after 20 weeks' gestation; typically resolves postpartum. - Both chronic and gestational hypertension can be managed with methyldopa (α2 agonist), labetalol, or nifedipine. Hydralazine can be used in acute settings.
Preeclampsia / eclampsia	- Preeclampsia = HTN + proteinuria after 20 weeks of pregnancy. - Eclampsia = preeclampsia + seizures. - True Obgyn definition of preeclampsia is more expansive to incorporate signs of damage to other organs, but for USMLE, 4/5 Qs will just be HTN + proteinuria. - In Qs that don't give you the strict HTN >20 weeks' gestation + proteinuria combo, they will usually just give straight-up HELLP syndrome (discussed below), which is considered to be a severe form of preeclampsia involving damage to the liver. - Mechanism for preeclampsia/eclampsia is uteroplacental insufficiency, where inadequate perfusion to the placenta causes a compensatory attempt by the cardiovascular system to increase/force perfusion, thereby resulting in systemic HTN. - The uteroplacental insufficiency can be idiopathic or due to secondary factors such as smoking and SLE. - "Severe preeclampsia" = BP > 160/110 (either number), proteinuria 3+ on dipstick, or any evidence of multi-organ system involvement (i.e., liver, CNS, lungs). Liver damage can be overt HELLP syndrome or just any elevation in hepatic enzymes. CNS involvement can refer to visual disturbance or headache. Pulmonary involvement refers to pulmonary edema or cyanosis. - Magnesium is only given prophylactically to prevent eclampsia (seizures) if patient has severe preeclampsia. - If patient already has overt eclampsia with seizures, first Tx is magnesium. - Definitive management for both preeclampsia and eclampsia is delivery of the fetus.
HELLP syndrome	- Hemolysis, Elevated Liver enzymes, Low Platelet count. - As mentioned above, a severe variant of preeclampsia. - Q might not even mention proteinuria. Doesn't matter. - Highest yield point for USMLE is that we see **schistocytes** on a blood smear. This is due to RBC shearing that occurs within hepatic damaged microvasculature. - Platelets fall as a result of microvascular endothelial cell dysfunction/damage, where microthrombi form in an attempt to mitigate damage. - The combination of schistocytosis and thrombocytopenia is called microangiopathic hemolytic anemia (MAHA).

SLE and antiphospholipid syndrome
- SLE causes uteroplacental insufficiency and recurrent miscarriage when patient has anti-phospholipid antibodies. The latter result in microthrombi within uteroplacental microvasculature leading to decreased blood flow and fetal compromise. If miscarriage does not occur, intrauterine growth restriction (IUGR) and oligohydramnios may result secondary to the reduced uteroplacental oxygen delivery. - Women with Hx of miscarriage due to phospholipid syndrome in SLE are treated with combination therapy of low-dose aspirin + LMWH in subsequent pregnancies. Warfarin is avoided because it is teratogenic.

- SLE resulting in uteroplacental insufficiency is a risk factor for preeclampsia, since the latter is caused by reduced uteroplacental perfusion.
- IUGR in pregnancy is best ascertained by looking at fetal abdominal circumference. Sounds weird (i.e., student says, "Why not head circumference or something."), but it's because abdominal circumference reflects liver size and overall fetal fat accumulation, which are notably affected in IUGR. When the placenta is not delivering adequate oxygenation and nutrients, the fetus often prioritizes brain over abdominal growth, leading to a reduced abdominal circumference.

Thyroid in pregnancy

- For pregnancy on USMLE, choose the combo of **no change TSH, no change free T4, ↑ total T4** for women who have no thyroid symptoms.
- **Estrogen** causes ↑ thyroid-binding globulin (TBG) production by the liver. TBG is the protein carrier molecule for thyroid hormone in the blood. An NBME Q asks for which hormone causes the ↑ TBG in pregnancy → answer = estrogen, not progesterone.
- Free T4 is the physiologically active form of thyroid hormone. T4 protein-bound to TBG (99%) has minimal effect. Free T4 + TBG-bound T4 = total T4.
- TBG will mop up free T4, causing free T4 to transiently decrease and TSH to rise (less negative feedback). This rise in TSH will stimulate more production of T4 by the thyroid gland, making total T4 go up. The absolute amount of free T4 will increase back to normal, thereby suppressing TSH back to normal. But the total amount of T4 is now increased – i.e., free T4 is normal again, but TBG-bound T4 is higher.
- T3 is normal because free T4 is normal. Free T4 is peripherally converted to T3. I've never seen anything about "free T3" on NBME material and I wouldn't worry about it.
- A student might say, "Wait, but why are you giving the above bold arrows if you just gave me all sorts of transient changes in the arrows based on TBG?." It's because the bold arrows are what the USMLE wants. Pregnant women who are euthyroid will have normal free T4 and increased total T4, and their TSH will be normal. The changes due to TBG rising are likely synchronous and slow enough that the patient's TSH and free T4 stay within reference ranges.
- Postpartum (silent) thyroiditis can result in either hypo- or hyperthyroidism following parturition. These arrows are unrelated to the aforementioned ones. The highest yield point you need to know is that ^{131}I uptake into the thyroid gland is low, *even if the patient is hyperthyroid.* This is the same for deQuervain and drug-induced thyroiditis, where uptake is always low. This is because with thyroiditis conditions, there is merely increased spacing between cells of the thyroid gland due to inflammation, allowing thyroid hormone to leak out into the blood. The gland itself is not excessively producing thyroid hormone. Then we have negative feedback causing low TSH, and in turn less stimulation of the thyroid gland, which is why uptake ↓.
- If USMLE asks you about levothyroxine dosing during pregnancy, the answer is "increase dose by 50%."
- Avoid methimazole in first trimester (teratogenic).
- PTU is the answer for thyroid storm during pregnancy, even though longer-term use in 2nd and 3rd trimesters isn't considered ideal because of hepatic toxicity risk. I've seen students get these Qs wrong saying, "Oh but I thought PTU was contraindicated in 3rd trimester." The answer is, it's what they want for acute thyroid storm, where the importance of the acute PTU Tx eclipses the hepatotoxicity concern.

Hyperemesis gravidarum (2CK/3 only)

- Vomiting in pregnancy that is so severe that ketones are present in the urine.
- hCG is responsible for nausea in early pregnancy. It peaks at 8-10 weeks, which is when the placenta takes over production of progesterone, obviating the need for the presence of hCG to maintain a corpus luteum.
- Vignette will give you a woman with severe vomiting at 8-10 weeks and then ask for next best step in diagnosis → answer = "urinary ketones."
- Sometimes they will already mention ketonuria, and then they want the management, which is "intravenous hydration and anti-emetic therapy." Women with HG clearly can't keep anything down, so oral rehydration and anti-emetics are not sufficient. Both metoclopramide and ondansetron are used for Tx.
- The arrows for vomiting in pregnancy are asked on both Steps 1 and 2CK Qs: ↓ K^+, ↓ Cl^-, ↑ HCO_3^-, ↑ pH.

Ectopic pregnancy points
- Implantation of conceptus outside the uterus, including the parametrium of the uterus and cervix. - β-hCG levels will be much lower than expected for gestational age. You don't have to be an obstetrician and know exact values. - β-hCG should double approximately every 2-3 days in early pregnancy, so the Q can also mention something about poor rate of change of increase. - Highest yield point for USMLE is that ectopic risk is ↑↑ in women who have Hx of PID. As discussed earlier, this is due to scarring of the Fallopian tubes and disruption of the cilia. Ectopic implantation most frequently occurs in the ampulla (70-80%). **(2CK/3 only):** - Methotrexate is given for Tx (asked once on 2CK form) for "small, stable ectopics" – i.e., mother is hemodynamically stable, there is no evidence of tubal rupture / fluid in the peritoneal cavity, the ectopic is <3.5 cm, and β-hCG is <5000 mIU/mL. For your Obgyn rotation, I would know those criteria. - Laparoscopic salpingostomy/salpingectomy is done if methotrexate cannot be given. - If the female is hemodynamically unstable (i.e., low BP), the answer is laparotomy, not laparoscopy.

Miscarriage types (2CK/3 only)	
- USMLE wants you to know that the most common cause of spontaneous abortion is "chromosomal abnormality," or "unbalanced chromosomal rearrangement." This is not specific to any abortion type.	
Threatened	- Means miscarriage may occur, but not inevitable. - Vaginal bleeding from closed cervix, with or without cramping, usually during first trimester. - Bleeding can be heavy. "Heavy" vaginal bleeding on USMLE can be described as passage of "clots" per vaginum. This is the same for menses. - Different from implantation bleeding, which is light vaginal bleeding that sometimes occurs 1-2 weeks after ovulation. - Classic Tx is bed rest, but new guidelines suggests evidence for benefit is lacking.
Inevitable	- As the name implies, means miscarriage that is ensuing / unavoidable. - Vaginal bleeding from open cervix, usually during first trimester. - Labor during first trimester usually presents as severe cramping. - Treatment is expectant management – i.e., allow for spontaneous first-trimester labor to occur. Misoprostol (PGE1 analogue) can be given to ripen the cervix. Surgical interventions like dilation & curettage (D&C) or suction curettage are also sometimes indicated.
Complete	- All products of conception have been expelled; no gestational sac seen. - As the name implies, the abortion is completed/finished.
Missed	- Miscarriage that the body has not detected or attempted to expel yet. - Picked up on ultrasound as absent fetal heartbeat. - Cervix is closed. - Treatment similar to inevitable abortion (i.e., expectant, prostaglandin, surgical).
Incomplete	- Only some of the uterine contents have been expulsed. - Most difficult abortion type on USMLE because it will sound like missed, where they say a fetus with no heartbeat is visualized on ultrasound, but the difference is that they will say there's been passage of clots through an open cervix. - As mentioned before, passage of clots can suggest "heavy" passage of material, indicating components of the fetal material has already passed. - Even though heavy bleeding / passage of clots can occur with threatened abortion as well, in the latter, there is a closed cervix and the fetus is viable. - Treatment is expectant, prostaglandin, surgical.
Septic	- Miscarriage as a result of chorioamnionitis (infection of the gestational sac during pregnancy). - USMLE Q will give fever and discharge in a pregnant woman and then say buzzy details such as that it was an unwanted pregnancy + there is a laceration visualized

	on the cervix, indicating an attempt to self-abort with a hanger, which inoculated / seeded the infection. - Treatment is usually IV antibiotics and surgical evacuation.
Blighted ovum	- Aka anembryonic pregnancy. - Obscure diagnosis, but is asked on 2CK NBME and confuses people. - Fertilized egg that implants onto uterine wall, a gestational sac starts to develop, but then the embryo fails to develop + resorbs, resulting in an empty gestational sac. - β-hCG starts to rise normally because the placenta is present but the body doesn't yet recognize that the embryo is unviable. - The answer on USMLE for early pregnancy where they say ultrasound shows a gestational sac but no yolk sac or embryo.
Recurrent/habitual	- Aka repeated pregnancy loss. - Defined as 2 or more consecutive spontaneous abortions. - Distinct from isolated miscarriages, which occur in 15-20% of pregnancies. - Most important cause on USMLE is anti-phospholipid syndrome in SLE.

HY organ system infections	
Pyelonephritis	- USMLE wants you to know pregnant women are at ↑ risk of pyelo due to two main reasons: - 1) Progesterone slows ureteral peristalsis, thereby ↑ risk of urinary stasis and backup to kidney. - 2) Compression of ureter(s) by uterus, notably in 3rd trimester, ↑ risk of stasis. - For this reason, asymptomatic bacteriuria is always treated in pregnancy, even though it is not treated in non-pregnant women. Nitrofurantoin is a classic drug used to treat asymptomatic bacteriuria, standard UTIs, and cystitis in pregnancy.
Cholecystitis	- Progesterone also slows biliary peristalsis, ↑ risk of biliary sludging and cholesterol stone formation. - Estrogen upregulates HMG-CoA reductase, ↑ hepatic cholesterol synthesis, thereby ↑ risk of stone formation. - Cholelithiasis will present as colicky epigastric pain, usually after meals. - Cholelithiasis + fever = cholecystitis. - Diagnose both with ultrasound. - For cholecystitis only, if it is strongly suspected but ultrasound is negative, the next best step is HIDA scan. - Ursodeoxycholic acid (ursodiol) can be given to pregnant women with cholelithiasis to ↓ stone size. The MOA of this drug is: "decreases cholesterol secretion into bile." - Laparoscopic cholecystectomy is done for cholecystitis in pregnancy.
Appendicitis	- Highest yield point is that appendicitis can present as RUQ pain in pregnancy because the 2nd-3rd trimester uterus pushes the appendix upward. - The Q will say ultrasound of the gallbladder shows no abnormalities, which is how you know the RUQ pain is not due to cholecystitis. - Laparoscopic appendectomy is performed in pregnancy.

Group B Strep prophylaxis (2CK/3 only)
- Adequate prophylaxis is considered to be IV penicillin or ampicillin administered within 4 hours of completion of Stage 2 (delivery of fetus; stages of delivery discussed later). - Oral amoxicillin + clavulanate (Augmentin) is wrong. This is listed as wrong answer choice on 2CK form. - When discussing indications for giving GBS prophylaxis, the first step is mentioning when you do **not** give it, which is when there is Hx of mere colonization with GBS in prior pregnancy. In other words, we do not automatically give GBS prophylaxis in the current pregnancy just because the woman had a (+) culture in the prior pregnancy. Knowing not to give it in this scenario is actually the highest yield point. - Indications for giving GBS prophylaxis:

- (+) Rectovaginal swab at 36 weeks.
- GBS bacteriuria at any point during the pregnancy, even if it was successfully treated. For example, if they say woman had 1st trimester GBS asymptomatic bacteriuria + she received ampicillin or penicillin + cultures are currently negative, we still give intrapartum prophylaxis.
- Hx of early-onset GBS disease in prior pregnancy (i.e., meningitis, pneumonia, or sepsis) in neonate. This is different from mere colonization in the mom while pregnant. In this scenario, the neonate actually went on to get a GBS infection.
- If mother's GBS status is unknown or equivocal, we give it if any one of the following is present:
    - 1) Rupture of membranes (ROM) > 18 hours.
    - 2) Maternal fever >38 C.
    - 3) Preterm delivery (<37 weeks).

- 2CK Obgyn form, for instance, says that a pregnant woman is in labor at 40 weeks, her temperature is 37 C, and her GBS status is unknown. They ask which of the following is the best indication for prophylaxis in this patient → answer = ROM > 18 hours.
- Another 2CK Obgyn Q asks for the Tx of GBS sepsis in the neonate → answer = ampicillin + gentamicin; vancomycin + ceftriaxone is wrong on the form.

- Should be noted that the three most common bacteria causing infections in neonates are: GBS, *E. coli*, and *Listeria*.
- If they say there is neonatal infection caused by gram-positive cocci, answer = GBS; if they say gram-negative rods, answer = *E. coli*; if they say gram-positive rods, answer = *Listeria.*
- *Listeria* can be contracted by the pregnant female via soft cheeses and deli meats.

Other HY Obgyn infections (2CK/3 only)	
Chorioamnionitis	- Infection of the uterus during pregnancy. - Greatest risk factor is ROM >18 hours. - Cause is usually polymicrobial. - Tx = ampicillin + gentamicin +/- clindamycin.
Endometritis	- Infection of the uterus *after* pregnancy. - Endometritis = "Postpartum endometritis" for all intents and purposes. - Can present as fever, uterine tenderness, and foul-smelling lochia within a few days of delivery. - Cause is usually polymicrobial. - Q will say woman had C-section two days ago + now has fever + lower abdo pain +/- foul-smelling lochia → answer = postpartum endometritis; chorioamnionitis is wrong answer because she's not pregnant anymore. - Tx = ampicillin + gentamicin +/- clindamycin. - USMLE also wants you to be aware of what is "normal post-operative course" versus endometritis. For example, the NBME Q can say something like, woman had Caesar two days ago + temp 99.3 F + incision site mildly erythematous + two inguinal lymph nodes are palpable and tender → answer = "normal post-operative course." Surgery is trauma, and the immune system will generate a natural response to it. Palpable lymph nodes can be a normal response to inflammation and surgery.
Septic pelvic thrombophlebitis	- Formation of infected blood clots in the pelvic veins (umbrella term for uterine and ovarian veins). - Endometritis is big risk factor. - Differentiated from endometritis in that it causes persistent pelvic pain and fever despite administration of antibiotics. - Answer choice shows up quite a bit on Obgyn forms, so know it exists.
Puerperal sepsis	- Generalized term for sepsis occurring after childbirth. - Presentation is variable; causes high fever, chills, tachycardia, and low BP. - Q might list both puerperal sepsis and septic pelvic thrombophlebitis as answer choices. Remember that SPT is pelvic pain unresponsive to antibiotics; puerperal sepsis is actual sepsis due to a number of causes, but often endometritis.

Toxic shock syndrome	- Caused by TSST superantigen of *S. aureus*. - Tampon use, or vulvovaginal/perineal skin infections are HY etiologies. For non-Obgyn Qs, cotton nasal packing also HY. - Toxin bridges MHC-II on macrophages with T cell receptor on CD4+ T cells, causing macrophages to release cytokines leading to shock. - Shows up on Obgyn material where they ask most likely cause of death in TSS, and the answer is ARDS. Slightly unusual answer, but it's a Q they ask.

Dermatology of pregnancy (2CK/3 only)	
Melasma	- Aka chloasma. - Normal/benign hyperpigmentation seen in pregnancy and with OCP use. - Mechanism is ↑ estrogen + progesterone causing ↑ melanin production. - Risk thought to be ↑ with sun exposure. - HY point is that "bronze diabetes" in hereditary hemochromatosis can be misdiagnosed as mere melasma. Remember that menses in women prolong time to onset of clinical signs / symptoms (due to menses ↓ iron). So during pregnancy when menses temporarily cease, hemochromatosis presentation can become clinical for the first time. Ferritin > 300 ng/mL is diagnostic on USMLE.
PUPPP	- Pruritic urticarial papules and plaques of pregnancy. - Benign skin condition; usually occurs in 3rd trimester or postpartum. - Presents as intensely itchy, red, raised patches and hives, usually starting on the abdomen **within stretch marks;** can spread to limbs and buttocks. - Exact cause unknown; believed to be related to stretching of the skin. - Tx = topical corticosteroids + oral antihistamines + emollients (lotion).
Pemphigoid gestationis	- Aka herpes gestationis. - Presents as intensely itchy, red papules and hives **around the umbilicus.** - Occurs during 3rd trimester or postpartum. - Mechanism is autoantibodies against hemidesmosomes (yes, same as bullous pemphigoid). - Associated with adverse fetal outcomes (i.e., low birth weight and preterm delivery) because the IgG autoantibodies can cross the placenta. - Tx = topical and/or oral corticosteroids.
Intrahepatic cholestasis of pregnancy	- ↑ estrogen during pregnancy causing ↓ secretion of bile acids by the liver, resulting in ↑ **serum bile acids,** but not necessarily ↑ direct bilirubin. - Presents classically as **pruritis of the palms and soles,** without body rash. - ↑ risk of 3rd trimester spontaneous abortion. - Tx = ursodeoxycholic acid (ursodiol) + recurrent fetal monitoring + potential early delivery.
Acute fatty liver of pregnancy	- Life-threatening disorder for both the mother and fetus presenting in 3rd trimester. - Due to long-chain 3-hydroxyacyl-coenzyme A dehydrogenase (LCHAD) deficiency in the fetus, leading to impaired mitochondrial fatty acid oxidation. Sounds highly pedantic and arcane, but it has been asked. - Presents with nausea, vomiting, abdominal pain, and jaundice that can progress to **encephalopathy,** coagulopathy, and multi-organ failure. - Labs can show impaired LFTs, RFTs, and coagulation tests. - Tx = immediate delivery of fetus.

Peripartum bleeding	
Uterine atony	- Most common cause of postpartum bleeding (70-80%). - Presents as vaginal bleeding in the context of large, boggy uterus. - Uterine massage immediately following Stage 2 delivery ↓ risk. - Refractory bleeding can be treated with oxytocin to enhance uterine tone and contractions. If ineffective, ergonovine/methylergonovine can be administered. - Ergonovine is contraindicated in HTN, preeclampsia, migraine with aura, smokers, and cardiovascular disease. - Surgery such as uterine artery embolization or hysterectomy is last resort.
Vaginal laceration	- Usually seen in the setting of fetal macrosomia (e.g., maternal diabetes), as discussed earlier. - Episiotomy can be performed to ↓ risk.
Placenta previa	- Placenta implants over the internal cervical os. - Causes **painless 3rd trimester bleeding.** - C-section recommended 37 weeks onward. - Prior to 36 weeks, placental implantation site can occasionally move off the cervical os as the uterus continues to grow. - Hx of prior interventions to the uterus (i.e., myomectomy, previous Caesars) ↑ risk of placenta previa due to ↑ probability of abnormal implantation (i.e., if the endometrial lining is disrupted in any form, then chance of normal implantation is ↓).
Abruptio placentae	- Placental detachment from the uterine wall prior to parturition. - Causes **painful 3rd trimester bleeding.** - Can present as intense cramping. - Classic causes are cocaine and deceleration injury (i.e., car accident).
Placenta **a**ccreta	- Placental **a**ttachment to the surface of the myometrium. - Causes postpartum bleeding.
Placenta **in**creta	- Placental attachment **into** the myometrium. - Causes postpartum bleeding.
Placenta **per**creta	- Placental attachment through the myometrium, sometimes onto adjacent structures such as the bladder (**per**creta = **per**forates). - Causes postpartum bleeding.
Vasa previa	- Fetal vessels overly the internal cervical os and are prone to shearing forces and bleeding. - Presents as triad of 1) painless third-trimester bleeding, 2) rupture of membranes, 3) fetal bradycardia. - Caused by conditions such as velamentous cord insertion or succenturiate lobe of placenta. - Velamentous cord insertion = portion of fetal vessels normally protected by Wharton jelly within umbilical cord are exposed within the fetal membranes before they insert onto the placenta → ↑ risk of shearing/trauma → ↑ risk of bleeding. - A succenturiate lobe is an accessory/extra lobe of the placenta that is separate from the main body of the placenta but remains connected to it by blood vessels. When vessels connecting the main placenta and the succenturiate lobe run across the cervical os, this can cause vasa previa.
Retained placental tissue	- Fragments of the placenta or membranes remaining in the uterus post-delivery. - Just be aware it's a cause of postpartum bleeding.
Heme disorders	- Conditions such as DIC, ITP, and vWD ↑ risk of postpartum bleeding.
Uterine inversion	- Uterus turns inside out following the delivery of the placenta. - Typically presents as postpartum bleeding and abdominal pain following/during delivery of the placenta + the Q will say the uterus is not palpable (abdominally). - The Q does not need to say vigorous cord traction was applied. There is a 2CK Obgyn Q where they literally say "gentle cord traction," and the answer is still uterine inversion.
Uterine rupture	- Wall of uterus tears; often associated with labor or previous uterine surgery like Caesars (especially classic vertical incision, rather than low, horizontal incision).

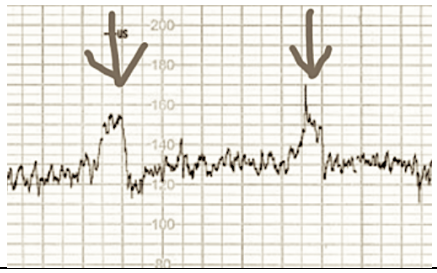
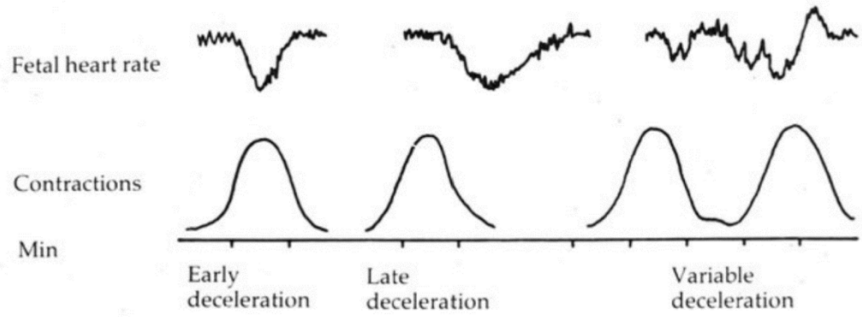
	- USMLE Q will tell you oxytocin was given + there were strong and persistent contractions + now fetal parts are palpable in the RUQ. Student says, "What do they mean there's fetal parts in RUQ?" Exactly. They shouldn't be there, which indicates there was a rupture of the uterus. - Oxytocin can cause tachysystole and hypertonus, which ↑ risk of rupture. - Tachysystole = abnormally high frequency of uterine contractions, usually more than five contractions in 10 minutes, averaged over a 30-minute window. - Hypertonus = prolonged contraction of the uterus > 2 minutes. - Contractions should normally occur once every 2-5 minutes and last 30-60 seconds. - I explain contractions more below with respect to Montevideo units.

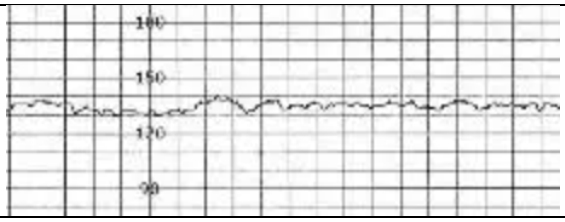
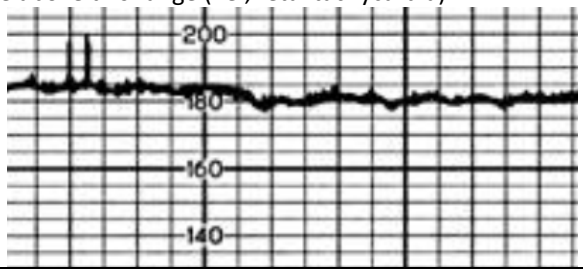
Montevideo units (2CK/3 only)
- Used to assess adequacy of force of uterine contractions during labor. - Annoying for boards purposes, but show up on 2CK Obgyn forms. - Calculated by subtracting baseline uterine pressure from peak uterine pressure during contractions. This difference = force of contraction. - Each force of contraction is then summed over a 10-minute period. - For example, if baseline uterine contractions are 10 mmHg and peak uterine pressure during contractions is 60 mmHg, each contraction is 50 MVU. If we have 4 contractions in 10 minutes, we have 200 MVU. - 200-250 MVU in 10 minutes is normal. - If less than this range, there is insufficient force of uterine contractions, and oxytocin is often indicated to augment labor.

"Power, Passenger, Pelvis" (2CK/3 only)
- Colloquialism is classic framework used in obstetrics to describe the three primary factors influencing the progression of labor and delivery of the fetus. - Power refers to the force exerted by uterine contractions. As discussed above, we want 200-250 MVU within 10 minutes. - Passenger refers to fetal size and orientation – i.e., normal or macrosomic; cephalic or breech. - Pelvis refers to the maternal bony pelvis, including its size and shape. Some pelvis shapes are more conducive to vaginal deliveries, while others might pose challenges, potentially necessitating interventions like a cesarean section. - **Cephalopelvic disproportion (CPD)** refers to when the fetal head is too large to fit through the mother's pelvis, preventing a safe vaginal delivery. When CPD is diagnosed, Caesar is often performed.

Breech delivery points (2CK/3 only)
- Delivery where the fetus in not cephalically engaged (i.e., usually limb first). - Although not mandatory, managed with Caesar in developed countries because of maternal/fetal risks. - If USMLE tells you fetal position prior to parturition is not known, next best step is ultrasound. - If the fetus is breech, **external cephalic version,** which is attempting to manually shift a fetus from breech to cephalic engagement is attempted at **36-38 weeks** before Caesar. After 38 weeks, the fetus is often sufficiently large where risk of uterine rupture ↑. - If ultrasound or Leopold maneuvers (abdominal palpation techniques to determine fetal lie) determine the fetus is breech or transverse after 38 weeks, NBME wants Caesar as next best step. - Internal podalic version will show up as a wrong answer choice quite often on NBME. I've never seen it as a correct answer. But in theory, it is performed to deliver a second twin during a multi-gestational delivery, where following delivery of the cephalically engaged first twin, the second twin is breech. In this case, the feet of the second twin can be manipulated to attempt to revert to vertex (head-down) positioning. - Breech delivery ↑ risk of complications like: - Erb-Duchenne palsy (upper brachial plexus injury; C5/6);

- Klumpke palsy (lower brachial plexus injury; C8/T1); - Cerebral palsy (hypoxia at birth → impaired brain development and "scissoring" of the limbs); - Cord prolapse (variable decelerations [discussed below] +/- bleeding; pelvic exam is done to palpate for the cord; C-section is performed).

Fetal non-stress testing	
- Mostly Steps 2CK/3, but known to show up on Step 1.	
Accelerations	- The non-stress test (NST) is a test that monitors fetal heart rate. - Accelerations reflect fetal well-being and refer to an increase in fetal heart rate of ~20 bpm that lasts ~20 seconds. This should occur 2-3 times within a 20-minute period. 
Early decelerations	- Mean fetal head compression. - Drop in the fetal heart rate at the same as uterine contractions (i.e., the nadir of the fetal bradycardia and peak of uterine contraction are in same y-plane). - Drop in fetal HR occurs as a result of ↑ vagal outflow. This is in part due to ↑ vascular resistance as a result of the compression. 
Late decelerations	- Mean fetal hypoxia, or uteroplacental insufficiency. - The fetal bradycardia occurs just after the uterine contraction. - Late decels are the "bad type." - Treatment for both late and variable decels is triad of 1) stop all oxytocin; 2) give maternal oxygen; 3) position the patient left-lateral decubitus. - If the above measures don't work, C-section is often performed. - Fetal scalp pH can be done in theory to assess degree of fetal hypoxia if decels are prolonged. Procedure involves obtaining a fetal blood sample from the scalp. If pH of the fetal blood is <7.2, this suggests acidosis and hypoxia. Even though <7.35 is acidosis, the <7.2 cutoff is used as a stronger threshold to determine whether more urgent interventions are warranted.
Variable decelerations	- Mean umbilical cord compression. - Amnioinfusion is an answer on USMLE as Tx for variable decelerations in conjunction with above triad. This is when sterile saline is injected into the uterine cavity to help relieve umbilical cord compression.
Fetal sleep state	- Normal fetal heart rate is 110-160 bpm. - The answer on USMLE if they give a non-stress test showing a straight line within this normal range.

	
Maternal fever	- The answer on USMLE if they give non-stress test showing a straight line for fetal heart rate above this range (i.e., fetal tachycardia). 

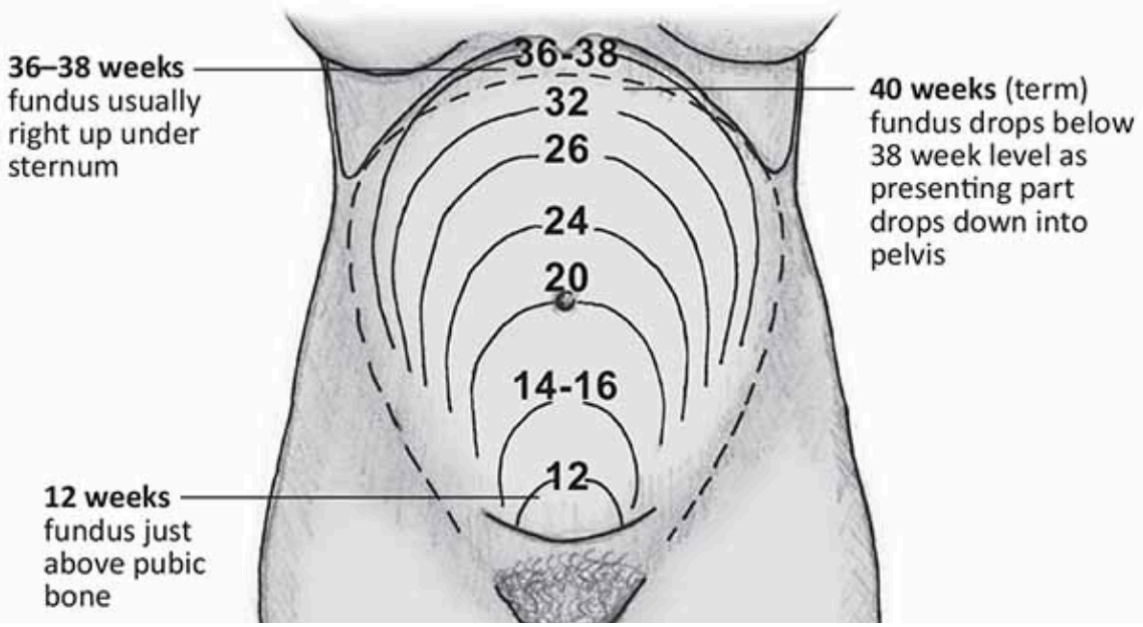
Biophysical profile (BPP) (2CK/3 only)
- Ultrasound evaluation to determine fetal well-being, particularly in high-risk pregnancies or instances where a fetus might be in distress. - Evaluates five parameters: 1) Fetal movement (3+ discrete body movements in 30 mins); 2) Fetal tone (1+ episode of active extension/flexion of a joint in 30 mins); 3) Fetal breathing movements (1+ episode lasting >30 seconds in 30 mins); 4) Amniotic fluid volume (>1 pocket of fluid measuring >1cm in 2 perpendicular planes); 5) Fetal heart rate pattern (>2 accelerations in 20 mins). - Each parameter is scored with either 0 or 2 points, providing a maximum total score of 10. - Score of 8-10 is considered reassuring, while a score below 8 might warrant further evaluation or intervention to ensure the well-being of the fetus. - BPP particularly crucial in cases where the mother has a condition capable of causing fetal compromise (i.e., preeclampsia or diabetes), when the fetus shows signs of potential distress, or for post-term gestation. Therefore, it is a vital tool to assist healthcare providers in making informed decisions regarding the management and timing of delivery in pregnancies with potential concerns. - For USMLE, you need to be aware of this test as an answer choice of exclusion (i.e., you eliminate to get there). In other words, the vignette might give you a 3rd trimester female with intrahepatic cholestasis of pregnancy, who hasn't felt fetal movements in the past 24 hours. You can eliminate nonsense answers such as amnioinfusion and immediate C-section, and select BPP as next best step.

Bishop scoring (2CK/3 only)
- Scoring system used to determine likelihood of successful vaginal delivery. - Scored from 0-13 points. - Score of 8+ suggests a ripe cervix; higher success rate if induction of labor with oxytocin is initiated. - Score 6 or lower suggests inadequate cervical ripening; induction of labor may not be successful and cervical ripening agents may be recommended – topical dinoprostone (PGE_2 analogue).

	Score			
	0	1	2	3
Position	Posterior	Middle	Anterior	--
Consistency	Firm	Medium	Soft	--
Effacement	0-30%	40-50%	60-70%	80%+
Dilation	Closed	1-2cm	3-4cm	5+cm
Station	-3	-2	-1/0	+1/+2

Uterine size per gestational week (2CK/3 only)

- You aren't expected to be an obstetrician and memorize the exact fundal heights per week of gestation, but USMLE has the expectation you know a couple points:
- Fundal height, measured in centimeters from the pubic bone to the top of the uterus, is usually equal to gestational age in weeks between 20 and 32 weeks of pregnancy, (McDonald rule).
- USMLE can say something like, "She's 30 weeks' gestation + fundal height is 36 cm," where you should say, "That sounds like polyhydramnios" (e.g., for GDM, etc.).
- At 20 weeks' gestation, fundal height should be at the level of the umbilicus. USMLE loves this landmark to hint at molar pregnancies (where the female presents big for gestational age). For example, they'll say a woman is 16 weeks' gestation + fundal height is measured at the level of the umbilicus.

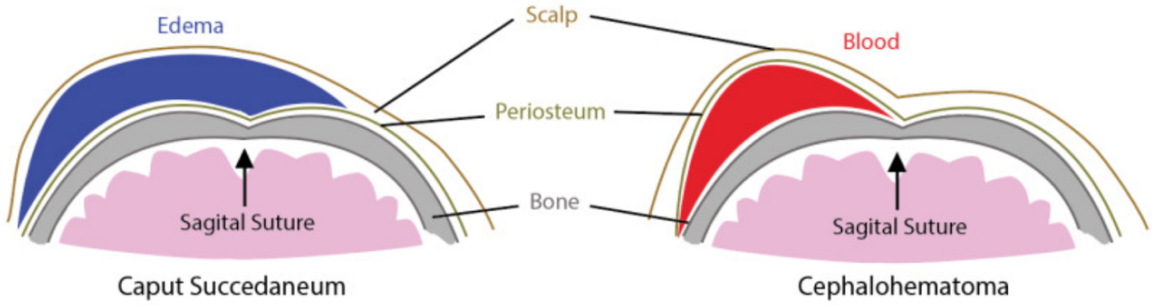


Braxton Hicks vs true labor contractions (2CK/3 only)	
Braxton Hicks contractions	- Can occur as early as 2nd trimester, but are usually 3rd trimester. - Irregular + infrequent. - Often subside with changes in activity or position. - Do not increase in intensity or duration.
True labor contractions	- Occur at regular intervals, gradual ↑ in frequency, duration, and intensity. - Associated with changes in the cervical effacement and dilation. - Usually persist and intensity despite movement or positional changes. - Often accompanied by other signs of labor such as the rupture of membranes or bloody show.

Stages of labor (2CK/3 only)	
1st stage	- Refers to dilation of the cervix from 0 to 10cm. - Latent phase of Stage 1: cervical dilation from 0-6 cm. - Apparently 4 cm used to be the old cutoff, but now it's 6 ("6 is the new 4"). - Latent phase is long (i.e., up to 20 hours for nulliparas women, and 14 hours for multiparas women). - Protracted latent phase is <1cm dilation over 4-6 hours, despite regular contractions. - Arrest of latent phase is no cervical change over 6 hours, despite regular contractions. - Active phase of Stage 1: cervical dilation from 6-10 cm. - Occurs much faster than latent phase. - >1.2cm/hr for nulliparas women; >1.5cm/hr for multiparas women. - Protracted active phase is slower than the above numbers. - Arrest of active phase is no cervical change in >2 hours.
2nd stage	- Starts once max cervical dilation (10cm) is achieved until delivery of fetus. - Nulliparas women: 4 hours with an epidural; 3 hours without an epidural. - Multiparas women: 3 hours with an epidural; 2 hours without an epidural. - Protracted labor is longer than the above numbers. - Arrest of labor is no fetal head descent for 1 hour despite adequate contractions. - Vacuum (suction cup) or forceps delivery may be done to expedite 2nd stage. - Vacuum delivery ↑ risk of caput succedaneum and cephalohematoma (discussed in table below); forceps delivery ↑ risk of fetal cranial nerve or sternocleidomastoid injury (leading to congenital muscular torticollis).
3rd stage	- Starts after the fetus has already been delivered. Refers to delivery of the placenta. - Takes 5-30 minutes. - Prolonged third stage is >30 minutes. - Placenta should be checked for all lobes attached (i.e., to prevent retained placental tissue). - Uterine massage should be done immediately after to ↓ bleeding risk from uterine atony.

PROM vs PPROM (2CK/3 only)
- Premature rupture of membranes (PROM) = rupture of the amniotic sac prior to labor >37 weeks. - Preterm premature rupture of membranes (PPROM) = PROM <37 weeks. - Can be confirmed through various tests – i.e., pooling of fluid, nitrazine paper test, or microscopic examination of vaginal secretions for ferning. (Tests discussed in below table). - PROM and PPROM bear increased infection risk if prolonged >18 hours; give prophylactic IV ampicillin, penicillin, or erythromycin. - PROM (i.e., 37 weeks or later) is usually treated with delivery, since fetus is already term. - PPROM is managed depending on how early it occurs, striking a balance between allowing for progression of gestational age vs infection risk. The mindset of "each day matters" applies to fetal gestational development when the child is preterm. - If PPROM occurs <24 weeks, the pregnancy is usually maintained as long as possible to achieve fetal viability, followed by monitoring fetal activity and infection risk to achieve ↑ gestational age.

Random tests for ROM (2CK/3 only)	
- List is not meant to be exhaustive. There are many such tests. I've seen these on NBME exams.	
Fetal fibronectin	- Protein produced by the chorion of the placenta. - Acts as a "glue" attaching the chorion to the uterus. - Its presence in vaginal secretions between 22 and 34 weeks of gestation is often associated with a higher risk of preterm birth. - The test may be utilized to help predict the likelihood of preterm labor delivering within the next week or two. - A (+) result indicates ↑ risk; a (-) result suggests ↓ risk.
Nitrazine paper test	- Used to detect presence of amniotic fluid in vaginal secretions; indicates ROM. - Nitrazine paper changes color in response to pH. - Normal vaginal pH is acidic (4.5-6.0); amniotic fluid is more alkaline (>7.1). - When the Nitrazine paper comes into contact with amniotic fluid, it changes from yellow/orange to blue.
Ferning pattern	- Used to assess ROM. - Sample of vaginal fluid is viewed on light microscopy and observed for the characteristic pattern that resembles fern leaves once it has dried. - This crystallization pattern results from the presence of amniotic fluid that leaves behind salts dry in a distinctive feathery/fern-like pattern. - Used alongside the Nitrazine test due to potential false-positives (i.e., low specificity) arising from cervical mucus, which can also create a fering pattern.

Caput succedaneum vs cephalohematoma (2CK/3 only)	
	
Caput succedaneum	- Swelling of an infant's scalp appearing shortly after delivery. - Soft, puffy, and sometimes bruised area that can cross suture lines. - Relatively common; can occur with vaginal deliveries due to pressure exerted on the fetal head by the vaginal canal. - Not threatening; resolves on its own within a few days post-birth without Tx.
Cephalohematoma	- Collection of blood between a baby's skull and the periosteum (membrane covering the skull). - Presents as a raised, fluctuant, and well-defined mass that is confined to suture lines (in contrast to caput succedaneum, which can cross them). - Typically caused by vacuum or forceps delivery. - Usually do not pose a risk to the infant and self-resolve within weeks to months. - ↑ risk of jaundice as the blood is reabsorbed, due to ↑ bilirubin levels.

Sheehan syndrome
- Maternal pituitary grows in size during pregnancy due to ↑ hormone production. - Traumatic labor (e.g., C-section with loss of considerable blood) ↑ risk of anterior pituitary ischemic infarction.

- Textbook vignette is ↓ ability to breastfeed due to ↓ milk production following labor where BP was ↓.
- Schools/resources will usually just teach that prolactin is ↓. But USMLE wants you to know that **all hormones** coming from the anterior pituitary are low.
- NBME wants a ↓ for prolactin, ACTH, and TSH, as well as an ↑ for aldosterone.
- The ↑ for aldosterone is weird, I agree. Take it up with NBME, not me. My suspicion is that ↓ cortisol from ↓ ACTH can lead to ↓ basal blood pressure, where aldosterone could go ↑ to compensate. The patient is not going to have overt hyperaldosteronism with ↑ Na^+, ↓ K^+, ↑ HCO_3^-, and ↑ pH. It's more that, in theory, basal level aldosterone could just go ↑ slightly to compensate.
- The vignette can mention fatigue, which reflects ↓ TSH (secondary hypothyroidism).

HY Obgyn cardiovascular DDx	
Amniotic fluid embolism	- The answer on USMLE for acute-onset shortness of breath and tachycardia 30 seconds to 2 minutes after delivery of the placenta. - Amniotic fluid leaks into maternal circulation + goes to pulmonary arterioles. - Can cause **disseminated intravascular coagulation** with bleeding from IV lines / catheter sites. - Tx = supportive.
Pulmonary embolism	- The answer on USMLE for acute-onset shortness and tachycardia two days post-Caesar when the mother gets up to go to the bathroom (presumably a DVT that formed while in hospital bed launched off to lungs). - Tx = heparin first, followed by CT of the chest (aka spiral CT or CT angio). - *If woman is pregnant,* then do V/Q scan before CT. This is in order to decrease potential radiation exposure to the fetus. - If V/Q scan is performed during pregnancy that shows "segmental perfusion defects" (i.e., suggestive of PE), the next best step in diagnosis is CT of the chest if it's listed. Student says, "Wait, but I thought you just said we don't do that in pregnancy because of radiation." I agree with you. That's why we did V/Q scan first. But if they force you to choose a NBS in Dx following the V/Q scan, the answer is still CT. Take it up with USMLE, not me.
Peripartum cardiomyopathy	- Dilated cardiomyopathy almost always during 3rd trimester or in the first few months postpartum. - Cause is multifactorial, but some studies have identified an autoantibody-mediated process. - Will present as gradually worsening shortness of breath on exertion (i.e., LHF at a minimum). Depending on severity, can also present with JVD and significant pitting edema (RHF findings). - Peripartum cardiomyopathy tends to get worse with each subsequent pregnancy. - Diagnosis is made with transthoracic echo looking for low ejection fraction. - Qs like to ask how to determine degree of maternal/fetal risk in subsequent pregnancies → answer = transthoracic echo (looking for EF). In other words, the next pregnancy is going to have an even lower EF than the current, so if the EF is already significantly low (NR 55-70%), we know it'll only be worse subsequently.
Normal peripheral edema	- Peripheral edema is common in pregnancy due to ↑ plasma volume + the uterus compressing the pelvic veins and IVC; both cause ↑ venous hydrostatic pressure.
IVC compression syndrome	- The answer on USMLE for 3rd trimester female who gets lightheadedness, nausea, palpitations, shortness of breath, and sometimes a drop in blood pressure while supine. - Due to compression of uterus on IVC, ↓ venous return. - Ameliorated with positioning/sleeping left-lateral decubitus.
Atelectasis	- Confusing word that refers to "lung collapse," or "collapse of alveoli." - Highest yield point for USMLE is that it is the most common cause of fever within 24 hours of post-surgery. If this is the first time you're reading this, that might sound weird, but this is pass-level and extremely important for 2CK.

	- There is one 2CK Q where they say a woman had a C-section two days ago and the answer is still atelectasis, so even though it's most common <24 hours, just be aware one Q exists where, oh em gee, it's 2 days later. - The mechanism is related to combo of pain meds + sedentation, where breathing becomes slower + shallower in hospital bed, leading to mild collapsing of some alveoli. This is why breathing exercises can be important post-surgery. - Will often present as **bibasilar shadows or opacities.** In other words, patient had surgery yesterday + now has fever + CXR shows mild opacity at the lung bases → answer = atelectasis. - Step 1 NBME assesses obstructive (aka resorptive) atelectasis. This is when an area of lung distal to an obstruction from, e.g., a tumor, can cause alveoli to collapse. This then increases the chance for pneumonia distal to the obstruction. - 2CK IM form has "endobronchial obstruction" as answer for distal area of lung collapse (i.e., atelectasis) in patient with lung cancer; "vascular occlusion by tumor" is wrong answer (makes sense, as the tumor obstructs the respiratory tree, not blood vessel, in this case, but I've seen students accidentally choose the latter).
Budd-Chiari syndrome	- Hepatic vein thrombosis due to hypercoagulable state, such as during/after pregnancy. - Presents as abdominal pain + hepatomegaly +/- ascites (latter indicates hydrostatic pressure backup to the portal vein). - Placenta produces plasminogen-activator inhibitor (PAI-2), which ↓ plasmin activity and fibrinonlysis → hypercoagulable state. In addition, ↑ estrogen and progesterone contribute to hypercoagulable state by ↑ fibrinogen, clotting factor, and vWF synthesis. - Hypercoagulable state in pregnancy is an evolutionary mechanism to ↓ hemorrhage risk at parturition.
Hemorrhoids	- Bleeding or protrusion from rectal veins. - Common during pregnancy, particularly in the second and third trimesters. - ↑ Pelvic blood flow and pressure from uterus on the pelvic and rectal veins. - ↑ Progesterone during pregnancy relaxes the walls of the veins, allowing them to swell more easily. - Constipation, which is also common during pregnancy, can cause straining during defecation, further contributing to the development of hemorrhoids. - Tx = dietary modifications to prevent constipation, using cushions or pillows to relieve pressure when sitting, and avoiding prolonged periods of standing or sitting.

HY neonatal conditions	
- I don't want to do an exhaustive list that ventures into pure Peds shelf content.	
Neonatal respiratory distress syndrome (NRDS)	- Aka hyaline membrane disease. - The answer on USMLE for respiratory distress in kid who is born <34 weeks' gestation. - Due to insufficient surfactant production by type II pneumocytes due to ↓ lamellar bodies (the specialized organelles that produce surfactant). - These kids have ↓ lecithin/sphingomyelin ratio (i.e., <2.0). Normally it is >2-2.4. - Another name for lecithin is dipalmitoyl phosphatidylcholine. This is asked on NBME. - USMLE can give simple vignette of NRDS and then ask for various manipulation of the ratio – i.e., "↑ sphingomyelin" might be an answer (makes sense, since this would ↓ the ratio). - ↓ surfactant production means ↓ alveolar compliance and ↑ **elastic recoil.** Surfactant is hydrophobic and normally prevents the alveoli from collapsing, so if it's deficient, the hydrophobic interactions of the alveolar walls enable ↑ collapsing/elastic recoil. - CXR shows a "reticulogranular" appearance. Very buzzy and HY.

	- In order to prevent NRDS, a pregnant woman giving birth <34 weeks' gestation must be given ***two*** boluses of corticosteroids within 24 hours of parturition, which accelerate fetal lung maturity. For example, there is a 2CK Obgyn Q where they tell you a woman giving birth at 33 weeks' gestation was given a bolus of corticosteroids 12 hours ago. They ask for next best step → answer = "bolus of corticosteroids." Sounds weird because they said it was just done, but she needs ***two*** boluses. - For management, 2CK wants tactile stimulation first, then place under warming lights. This sequence is HY. Then give exogenous surfactant and oxygen.
Transient tachypnea of the newborn (TTN)	- The answer on USMLE when the vignette sounds like NRDS but the kid is term. For instance, they'll say neonate has difficulty breathing, but then you'll notice he's >37 weeks. - Usually seen **following C-section** or fast vaginal delivery. - Mechanism is insufficient time for the pulmonary lymphatics to clear amniotic fluid from the lungs. In addition, the pressure exerted on the fetus by the birth canal can help expulse fluid from the fetal lungs, so in the setting of a C-section, we don't have this. - CXR shows "fluid within fissure lines."
Bronchopulmonary dysplasia	- Fibrotic lung disease in an infant caused by continued use of supplemental oxygen. - Vignette will give you a kid who was born at 26 weeks' gestation who was in ICU on oxygen + now is 4 months old and is on home oxygen. They will say ECG shows right-axis deviation +/- CXR shows increased pulmonary vascularity → this reflects RVH (ensuing cor pulmonale) from pulmonary hypertension caused by the bronchopulmonary dysplasia. - Retinopathy of prematurity can also occur due to ↑ oxygen exposure in neonate.
Meconium aspiration syndrome	- Newborn aspirates a mixture of meconium (the infant's first stool) and amniotic fluid into their lungs during parturition. - Can cause airway obstruction, infection, inflammation, and dyspnea. - Q will mention "dark, meconium-stained fluid." - Highest yield risk factor for USMLE is post-term birth (i.e., after 40 weeks). - Can cause persistent fetal hypertension (below).
Persistent fetal hypertension	- Aka persistent fetal circulation. - Q will give a post-term birth at 42 or 43 weeks + meconium-stained fluid + echo of the neonate shows a R→L shunt across the foramen ovale. Student says, "Wait, but isn't the foramen ovale between the atria, and that's only open in the fetus but is supposed to close after birth?" Correct. Hence we have *persistent* fetal circulation. - Answer on USMLE will be "failure of pulmonary vasodilation." Meconium aspiration syndrome can ↓ opening of the lung vasculature, leading to ↑ right heart pressure and ↑ risk of persistent fetal circulation.

Repro/Obgyn Psych conditions	
Depression during pregnancy	- Sertraline is first SSRI used due to lower risk profile. - Electroconvulsive therapy (ECT) is used for women who have suicidal/homicidal ideation, or who are refusing to eat or drink. - ECT is generally considered to be safe for the fetus when performed appropriately and with careful precautions.
Bipolar during pregnancy	- Avoid lithium because of its teratogenic effects (Ebstein anomaly). - Avoid valproic acid because it causes neural tube defects and cognitive impairment. - Bipolar is Tx in women planning to get pregnant, or during pregnancy, with agents such as lamotrigine, olanzapine, or quetiapine.
Postpartum blues	- Aka "baby blues"; normal, transient postpartum psychologic state characterized by sadness, anxiety, mood swings, and insomnia. - Peaks in first week. If >10 days on USMLE, think postpartum depression instead.

	- Resolves without intervention. - Thought to be related to the sudden hormonal shifts and physical and emotional adjustments postpartum.
Postpartum depression	- Characterized by persistent sadness, anxiety, and exhaustion that can hinder daily care activities and bonding with the baby. - The answer on USMLE if the vignette mentions thoughts of worthlessness or guilt, especially after 10 days postpartum. - Vignette might say the woman leaves her baby in crib alone crying for long periods of time, or the child has soiled diapers (i.e., not catered to) → answer = "immediate psychiatric referral." - If SSRI is used, sertraline or paroxetine are often used because of ↓ concentrations in breast milk.
Postpartum psychosis	- Extreme mood swings, hallucinations, paranoia, and attempts to harm oneself or the baby; occurs in first few weeks postpartum. - Answer = immediate psychiatric referral. - Medications used are anti-psychotics, usually olanzapine or risperidone.
Premenstrual syndrome	- Mood swings, breast tenderness, and irritability occurring in the luteal phase (i.e., the two weeks prior to menses). - Common; benign; thought to be due to hormonal fluctuations.
Premenstrual dysphoric disorder	- Severe, sometimes incapacitating, form of premenstrual syndrome. - Characterized by significant mood disturbances and physical symptoms that dramatically interfere with socio-occupational functioning. - Also thought to be due to hormonal fluctuations. - SSRIs are often first-line.

Polycystic ovarian syndrome (PCOS)
- High BMI female → insulin resistance. - Insulin resistance causes abnormal GnRH pulsation → leads to ↑ LH and ↓ FSH. This is often truncated as just saying there's an ↑ LH/FSH ratio. Some students think FSH is also ↑, but it's just the ratio that's ↑. That's wrong. FSH is low. - LH normally acts on the theca lutein cells of the ovaries to make androgens. Since LH is ↑, we get hirsutism. - FSH normally stimulates follicular development. Since FSH is ↓, we have poor follicular development, leading to failure of a Graafian follicle to rupture during ovulation. The unruptured follicle is retained as a follicular cyst. - Failure of ovulation is called anovulation. This term is exceedingly HY on USMLE. It presents as a female with irregular periods. - Anovulation is a broader term than PCOS, as it can be due to other conditions as well, such as hypothyroidism and Cushing syndrome. But when you hear the word "anovulation" alone, it is usually used as synonymous for the same mechanism as PCOS – i.e., high-BMI female who has irregular periods due to abnormal GnRH pulsation causing an ↑ LH/FSH ratio, with the ↓ FSH causing failure of follicular rupture. - So high-BMI female + irregular periods = anovulation till proven otherwise. - Anovulation + hirsutism = PCOS. - In clinical practice, part of the PCOS diagnosis requires ultrasounds showing 11+ cysts bilaterally (Amsterdam criteria). But USMLE doesn't assess this. You just need to know anovulation + hirsutism = PCOS. - Normally when a follicle ruptures, the remnant is called the corpus luteum, which secretes progesterone. - Progesterone inhibits growth of endometrium; estrogen stimulates growth of endometrium. - Women who have anovulation have ↓ progesterone production because they don't form a corpus luteum. - This means they have ↑ estrogen in comparison to progesterone. We call this **unopposed estrogen.** This is one of the highest yield phrases for USMLE. - Unopposed estrogen means ↑ risk of endometrial hyperplasia and, in turn, ↑ risk of endometrial carcinoma. - The Q can give you a high-BMI female who's post-menopausal + has vaginal bleeding. Answer is just straight-up endometrial biopsy. Student asks how we know it's endometrial cancer. My response is, if she's

overweight, this implies she was probably overweight in the past, which implies she's had history of anovulatory cycles and endometrial hyperplasia, leading to ↑ endometrial cancer risk. - Because insulin resistance is the basis for PCOS, patients are at ↑ risk of developing type II diabetes.
- As mentioned above, hypothyroidism and Cushing syndrome are also HY causes of anovulation on USMLE. - ↑ Glucocorticoids in Cushing syndrome can cause insulin resistance and anovulation, where the diagnosis can appear like PCOS. The difference is PCOS is idiopathic in response to high BMI – i.e., it is not caused by a known secondary etiology like Cushing syndrome, even though the presentations can be similar. - Hypothyroidism leads to ↑ thyroid-releasing hormone (TRH), which stimulates prolactin, which causes abnormal GnRH pulsation. Even though this is the mechanism, the caveat I issue is that USMLE does not directly assess this, and it is infinitely more important you know that the mechanism for prolactin secretion is ↓ dopamine, or overt dopamine-2 receptor antagonism (i.e., for severance of pituitary stalk Qs).

Progesterone withdrawal test
- Progesterone withdrawal test is the process of giving progesterone followed by seeing if bleeding occurs following its withdrawal. It can be used to help diagnose the cause of oligo- or amenorrhea. - As discussed earlier, menses are caused by the presence of progesterone followed by its withdrawal. - If bleeding occurs (i.e., positive progesterone withdrawal test), it means estrogen levels are normal and the endometrial lining builds up just fine, and that anovulation is the cause of the irregular menses. - In PCOS, for instance, since we don't have a corpus luteum and progesterone secretion, we don't have the sequence of "progesterone present followed by withdrawal," so we can't get regular menses. However, since estrogen levels are normal in PCOS, the endometrial lining can grow without a problem, so if progesterone is given exogenously, followed by its withdrawal, we will see bleeding. - In conditions like Turner (where estrogen is low) or premature ovarian failure, giving progesterone won't induce bleeding since the endometrial lining hasn't developed. The bleeding will only occur if estrogen is present in normal amounts to allow for endometrial growth, where the administration, followed by withdrawal, of exogenous progesterone essentially induces forced menses. - In Asherman syndrome, even though estrogen levels are normal, the progesterone withdrawal test won't result in bleeding because the scarred uterus prevents the lining from growing.

Primary dysmenorrhea vs endometriosis vs adenomyosis	
Primary dysmenorrhea	- Dysmenorrhea = period pain. - Primary dysmenorrhea = "normal period pain"; benign. - Mechanism is prostaglandin PGF2α hypersecretion. - During menstruation, the endometrial cells release high levels of prostaglandins, which cause uterine contractions and pain. The prostaglandins can also make their way into the systemic circulation, leading to headache and nausea. - This is the answer on USMLE for period pain under the age of 20. The Q can say, girl who is 17 has menstrual pain so bad she has to miss class + physical exam shows no abnormalities → answer = primary dysmenorrhea. - The answer can also be written simply as "prostaglandin secretion." - The key detail regarding primary dysmenorrhea is that **the physical exam is normal.** - In endometriosis, the physical exam is abnormal + is always over age 20. - Treatment is NSAID. OCP is wrong answer if listed alongside NSAID (on NBME).
Endometriosis	- Growth of endometrial tissue outside the uterus. Most common location is the ovary, but can also grow in other locations like the pouch of Douglas. - Mechanism is thought to be retrograde menstrual flow through the Fallopian tubes + seeding onto the ovaries or peritoneum. - Causes severe menstrual pain over age 20 + **an abnormal physical exam.** - The abnormal physical exam can be any miscellaneous finding – i.e., there is no specific finding you have to memorize. They can say findings such as nodularity of the uterosacral ligaments, a fixed retroverted uterus (due to adhesions from lesions), or pelvic tenderness.

	- In contrast, severe period pain + normal physical exam = primary dysmenorrhea. - In other words: - 23F + period pain so bad she has to miss work + normal physical exam → answer = prostaglandin secretion / primary dysmenorrhea. - 23F + period pain so bad she has to miss work + physical exam shows nodularity of the uterosacral ligaments → answer = endometriosis. - Descriptors such as pain with defecation (due to pouch of Douglas lesions) or dyspareunia (pain during sex) are highly buzzy and pass-level but too easy, so are often omitted from NBME questions, as per my observation. - USMLE wants diagnostic laparoscopy as next best step. - NSAIDs and OCPs are short-term measures, but definitive Tx is laparoscopic removal of lesions.
Adenomyosis	- Growth of endometrial tissue within the myometrium. - Presents as a diffusely enlarged uterus + vaginal bleeding in woman 30s-50s. - May or may not be painful. - Q will say woman had tubule ligation two years ago + now has uterus that is 8 weeks' gestation in size + vaginal bleeding → answer = adenomyosis. Student is confused and says "how can she be pregnant if she had tubule ligation?" She's not. Use your head. It's just how the Q can describe the ↑ uterine size. - Tx = NSAIDs + OCPs. - Leuprolide can be used in theory, but I haven't seen it assessed.

Other repro/obgyn terms	
Mittelschmerz	- Ovulatory pain; benign; common; can present as sharp adnexal pain mid-cycle.
Menorrhagia	- Heavy periods (>80 mL; can present with clots); and/or menses lasting >7 days.
Metrorrhagia	- Mid-cycle bleeding (i.e., any bleeding between periods). - Slightly different from "breakthrough bleeding," which refers to mid-cycle bleeding in women using hormonal contraception.
Dyspareunia	- Persistent or recurrent pain during sex; no one specific etiology. - Can sometimes be seen in endometriosis.
Vaginismus	- Dyspareunia specifically caused by involuntary contraction of the pelvic floor muscles, especially the pubococcygeus (PC), presenting as spasmodic pain during sex.

Miscellaneous repro/obgyn DDx (2CK/3 only)	
Asherman syndrome	- Scarring of the uterus by fibrous adhesions, **aka uterine synechiae.** - Caused by Hx of instrumentation – i.e., D&C. - **Can cause amenorrhea.**
Cervical stenosis	- Q will say female has a Hx of LEEP for cervical CIN2 or 3 six months ago + now has pelvic pain with menses + examination shows a small, scarred cervical os. - Pain can occur in theory if outflow of menses is ↓, similar to imperforate hymen.
Ovarian hyperstimulation syndrome (OHSS)	- Can occur in some women who take fertility meds (e.g., clomiphene) to stimulate egg production. - Results from over-response to the drug, leading to swelling of the ovaries and fluid accumulation in the abdomen, sometimes also affecting the chest. - Symptoms of OHSS can range from mild (i.e., bloating and mild pain) to severe, involving rapid weight gain, severe pain, and shortness of breath. - Usually self-limiting / self-resolves. Just know the Dx exists.
Premature ovarian failure	- Aka primary ovarian insufficiency. - Term used to apply to idiopathic menopause age 40 or younger.
Atrophic vaginitis	- Vaginal atrophy, leading to thin, dry, itchy, and inflamed vaginal walls.

	- Usually due to menopause, but can in theory be caused by any etiology of low estrogen. - Can cause dyspareunia. - Topical lubricants are tried first, followed by topical estrogen.
Chemo- / radiotherapy-induced amenorrhea	- As the name implies. You just need to be aware that chemo- and radiotherapy can cause amenorrhea. - Shows up on NBME as "primary hypogonadism" as the answer – i.e., the ovaries themselves are hypo-secreting hormones as reason for amenorrhea.
Autoimmune oophoritis	- Idiopathic autoantibody-mediated destruction of ovarian tissue in women of child-bearing age. Rare. Just know it exists.
Carpal tunnel syndrome	- Compression of median nerve within carpal tunnel. - Normal in pregnancy due to edema in 2nd and 3rd trimesters.
Chronic interstitial cystitis	- Unexplained pelvic pain, bladder pain, and/or dysuria >6 weeks, where all laboratory studies are normal. - Can present with anterior vaginal wall pain (bladder is anterosuperior to vagina). - Q will say 25F + 6 months of dysuria + pubic pressure + pregnancy test negative + labs all normal → Dx = interstitial cystitis straight-up. Not hard. - Treatment is mundane, soft porn stuff like lifestyle modification and dietary changes. Steroids are a **wrong** answer.

APGAR scoring (2CK/3 only)

SCORE	0 points	1 point	2 points
Appearance (Skin color)	Cyanotic / Pale all over	Peripheral cyanosis only	Pink
Pulse (Heart rate)	0	<100	100-140
Grimace (Reflex irritability)	No response to stimulation	Grimace or weak cry when stimulated	Cry when stimulated
Activity (Tone)	Floppy	Some flexion	Well flexed and resisting extension
Respiration	Apneic	Slow, irregular breathing	Strong cry

- The 1-minute APGAR score is used to evaluate conditions at birth, evaluating heart rate, respiratory effort, muscle tone, reflexes, and color, with scores ranging from 0 to 10, indicating the need for immediate medical attention if low.
- The 5-minute APGAR score reflects response to resuscitation; more accurate indication of the baby's overall health and adjustment to life outside the womb.
- Scores >7 are typically considered normal.
- **Acrocyanosis** = bluish discoloration of neonatal extremities at birth (i.e., 1 point) for appearance. USMLE wants you to know this term. Next best step is place under warming lights, followed by tactile stimulation.

Notable Teratogens	
- Agent that causes malformations in the fetus. - Not an exhaustive list. Point is to be HY, not superfluous. - **Fetus is most susceptible at 3-8 weeks' gestation.**	
Agent	**Defect(s) in the fetus**
ACE inhibitors	- Renal issues.
Aminoglycosides	- Hearing issues.
Anti-epileptics	- Neural tube defects (disrupt folate metabolism). - Fetal hydrantoin syndrome (fingernail/digital hypoplasia, facial anomalies).
Carbimazole/methimazole	- Aplasia cutis congenita, choanal atresia, esophageal atresia.
Diethylstilbestrol (DES)	- Vaginal clear cell carcinoma 30-40 years later.
Isotretinoin	- Craniofacial and cardiac abnormalities; cleft lip/palate.
Lithium	- Ebstein anomaly (atrialization of right ventricle).
Methotrexate	- Neural tube defects.
NSAIDs	- Premature closure of ductus arteriosus; oligohydramnios; inhibited labor.
Ribavirin	- Craniofacial anomalies; growth deficits.
Tetracyclines	- Fetal teeth discoloration; inhibition of bone growth.
Thalidomide	- Phocomelia (severe flipper-like malformation of limbs). - Was used frequently in the 1950s-60s to treat nausea in pregnancy.
Warfarin	- Bone and facial anomalies. - CNS defects like Dandy-Walker malformation (HY Neuroanatomy PDF). - Bleeding diathesis in fetus.

Substance abuse in pregnancy	
Agent	**Defect(s) in the fetus**
Alcohol	- Fetal alcohol syndrome; most common cause of mental retardation. - Heart/lung defects (defective neural crest migration). - **Smooth, flat, and/or elongated philtrum.** - Thin vermillion border (thin upper lip). - Widely spaced eyes (hypertelorism), midface hypoplasia, short nose.
Amphetamines + cocaine	- IUGR; cognitive deficits. - NBME can give long vignette of drug use, followed by ask what is most likely to occur in the pregnancy → answer = "preterm delivery and birth." Sounds non-specific and vague, but it's still *most likely* in comparison to most other things. - Cocaine in particular also increases risk of abruptio placentae.
Anti-epileptics	- Neural tube defects (disrupt folate metabolism). - Fetal hydrantoin syndrome (fingernail/digital hypoplasia, facial anomalies).
Heroin/opioids	- Neonatal heroin/opioid withdrawal. - Piloerection (goosebumps), yawning, rhinorrhea.
MDMA (ecstasy)	- IUGR; cognitive deficits.
Smoking	- IUGR and reduced IQ.

Contraception (2CK/3 only)	
Barrier (condoms)	- Sheaths made of latex, polyurethane, or lambskin worn over the penis (male condom) or inside the vagina (female condom). - Self-explanatory, but apart from abstinence, male condoms are #1 way to ↓ pregnancy and STD risk. - Recommended for use in women on isotretinoin, even if they are already on other contraceptive method.

	- OCPs ↑ risk of cervical cancer slightly, not as a direct effect, but because of ↓ barrier contraception use, where HPV exposure is ↑.
Combined oral contraceptive pills	- Contain both estrogen and progesterone. - Estrogen helps to ↓ breakthrough bleeding risk (i.e., metrorrhagia). - Estrogen ↓ probability of ovulation; progesterone ↓ penetration of sperm by ↑ thickening of cervical mucous. In some women, and depending on dose, progesterone can also ↓ ovulation. - Contraindicated in smokers >35 (HY on USMLE), migraine with aura, Hx of thrombotic disorders, or current breast/gynecologic cancer due to the estrogen-containing component. - Obgyn forms can be nebulous and, rather than writing OCPs as the answer, they can write “synchronization of endometrium,” or “triphasic oral contraceptive pills,” which mean the same thing. - ↓↓ Risk of ovarian cancer the most due to synchronization of cycles (asked on NBME). Also ↓ risk of endometrial (probably also due to synchronization). - ↑ risk of cervical cancer due to ↓ barrier contraception use (as mentioned above); some studies have suggested slight ↑ in breast cancer risk, albeit without significance.
Progestin-only pills	- Aka “mini-pill”; only contain progesterone analogue. - Thicken cervical mucous. - Must be taken precisely at same time every day, so require female is well-adherent / compliant.
Transdermal patch	- Releases both estrogen and progesterone and acts same as combined OCP. - Worn for one week at a time.
Vaginal contraceptive ring	- Aka NuvaRing; releases both estrogen + progesterone. - Worn monthly.
Copper IUD	- Copper intrauterine device; releases (you wouldn’t have guessed it) copper ions into the uterine cavity, creating an environment that is toxic to sperm + inhibits their motility, thereby preventing fertilization. - Known for being one of the most effective forms of emergency contraception when inserted within five days post-coitally and can provide up to 10 years of contraceptive protection. - IUDs have risk of migration through uterine wall.
Levonorgestrel IUD	- Release progesterone analogue; usually last 3-7 years and are ideal in women who desire longer-term contraception or have poor medication compliance. - Thicken cervical mucous. - IUDs have risk of migration through uterine wall.
“Depo shot”	- Depot medroxyprogesterone acetate injection form of contraception. - Administered every 3 months. - Thickens cervical mucous. - Known to ↓ bone mineral density; rebounds with discontinuation. - Known to cause erratic bleeding in 1/3, no changes in cycle bleeding in 1/3, and complete amenorrhea is 1/3. - The answer on Obgyn form 5 for woman who normally forgets to take contraception + needs something more reliable + *has had STI within the past 3 months*. This is because IUD is contraindicated if STI within past 3 months.
Implantable rod	- Aka Implanon/Nexplanon. - Releases progesterone analogue; thickens cervical mucous; lasts 3 years.
Diaphragm	- Cervical cap that covers the uterus; often inserted with spermicidal jelly. - Prevents sperm from entering uterine cavity.

Abortion (2CK/3 only)	
Plan B	- Emergency contraceptive meds. - Levonorgestrel (progesterone analogue); effective within 3 days. - Ulipristal (selective-progesterone receptor modulator; SPRM); effective within 5 days. - Once again, copper IUD can be used for emergency contraception and is most effective if inserted.
Plan C	- Abortion meds. - Mifepristone (progesterone receptor antagonist) is used up to 10 weeks post-intercourse. - This is followed by misoprostol (PGE1 analogue) 1-2 days later.
Plan D	- Surgical abortion. - Vacuum aspiration can be done 6-16 weeks. - Dilation and evacuation (D&E) done >16 weeks.

Hormone replacement therapy (HRT) (2CK/3 only)
- Estrogen combined with progesterone; only approved for severe perimenopausal vasomotor symptoms (i.e., hot flashes, urge incontinence, atrophic vaginitis). - Not given to help preserve bone density or for positive role on mood and neurocognition, since ↑ absolute estrogen exposure in women ↑ risk of breast cancer, MI, and thromboembolic events (i.e., DVT, PE, stroke). The latter is because estrogen upregulates fibrinogen and factors V and VIII. - Can cause unopposed estrogen and ↑ endometrial cancer risk only if the woman inadvertently stops taking the progesterone component (i.e., endometrial hyperplasia → ↑ endometrial adenocarcinoma risk). - Estrogen alone can be considered only if the woman has Hx of hysterectomy (i.e., no ↑ endometrial adenocarcinoma risk). - **Can cause ↓ libido due to ↓ endogenous androgen production** (i.e., ↑ negative-feedback at hypothalamus / anterior pituitary → ↓ GnRH → ↓ LH and FSH).

Repro/obgyn drugs	
- Not meant to be exhaustive list; just HY ones for the USMLE.	
Aromatase inhibitors	- ↓ Conversion of androgens to estrogens. - Letrozole is used to stimulate ovulation by ↓ negative-feedback at hypothalamus / anterior pituitary. Some students ask about this agent in comparison to clomiphene (discussed below). The literature seems to be split on it, and I haven't seen NBME give a fuck. - Anastrozole and exemestane are used for breast cancer in post-menopausal women.
BPH meds	- Finasteride is 5α reductase inhibitor used for BPH. ↓ conversion of testosterone to DHT. Since DHT causes prostatic growth, ↓ DHT means ↓ prostatic growth. - Tamsulosin/terazosin are α1 blockers that ↓ constriction of internal urethral sphincter of the bladder and help promote urinary outflow.
Clomiphene	- Selective-estrogen receptor modulator (SERM) used to stimulate ovulation in those with irregular cycles (in particular those PCOS). - Has partial-agonist effects at the hypothalamus that are weaker than endogenous estrogen, so the hypothalamus interprets this as ↓ estrogen is present, so negative-feedback also ↓ → GnRH ↑.
Danazol	- Androgen-receptor partial agonist. - Theoretically a med that can be used for endometriosis, but USMLE doesn't assess this use-case. - Used for hereditary angioedema, where it stimulates the liver to synthesize more C1-esterase inhibitor. - Can cause hirsutism and pseudotumor cerebri.
GnRH receptor agonists	- Leuprolide, goserelin, and nafarelin are all GnRH receptor agonists. When administered continuously, they cause desensitization of the GnRH receptor at the

	anterior pituitary, leading to ↓ LH and FSH secretion (i.e., even though the drugs are pharmacologic agonists, they function clinically as antagonists).
HTN meds	- Methyldopa (α2 receptor agonist), labetalol, and nifedipine classically used. - Hydralazine used for HTN emergencies in pregnancy.
Magnesium	- Used in severe preeclampsia to prevent seizures (eclampsia). - Used to treat seizures in eclampsia. - Given to women giving birth <32 weeks' gestation as a neuroprotective agent for the neonate. - Can in theory be used as tocolytic (discussed below).
Methotrexate	- Given for small, stable ectopic pregnancies – i.e., mother is hemodynamically stable, there is no evidence of tubal rupture / fluid in the peritoneal cavity, the ectopic is <3.5 cm, and β-hCG is <5000 mIU/mL. For your Obgyn rotation, I would know those criteria for 2CK. You don't need to know those specifics for Step 1, but should instead just know that it's a dihydrofolate reductase inhibitor (i.e., interferes with folate metabolism).
Minoxidil	- Arteriolar dilator that promotes hair growth in androgenetic alopecia. - USMLE wants you to know the latter is polygenic and risk is ↑ with anabolic steroids.
SERMs	- SERMs are agents that have different effects depending on the tissue. - Tamoxifen and raloxifene are SERMs used for estrogen-receptor (+) breast cancer, where they both are antagonistic at breast and agonistic at bone. - Tamoxifen causes ↑ risk of endometrial cancer due to agonistic effects at endometrium.
Tocolytics	- Suppress labor by ↓ uterine contractions; often utilized to delay premature labor to allow for fetal lung maturity while two boluses of corticosteroids are administered. - Terbutaline (mixed β1/2 agonist), ritodrine (β2 agonist), and nifedipine are HY ones. - Indomethacin and magnesium can also technically be used for this reason.

Original bullet point structure (You can use the following for continued/rapid review, or move to next PDF):

- "What do I need to know about embryologic development (i.e., # of weeks certain things develop, etc.)?"
    - At **2** weeks' gestation, the embryo has **2** layers (epiblast + hypoblast).
    - At 3 weeks, neural tube development has commenced. This is completed by week 4.
    - Between 3-8 weeks, most organogenesis is occurring. Fetal heart beat doesn't commence until week 4.
    - What this means for USMLE: the range of 3-4ish weeks is when the fetus is most susceptible to neural tube defects (i.e., spina bifida) if there is folate deficiency, or exposure to drugs such as valproic acid or other anti-epileptics (which cause folate malabsorption). In addition, if they Q asks you when the fetus is most susceptible to teratogens in general, select the answer that is 3-4 weeks as priority; if that tight range isn't listed, select the broader one that encompasses it, e.g., 3-8 weeks. This is all over NBME exams.

- "What do I need to know about which bodily structures/organs originating from certain germ layers, i.e., ectoderm, etc.?"
    - Most embryologic derivative memorization is nonsense, especially now that Step 1 is P/F.
    - A good rule of thumb is: if you literally have no idea on a USMLE question what the answer is, **neural crest** is usually correct. I'd say this is the case in at least 3/4 of questions.
    - "Failure of neural crest migration" is answer for heart defects due to DiGeorge syndrome and fetal alcohol syndrome. It's also answer for Hirschsprung disease.
    - Additionally, you should be aware that pheochromocytoma, melanoma, and neurofibromatosis I/II are neural crest.
    - Craniopharyngioma = derived from Rathke pouch, which is the "roof of the primitive pharynx"; considered ectoderm.
    - Thyroglossal duct cyst = derived from "endoderm of foramen cecum"; the latter is the base of the tongue; in other words, the thyroid gland starts off embryologically at base of tongue and descends.
    - CAP = Clefts, Arches, Pouches; clefts (aka grooves) become ectoderm; arches become mesoderm; pouches become endoderm.
    - Ectoderm → highest-yield structures are: skin + anal canal below pectinate line.
    - Mesoderm → highest-yield structures are: muscle + connective tissue; spleen + kidneys.
    - Endoderm → esophagus + lining of GI tract until the pectinate line; parathyroids + thymus.
    - 3rd + 4th pharyngeal pouches are highest yield of the CAP on USMLE:
        - 3rd pouch = the two inferior parathyroids + thymus (they form a triangle; so 3).
        - 4th pouch = the two superior parathyroids.
        - Agenesis in DiGeorge syndrome. USMLE can also ask about, e.g., a missing parathyroid gland, or a parathyroid adenoma, and you need to know whether it's the 3rd or 4th. It's not hard, but you need to know these structures.
    - Pharyngeal arch #1 → things innervated by CN V (muscles of mastication).
    - Pharyngeal arch #2 → things innervated by CN VII (muscles of facial expression, stapedius, platysma).
    - Pharyngeal arch #3 → things innervated by CN IX (stylopharyngeus).

- Pharyngeal arch #4 → things innervated by superior laryngeal nerve branch of CN X (cricothyroid).
- Pharyngeal arch #6 → things innervated by recurrent laryngeal nerve branch of CN X (laryngeal muscles, but not cricothyroid). 5th arch has no major contributions.
- Back in the numerical Step 1 days, memorizing every structure had utility when our aim was to get a 280+. But now that the exam is Pass/Fail, the above is literally enough to get the vast majority of embryo. Remember, the point here is yieldness, not superfluousness.
- HY endoderm stuff regarding foregut, midgut, hindgut, I discuss in the Gastro PDF, but this stuff is so HY that I will again quickly mention HY points here:
  - Foregut → supplied by Celiac trunk (T12); spans esophagus to 1st part of duodenum.
  - Midgut → supplied by SMA (L1); spans from 2nd part of duodenum to distal 2/3 of transverse colon.
  - Hindgut → supplied by IMA (L3); spans from last third of transverse colon to the anal canal at the pectinate line.
  - L2 (between the SMA and IMA, clearly) → renal arteries and gonadal arteries (testicular arteries in males; ovarian arteries in females) come off here.
  - Weird factoid USMLE likes: "Which organ is supplied by an artery of the foregut but is not itself derived from the foregut" → answer = spleen; supplied by Celiac trunk but derived from midgut).

- "What do I need to know about terms like malformation vs deformation, etc.?"
  - Malformation = developmental defect as a result of genetics or teratogens.
    - Example is cleft lip (polygenic) or heart defect (polygenic or teratogen).
  - Deformation = developmental defect as a result of extrinsic/mechanical force.
    - Example is clubbed feet in Potter sequence (due to oligohydramnios).
  - Disruption = distortion of previously normal tissue (i.e., originally formed OK).
    - Example is amniotic band syndrome (fibrous bands in amniotic sack compress limbs of the fetus).
- "What are specific HY teratogens I need to know for USMLE?"

- Lithium → Ebstein anomaly ("atrialization of right ventricle" → the right ventricle is tiny and right atrium massive).
- Anti-epileptics → valproic acid, phenytoin, and carbamazepine are all known to cause neural tube defects, but especially valproic acid.
- Isotretinoin → high-dose vitamin A used for acne that can cause cleft lip/palate in neonate; USMLE cares less about "what" isotretinoin causes, and more just that you know β-hCG needs to be done in any female commencing it.
- Warfarin → bone anomalies or bleeding in fetus.
- Fluoroquinolones (e.g., ciprofloxacin) → cartilage abnormalities.
- ACE inhibitors → renal defects.
- Aminoglycosides → congenital hearing loss.
- Diethylstilbestrol (DES) → vaginal clear cell carcinoma.
- Alcohol → fetal alcohol syndrome; most common cause of mental retardation; philtrum changes are highest yield (i.e., long, smooth philtrum); hypertelorism; heart/lung defects.
- Cocaine and smoking → intrauterine growth restriction (IUGR) due to reduced blood flow.

- "What do I need to know about placental anatomy?"
  - Two components: fetal (trophoblast) and maternal (decidua).
    - Trophoblast = syncytiotrophoblast (deep layer) + cytotrophoblast (superficial layer).
    - Syncytiotrophoblast secretes hCG, which functions to maintain the corpus luteum (ruptured follicle). The corpus luteum secretes progesterone in order to maintain the endometrial lining / pregnancy. At 8-10 weeks, hCG peaks. This is because after this point, the placenta takes over production of progesterone, so we no longer need hCG to maintain the corpus luteum.
    - Fetal and maternal circulations do not mix and merely exchange gas and nutrients across placenta. Fetal hemoglobin (alpha-2 gamma-2) has stronger affinity for oxygen and can pull it off of the maternal hemoglobin (alpha-2 beta-2) despite membrane separation.
    - IgG from the mom can cross placenta; IgA is passed through breast milk.
- "What do I need to know about poly- vs oligohydramnios?"

  - We want to think of fetal versus maternal causes.
  - Polyhydramnios → maternal diabetes (insulin does not cross placenta; high glucose crosses placenta, leading to polyuria in fetus); fetal anencephaly or tracheoesophageal fistula.
  - Oligohydramnios → maternal smoking / cocaine use; maternal SLE (antiphospholipid syndrome); these all result in "uteroplacental insufficiency" as the HY cause of the oligohydramnios; fetal Potter sequence; fetal posterior urethral valves; these both cause decreased urination.

- "What do I need to know about twinning?"
  - Number of placentas = number of chorions; number of amnions = number of yolk sacs.
  - The yolk sac comes from hypoblast; the amnion comes from epiblast.
  - Splitting of embryo days 0-4 post-fertilization→ dichorionic-diamniotic; two placentas with two yolk sacs, showing thick, dividing membrane (chorion) on ultrasound.
  - Splitting at days 5-8 → monochorionic-diamniotic; fetuses share one placenta; there is absence of thick, dividing membrane on ultrasound, but two distinct amniotic sacs, and hence two yolk sacs.
  - Splitting at days 9-12 → monochorionic-monoamniotic; the fetuses share single placenta and yolk sac; risk of twin-twin transfusion syndrome is greatest.
  - Splitting after day 12 → conjoined twins.
- "What do I need to know about umbilical cord anatomy?"
  - Contains one umbilical vein (oxygenated), two umbilical arteries (deoxygenated), and the allantois (tube for fetal urine to go back to mom); these are surrounded internally within the cord by Wharton jelly.
  - The deoxygenated umbilical arteries are derived from the fetal internal iliac arteries (not veins).
- "What are the allantois, urachus, etc.?"
  - Allantois = tube that carries urine from fetal bladder back to placenta; it runs from the fetal bladder, through the umbilical cord, and all the way to the placenta.

  - Urachus = thicker, fibrous part of the allantois that runs from the fetal bladder to the umbilicus (fetal belly button); in other words, urachus just = the name of the part of the allantois still inside the fetal body.
  - Post-birth, the urachus closes and is known as the median umbilical ligament.
  - If the urachus remains patent or partially open, it can be known as a urachal diverticulum, or urachal cyst, or just patent urachus. The latter, for instance, could present as the neonate's urine coming out through the umbilicus.

- "What is the vitelline duct (omphalomesenteric duct)?"
  - Connects the fetal midgut lumen to the yolk sac.
  - What you need to know: failure to fully involute/obliterate causes Meckel diverticulum.
- "What do I need to know about cleft lip/palate?"
  - Answer on USMLE is "polygenic" or "multifactorial."
  - If couple has child with cleft lip/palate, chance of having another child with it is 3-4% (this statistic is similar for things like ASD/VSD and pyloric stenosis).
  - Cleft lip embryo = "failure of fusion of maxillary and medial nasal processes" on NBME.
  - Cleft palate embryo = "failure of fusion of lateral palatine shelves."
- "What do I need to know about Mullerian vs Wolffian duct stuff, etc.?"
  - Mullerian (paramesonephric) duct → becomes female internal structures → Fallopian tubes, uterus, upper portion of vagina.
  - Wolffian (mesonephric) duct → becomes male internal structures → SEED → Seminiferous tubules, Epididymis, Ejaculatory duct, Ductus deferens.
  - SRY gene on Y chromosome produces testis-determining factor → causes testes development → testes are composed 90% of seminiferous tubules (coiled tubes for sperm production) → Leydig cells produce testosterone (necessary for Wolffian development / internal male structures) → converted to DHT via 5α-reductase (necessary for prostate + external male structures – i.e., penis).
  - Sertoli cells produce Mullerian inhibitory factor (MIF) → shuts off development of female structures. Sertoli cells also produce androgen-binding protein (keeps local testosterone

concentration high for sperm production), inhibin B (induces negative feedback at hypothalamus for GnRH), and aromatase (converts androgens to estrogens).

- "What do I need to know about LH and FSH for basic repro physiology?"
    - LH stimulates the Leydig cells (in males) and theca interna cells (in females) to make androgens (androstenedione and testosterone).
    - FSH stimulates the Sertoli cells (in males) and granulosa cells (in females) to make aromatase.
    - The androgens from the Leydig cells / theca interna cells are then converted to estrogens via the aromatase produced by the Sertoli cells / granulosa cells.
    - Both androgens and inhibin B can shut off GnRH production at the hypothalamus, but androgens have a stronger effect shutting off LH; inhibin B has a stronger effect shutting off FSH.
    - Low estrogen production by the ovaries in Turner syndrome, premature ovarian failure, and menopause leads to high LH in the female due to lack of negative feedback; low inhibin B results in high FSH. **Specifically, high FSH is exceedingly HY as a marker of menopause.**
- "What do I need to know about hysterosalpingograms?"
    - USMLE loves hysterosalpingograms (dye injected into uterus via the cervix + visualization by x-ray). By far the highest yield point you need to know is that since the Fallopian tubes are normally open on both ends, **spillage of dye into the peritoneal cavity is normal.** Do not select answers such as "rupture of Fallopian tubes," etc. When the ovum is released from the ovary, it will be drawn into the Fallopian tube, which is open at its lateral end.
    - If USMLE shows you a hysterosalpingogram where dye **does not** spill into/enter the peritoneal cavity, this can be reflective of Hx of pelvic inflammatory disease, where there is scarring of the Fallopian tube.
    - If USMLE shows you image of a uterus with a septum running down the middle of it, this is called a bicornuate uterus → causes increased risk of premature delivery + miscarriage. The mechanism of this on NBME = "failure of paramesonephric ducts to fuse."
    - An actual complete double uterus is called uterine didelphys.
- "What do I need to know about male vs female embryologic equivalent structures?"

  - Genital tubercle becomes:
    - Glans penis (males) = glans clitoris (females).
    - Corpus cavernosum/spongiosum (males) = vestibular bulbs (females).
  - Urogenital sinus becomes:
    - Prostate gland (males) = urethral/paraurethral glands of Skene (females).
    - Bulbourethral (Cowper) glands (males) = greater vestibular (Bartholin) glands (females).
  - Urogenital folds become:
    - Ventral shaft (underside) of penis (males) = labia minora (females).
  - Labioscrotal swelling becomes:
    - Scrotum (males) = labia majora (females).

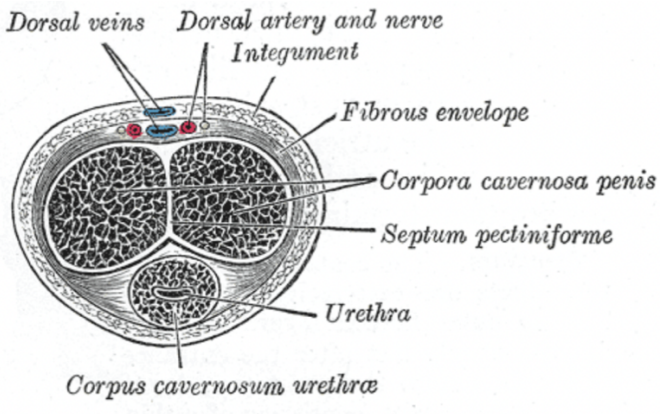
- "What do I need to know about abnormal urethral/penile development?"
  - Hypospadias = urethral meatus opens on the ventral shaft of penis (pointing downward).
  - Epispadias = urethral meatus opens on dorsal shaft (top) of penis (pointing upward).
  - Posterior urethral valves = most common genitourinary (GU) abnormality in fetal/neonatal males, where valves within the urethra that normally prevent backflow of urine are pointing the opposite direction, therefore preventing the excretion of urine. Severity can vary, where some cases result in oligohydramnios; other cases present as a newborn male who hasn't urinated (suprapublic mass = full bladder), or as infant male who has recurrent UTIs or pyelonephritis. Diagnosis is with ultrasound first, followed by voiding cystourethrogram. Don't confuse with retrograde urethrogram (used to Dx urethral injury) or retrograde cystourethrogram (used to Dx bladder injury). Treatment is often surgical.
  - Peyronie disease = fibrosis of tunica albuginea; results in abnormal curvature/bent penis; surgery not typically done for cosmetic purposes; reserved for functional impairment.
- "What are phimosis and paraphimosis?"
  - Phimosis = inability to retract the foreskin (prepus).
  - Paraphimosis = inability to reduce (put back) the foreskin. Paraphimosis is considered an emergency since ischemia can result.
- "What do I need to know about hydrocele / varicocele?"

  - Hydrocele = failure of closure of processus vaginalis → leads to fluid buildup within testis that transilluminates; treatment = observe under the age of 1.
  - Varicocele = congestion of the pampiniform plexus (venous plexus) draining the testes; can cause ↓ sperm production due to ↑ scrotal temperature; does not transilluminate; Tx is usually elective surgical repair.
  - Mechanism for varicocele is high-yield. It almost always occurs on the left because of the venous drainage. The left testicular vein enters left renal vein at 90 degrees. This creates pressure and congestion on the left side. The left renal vein will then go to the IVC. In contrast, the **right** testicular vein goes "**right** to the IVC," where there is no pressure effect.
  - There is Q on 2CK Peds CMS form where bilateral varicocele is the answer, where you have to eliminate to get there. In other words, just know that it is technically possible / is asked.
- "Do I need to know anything about lymphatic drainage or arterial supply?"
  - Lymphatic drainage of testes and ovaries is to para-aortic lymph nodes.
  - The scrotum is drained by the superficial inguinal nodes, not the para-aortic.
  - The testicular and ovarian arteries come directly off the abdominal aorta at L2.
  - The "gonadal arteries/veins" is a generic term that means testicular arteries/veins in males and ovarian arteries/veins in females.
- "What is cryptorchidism?"
  - Undescended testis. Tx = observe within the first 6 months of life; most will spontaneously descend; after 6 months, orchidopexy can be performed (surgery to move the testicle down into the scrotum). USMLE wants you to know that any Hx of cryptorchidism means the patient has an increased risk of testicular cancer (usually seminoma) in the future.
- "What do I need to know about epididymitis vs testicular torsion?"
  - Epididymitis will have intact cremasteric reflex; it is absent in torsion. This reflect is retraction of scrotal skin with direct palpation or palpation of medial thigh; this is mediated by the genitofemoral nerve.
  - Epididymitis has a positive Prehn sign; it is negative in torsion. This sign is relief of pain upon elevation of the testis.

  - Epididymitis is usually chlamydia or gonorrhea in younger males; males who are 40s and older, E. coli should be considered. This also applies to organisms causing prostatitis, where on NBME, E. coli is correct over chlamydia in a 45-year-old male.

- "What is torsion of appendix testis?"
  - This is **not** the same as torsion of testis. There is a structure called the appendix testis that can also torse. This is asked on one of the 2CK pediatrics forms, but you could be aware of it for Step 1 for the sake of it.
  - The question will tell you a kid has acutely painful testis, where the superior pole is blue; they will say cremasteric reflex is normal/intact; answer = torsion of appendix testis.
- "What about orchitis?"
  - Inflammation of testis; exceedingly rare on USMLE. In theory, classically caused by mumps.
- "Do I need to know about all of the ligaments relating to the uterus/ovaries, etc.?"
  - Unfortunately, USMLE cares. But I'll tell you exactly the HY points:
  - Suspensory ligament of ovary = connects lateral ovary to abdominal wall.
    - Contains ovarian vessels.
    - Ligation of ovarian vessels during oophorectomy risks injury to the ureter.
  - Ovarian ligament = connects medial ovary to uterus.
  - Round ligament of uterus = connects uterine horns to labia majora.
    - Uterine horns are the superolateral parts of the uterus that connect to the Fallopian tubes.
  - Cardinal ligament = connects the lateral cervix to pelvic wall.
    - Contains uterine vessels.
    - Ligation of uterine vessels during hysterectomy risks injury to ureter.
  - Broad ligament = large ligament that connects uterus, Fallopian tubes, and ovaries to pelvic wall.
    - Contains the ovaries, Fallopian tubes, and round ligaments of uterus.
  - Parametrium of uterus = fat and connective tissue surrounding the uterus.
    - 2CK Obgyn form mentions embryo developing within parametrium of the uterus; answer = ectopic pregnancy. So just be aware of this term.

- "Do I need to know vagina/perineal anatomy?"
    - The answer is not really. But there are a couple HY points you could be aware of.
    - If an episiotomy is performed posteriorly in the midline, if the obstetrician cuts too far, USMLE wants you to know that you cut into the external anal sphincter.
    - For Kegel (pelvic floor) exercises, the USMLE wants you to know that the internal anal and urethral sphincters are **not** strengthened. This might sound a bit unusual, as you could say, "Well there are tons of muscles not strengthened, e.g., the deltoids." But the point here is that internal sphincters are under sympathetic control (i.e., they're not voluntary/somatic), so clearly they can't be strengthened by a voluntary exercise. A key muscle that is strengthened would be levator ani.
- "Do I need to know penile anatomy?"
    - Unfortunately yes. You need to know the 2D-cross-section of the penis, where you have to identify the erectile muscle (i.e., they ask you where sildenafil would help, and you would just choose whichever muscular compartment is labeled with a letter).



- "What do I need to know about nerves/innervation about erection, etc.?"
    - Erection = parasympathetic = S2-4 ("S2, 3, 4 keeps the penis off the floor.") = pelvic splanchnic nerves.
    - Emission = sympathetic = hypogastric nerves = T11-L2.
    - Ejaculation = somatic (i.e., voluntary) = pudendal nerve.
- "Do I need to know anything about reproductive histology?"
    - USMLE wants you to know that the endometrium during the proliferative/follicular phase of the menstrual cycle will have straight/tubular glands.

- The secretory/luteal phase will have coiled glands.
  - Early-luteal phase will have coiled glands + no secretions within lumens.
  - Late-luteal phase will have coiled glands + secretions with lumens.
- Vagina and ectocervix are stratified squamous epithelium.
- Uterus and Fallopian tubes are simple columnar epithelium. Fallopian tubes are ciliated.
- The transformation zone of the cervix = squamocolumnar junction between the stratified squamous of the ectocervix and simple columnar of the endocervix.
- Ovaries are simple cuboidal.
- The Sertoli cells in males are more linear and form the blood-testes barrier.
- Leydig cells (aka interstitial cells) are more randomly distributed (i.e., the cells that are not linear).

- "What do I need to know about spermatogenesis, oogenesis, and gametes, etc.?"
  - Spermatogonia = stem cells that will differentiate into sperm.
  - Oogonia = stem cells that will differentiate into ova.
  - Process of spermatogenesis is low-yield. End-result is spermatids that are haploid (contain half the number of chromosomes of normal cell – i.e., 23 instead of 46).
  - Sperm require cilia for motility; motility is impaired in Kartagener syndrome (primary ciliary dyskinesia).
  - Sperm are absent in cystic fibrosis (CBAVD; congenital bilateral absence of vas deferens).
  - Ooogonia (stem cells) mature into primary oocytes that are locked in prophase I until ovulation.
  - At ovulation, the released ova are known as secondary oocytes and are locked in metaphase II until fertilization.
  - Once fertilization has occurred, they complete meiosis.
- "What do I need to know about hydatidiform moles?"
  - A mole is a nonviable conceptus due to abnormality with chromosomes/fertilization.
  - Complete mole = empty egg fertilized by two sperm, or when ovum is fertilized by a single sperm that then duplicates; all genetic material is paternal; chromosome number = 46; no fetal parts are presents; ultrasound shows a "snowstorm" or "bunches of grapes"

appearance; high risk of becoming choriocarcinoma (cancer of placental/syncytiotrophoblastic tissue).

- Incomplete/**parti**al mole = normal ovum fertilized by two sperm; chromosome number is 69; fetal **parts** are present; can lead to choriocarcinoma, but not as high-risk as complete mole.
- β-hCG will be abnormally high in both types of moles (i.e., hundreds of thousands).
- Women present large for gestational age – e.g., Q will say fundal height is measured at level of umbilicus when woman is only 16 weeks' gestation (this is normally level of fundus at 20 weeks).
- Can present similarly to preeclampsia (i.e., HTN + proteinuria), but **before** 20 weeks' gestation; after 20 weeks' preeclampsia is most likely diagnosis.
- Tx for both is dilation and curettage (D&C), or suction curettage.

- "What do I need to know about gynecologic tumors/cancers?"
  - It is in my view that resources vastly overemphasize certain details regarding this stuff. I'll tell you exactly what you need to know for the USMLE.
  - Choriocarcinoma = cancer of placental/trophoblastic tissue; Q will give very high β-hCG; likes to metastasize to the lungs (nodules on CXR) or brain (presents like stroke); appears grossly like "bloody mess."
  - Serous cystadenoma of ovary = benign; contains Fallopian tube-like epithelium.
  - Serous cystadenocarcinoma of ovary = malignant variant; has psammoma bodies (calcium rings).
    - Both serous cystadenoma and adenocarcinoma can be bilateral.
  - Mucinous cystadenoma of ovary = benign; contains mucous-like material; "loculated" (i.e., honeycomb-like).
  - Mucinous cystadenocarcinoma of ovary = malignant variant; can cause pseudomyxoma peritonei (inflammation of peritoneal cavity due to mucous production/spillage onto adjacent structures).
  - Dermoid cyst (aka mature cystic teratoma) = classically the "skin, hair, teeth tumor," since it is derived from all three germ layers; can calcify (an NBME Q mentions this as only finding); almost always benign (unless they specifically say "immature" histo).

- Dysgerminoma = tumor of ovary; can present with high LDH and pulling sensation in groin.
- Yolk sac tumor = pediatric ovarian tumor; secretes AFP.
- Leydig-Sertoli cell tumor (yes, in a female) = secretes androgens → virilization.
- Granulosa cell tumor = secretes estrogens → can cause endometrial hyperplasia.
- Struma ovarii = ovarian germ cell tumor that secretes thyroid hormone.
- Krukenberg tumors = bilateral gastric cancer metastases to ovaries; have signet ring cells on biopsy; these cells contain mucin.
- Brenner tumor = ovarian tumor containing bladder (transitional) epithelium.
- Endometrial adenocarcinoma = biggest risk factor is unopposed estrogen in setting of patient with prior anovulation (ovulation normally leads to corpus luteum that secretes progesterone, which limits overgrowth of endometrium); Hx of high BMI or abnormal periods can imply endometrial hyperplasia and risk of endometrial cancer; can present as midcycle bleeding in perimenopausal woman, or any bleeding in postmenopausal woman; must do endometrial biopsy.
- Uterine leiomyoma (aka fibroid) = most common tumor in women; benign; stains positive for muscle markers; can be described as white/whorled appearance grossly; highest yield point on USMLE is that these are almost always just simply observed – i.e., don't do myomectomy etc., even if the Q tells you many are present and she's going to get pregnant; if they bleed, patient can be given OCPs and/or NSAIDs initially.
- Leiomyosarcoma = malignant variant; only point you need to know is that this is not derived from leiomyoma; presumably this point is important because it justifies why we almost always just observe leiomyomas (i.e., they won't become malignant).
- Cervical cancer = squamous cell carcinoma; HY causes are HPV 16+18; Pap smear discussion, etc., is earlier in this document.
- Clear cell vaginal carcinoma = history of maternal use of diethylstilbestrol (DES).
- Sarcoma botryroides = rare rhabdomyosarcoma seen in pediatrics.
- Gynecologic cancers in general demonstrate increased risk in BRCA1/2 and HNPCC patients.

- "What is lichen sclerosus?"

  - Described as white/grey parchment-like, rough area of vulva in woman over 50; next best step is biopsy to rule out squamous cell carcinoma; if histo confirms lichen sclerosus, Tx is topical steroids; if SCC, must excise.

- "What is Bartholin gland cyst/abscess?"
  - Presents as tender/painful bump at the 4 or 8-o'clock position on the labia majora; can treat with warm compresses or Sitz bath; if lesion is warm, erythematous, and tender, can be drained immediately; USMLE wants you to know this is polymicrobial.
- "What do I need to know about testicular cancers?
  - Seminoma = most common; ages 15-35 classically; can present as hard nodule or mass that does not transilluminate; can be discovered incidentally after trauma (in an NBME question); increased risk in cryptorchidism or Klinefelter; histo can show large, clear cells; highly radiosensitive (i.e., responds well to radiotherapy, even If it's metastasized); can produce placental alkaline phosphatase (placental ALP) as tumor marker, but not mandatory.
  - Yolk sac tumor = pediatric testicular tumor; secretes AFP.
  - Choriocarcinoma = secretes β-hCG; same as described prior.
  - Leydig-Sertoli cell tumor = can present with gynecomastia in males – i.e., the androgens can be aromatized into estrogens.
- "What do I need to know about breast cancers?"
  - Fibroadenoma = benign; most common; rubbery, mobile, painless mass in woman 40s or younger generally; do FNA to diagnose; if diagnosed, surgically remove, even though benign.
  - Ductal carcinoma in situ (DCIS) = has malignant potential, but hasn't yet crossed basement membranes; presents with clusters of microcalcification; management step is "needle-guided open biopsy" (on NBME); FNA is wrong answer for that same question; Paget disease of breast often presents with underlying DCIS (i.e., eczematoid nipple in woman over 50 with underlying nipple mass = underlying DCIS).
  - Intraductal papilloma = unilateral bloody nipple discharge; don't confuse with DCIS.
  - Invasive ductal = same as DCIS but has already crossed basement membranes; can be described as having "stellate morphology" on histo.

  - Lobular carcinoma in situ = malignant, but hasn't crossed basement membranes; can be described as linear, or "Indian rows" of cells.
  - Invasive lobular carcinoma = same as LCIS, but has crossed basement membranes. Both lobular carcinoma types can occur bilaterally.
  - Cystosarcoma phyllodes = "leaf-like" tumor that is fast-growing.
  - Inflammatory carcinoma = cancer of breast that can appear red/inflamed and with pain; classically associated with peau d'orange, or mottling of skin due to tethering of edematous skin by Cooper ligaments of breast
  - Comedocarcinoma = "cheese-like" breast cancer.

- "What do I need to know about fibrocystic change?
  - Benign; can be unilateral or bilateral; classic textbook description is "bilateral breast pain/tenderness that waxes/wanes with menstrual cycle," but Obgyn forms can have it presenting as unilateral pain, or as a unilateral painless cyst that drains dark fluid; no treatment is necessary most of the time; if patient has a singularly enlarged cyst that appears after starting hormone-replacement therapy, FNA is the answer.
- "What do I need to know about menstrual cycle / pregnancy hormones?"
  - First half of menstrual cycle = proliferative/follicular phase; second half = luteal/secretory phase. The luteal phase is always 14 days; if menstrual cycle changes length, it's because of variation in the follicular phase.
  - Estrogen gradually increases throughout the follicular phase and is highest just prior to ovulation, then it declines after. The high estrogen causes an LH spike that triggers ovulation.
  - The corpus luteum is the follicular remnant and produces progesterone that maintains the endometrial lining during the luteal phase.
  - If pregnancy occurs, β-hCG will maintain the corpus luteum, which will enable continued progesterone production so the pregnancy can be maintained. If pregnancy does not occur and β-hCG is not present, the corpus luteum degrades, progesterone production ceases, and sloughing of the endometrium occurs (menstruation).

  - As discussed earlier, β-hCG peaks at 8-10 weeks of pregnancy. After this point, the placenta takes over production of progesterone, so we no longer need hCG to maintain the corpus luteum.
  - Human placental lactogen (hPL) is a hormone that increases during third trimester of pregnancy and causes insulin resistance in the mother. This ensures that glucose levels are high enough so that brain development in the fetus occurs properly. The tradeoff is that this is the hormone that increases risk of gestational diabetes.
  - Prolactin produced by the anterior pituitary stimulates milk production.
  - Oxytocin produced by the supraoptic nucleus of the hypothalamus (and stored in the posterior pituitary) causes milk letdown (release). It also stimulates uterine contractions.

- "Do I need to know anything about Tanner stages?"
  - Tanner stages 1-5 are a system for genital/breast development. You don't need to know the strict Tanner stages, just a few key factoids.
  - For whatever reason, it's exceedingly HY on 2CK Obgyn forms that you know once a female hits Tanner stage 3, menarche is imminent (meaning, will occur very soon); they ask this directly in one Q; they also incorporate it into other Qs. For instance, they'll say a 14-year-old girl who's never had a menstrual period is brought in by her mom + she is Tanner stage 3 + they ask for next step in management → answer = follow-up in 6 months (since she's Tanner stage 3, we know menarche is imminent, so we'll just wait it out).
  - Low Tanner stage (i.e., 1 or 2) can be the USMLE's way of telling you a boy or girl has constitutional short stature (i.e., will achieve normal height, but has growth curve that is delayed / shifted to the right). For instance, they can say a boy is shortest in his class freshman year of high school + is Tanner stage 1 → answer = constitutional short stature. This diagnosis is also made where bone age is less than chronologic age. If bone age = chronologic age, then the short stature is genuine.
  - Turner syndrome classically has Tanner stage 1-2 breasts (i.e., "shield chest"), but it is not mandatory the Tanner stage is low on Obgyn material.

- "What do I need to know about precocious puberty?"

  - Emergence of secondary sex characteristics in females age 8 or younger, or males 9 or younger.
  - Childhood height will be increased; adult height will be decreased.
  - Question might ask how we know if the cause of the precocious puberty is due to the hypothalamus/pituitary axis or not → answer = if LH is high (if LH low, we know cause is primary testicular production of testosterone).
  - If DHEA-S is abnormally high, we know the adrenal gland is the cause (the zona reticularis of the adrenal cortex secretes DHEA-S and androstenedione).
- "What do I need to know about imperforate hymen?"
  - Will present as bluish bulge behind hymen in female who's never had a menstrual period; they can describe Hx of cyclical pain (due to menses with blood backup behind the hymen).
  - Hematocolpos = blood backed up in the vaginal canal behind the hymen.
  - Hematometra = blood backed up all the way to the uterine cavity, precipitating and vagal response and low blood pressure.
- "What is placenta previa?"
  - Placental implantation site over the internal cervical os.
  - Causes third-trimester **painless** bleeding.
  - Can spontaneously move off the os prior the 36 weeks' gestation; after this point, C-section must be done, otherwise patient may experience hemorrhagic shock during parturition.
  - USMLE wants you to know that prior C-section is a risk factor for placenta previa (i.e., if the endometrial lining has been disturbed in the past in any way, then that simply increases the risk of an abnormal implantation).
- "What is abruptio placentae?"
  - Abruption (separation) of the placenta *in utero*.
  - Causes third-trimester **painful** bleeding, or painful cramping.
  - Deceleration injury (i.e., car accident, fall) and cocaine use are known risk factors.
- "What is placenta accrete/increta/percreta?"
  - Presents as **postpartum** bleeding/hemorrhage.
  - Accreta = placenta attaches to surface of myometrium. A = attaches.

  - Increta = placenta inserts into/inside myometrium. I = inserts inside.
  - Percreta = placenta perforates through myometrium and attaches onto external structures, such as the bladder. P = perforates.

- "What is most common cause of postpartum bleeding?"
  - Uterine atony → presents as boggy uterus postpartum; Tx with uterine massage, followed by intra-myometrial oxytocin injection, followed by ergotamine injection (avoid the latter if HTN or migraine Hx).
  - Less common causes are retained placental parts (if they tell you all lobes of placenta are not present), vaginal lacerations (e.g., from macrosomia in maternal diabetes, where the fetus experienced shoulder dystocia), or thrombotic disorders (e.g., von Willebrand).
- "What is vasa previa?"
  - When the fetal vessels overly the internal cervical os. Normally, the vessels are protected by Wharton jelly within the umbilical cord, but sometimes the vessels can be abnormally exposed in the setting of velamentous cord insertion, or if there is the presence of a succenturiate lobe of the placenta.
  - Presents as triad of 1) rupture of membranes; 2) vaginal bleeding; 3) fetal bradycardia.
- "What do I need to know about ectopic pregnancy?"
  - Can present as LLQ or LRQ pain in female who has a missed menstrual period.
  - β-hCG will be positive, but the numerical value will be described as a lot lower than expected. This is in contrast to hydatidiform mole or choriocarcinoma, where the β-hCG is much higher than expected.
  - Methotrexate can be given for small, stable ectopics. Otherwise, laparoscopic salpingostomy is performed. If the patient is unstable (i.e., low BP in ruptured ectopic), laparotomy is the answer.
- "What is preclampsia, eclampsia, and HELLP syndrome?"
  - Preeclampsia = HTN and proteinuria after 20 weeks' gestation. That is the most simplified definition and sufficient for USMLE.
  - Eclampsia = preeclampsia + seizure.
    - Tx = magnesium as Tx (and prophylaxis in severe preeclampsia).

    - HELLP syndrome = Hemolysis, Elevated Liver enzymes, Low Platelet count.
        - Severe manifestation of preeclampsia. **Schistocytes are seen on a blood smear.**

- "What is supine hypotensive syndrome?"
    - Low blood pressure in woman >20 weeks' gestation due to compression of IVC.
- "What do I need to know about endometriosis?"
    - Endometrial tissue growing outside the uterus, usually on the ovary; can cause severely painful periods; descriptors such as pain with defecation or dyspareunia are often too buzzy and omitted from questions. Physical examination will be abnormal (e.g., nodularity of uterosacral ligaments); patient can get hemorrhagic ("chocolate") cysts; diagnosis is done via exploratory laparoscopy. Treatment is with OCPs +/- NSAIDs. Definitive is laparoscopic removal of lesions.
- "What do I need to know about primary dysmenorrhea?"
    - This is "normal period pain" due to prostaglandin secretion; physical examination is normal, in contrast to endometriosis. Treatment is with NSAIDs. Don't confuse with mittelschmerz, which is ovulatory pain mid-cycle.
- "What is adenomyosis?"
    - Diffusely enlarged uterus in woman generally 30s-40s, often with vaginal bleeding. They can say a woman had a tubule ligation 2 years ago, but now has vaginal bleeding with a uterus that is 8 weeks' gestation in size. Treatment is with OCPs + NSAIDs.
- "What do I need to know about BPH?"
    - Prostatic enlargement occurs as a result of DHT stimulation.
    - USMLE loves post-renal obstruction due to BPH causing "increased tubular hydrostatic pressure," or "increased Bowman capsule hydrostatic pressure" in the setting of high creatinine.
- "What do I need to know about HY Repro pharm?"
    - Tamoxifen + raloxifene are selective estrogen receptor modulators (SERMs). They can be used in ER(+) breast cancer. They are antagonists at breast + agonists at bone. Highest yield point is that tamoxifen is partial agonist at endometrium → increased risk of endometrial cancer. Never give tamoxifen to woman who has a uterus. Give raloxifene instead.

- Anastrozole + exemestane are aromatase inhibitors. These can be used in breast cancer.
- Trastuzumab (Herceptin) targets HER2/neu(+) breast cancer. It is cardiotoxic.
- Clomiphene is partial agonist at the hypothalamus (the effect is as though it's an antagonist). This stimulates GnRH secretion → promotes ovulation. It is used in women who have abnormal periods, where ovulation is difficult to predict. Or it can be used in IVF.
- Mifepristone is a progesterone receptor antagonist used as an abortifacient.
- Misoprostol is a prostaglandin E1 analogue used with mifepristone as an abortifacient.
- Ulipristal is a progesterone receptor modulator used as emergency contraception.
- Dinoprostone is a prostaglandin E2 analogue used to soften the cervix.
- Ritodrine is a beta-2 agonist used as a tocolytic.
- Danazol is an androgen receptor partial agonist used as a tertiary agent in endometriosis. However it is one of the first-line agents for hereditary angioedema (causes liver to produce more C1 esterase inhibitor).
- Combined estrogen oral contraceptive pills → contraindicated in women who are smokers over 35, have migraine with aura, active breast cancer, or Hx of thrombotic disorders / DVT.
- Finasteride is a 5α-reductase inhibitor that prevents the conversion of testosterone to DHT in the treatment of BPH.
- Tamsulosin and terasozin are α1-antagonists used in the treatment of BPH. They relieve constriction of the internal urethral sphincter.
- Leuprolide is a GnRH receptor agonist that, when given continuously, causes desensitization of the GnRH receptor, thereby effectively acting as an antagonist. This causes a reduction in LH and FSH. It is used for prostate cancer. It can also be used for adenomyosis and fibroids, albeit not first line.
- Flutamide is an androgen receptor antagonist used in the treatment of prostate cancer. This is given prior to leuprolide, since the latter will cause a transient increase in LH and FSH prior to desensitization of the GnRH receptor. The transient increase in LH can theoretically cause a transient increase in testosterone, so flutamide must be given first to block receptors.
- Oxybutynin is a muscarinic receptor antagonist used in the treatment of urge incontinence.

  - Bethanechol is a muscarinic receptor agonist used in the treatment of overflow incontinence due to diabetes (neurogenic / hypotonic bladder). If the cause of the overflow incontinence is BPH, however, the BPH itself must be treated first as per above.

- 32F + not breastfeeding + upper-outer quadrant warm, tender, red non-fluctuant mass +/- fever; Dx? → answer on Obgyn NBME = **mastitis, not breast abscess;** the key here is **non-**fluctuant mass; abscess is identical presentation but fluctuant. For mastitis, the easier, Step 1 presentation is the standard red, cracked, fissured nipple in a breastfeeding woman → *S. aureus* → Tx = continue breastfeeding through the affected breast; can give **oral dicloxacillin (answer on newer Obgyn form)** or cephalexin for mastitis; for abscess, answer = always drain before Abx.
- 32F + recently stopped breastfeeding + temp 99.5F + tender, fluctuant mass in lateral breast + not warm + not erythematous; Dx? → answer on Obgyn NBME = galactocele (milk retention cyst); classically subareolar or in lateral breast; Tx on Obgyn NBME is warm compresses ("application of heat to the area").
- 31F + gave birth two days ago + exclusively bottle-feeding neonate + breasts are engorged and tender + fever of 101F + Sx of dysuria + suprapubic tenderness + urinalysis normal; Dx? → answer on Obgyn NBME = breast engorgement → every student gets this wrong because it sounds like obvious infection; learning point is: can present with fever; occasional Sx of dysuria + normal U/A are not atypical in women.
- 24F + amenorrhea since D&C 13 months ago for postpartum hemorrhage + progestin withdrawal test shows no withdrawal bleeding; Dx? → answer = Asherman syndrome or "uterine synechiae" on Obgyn shelf.
- 27F + spontaneous abortion at 10 weeks' gestation complicated by postpartum endometritis + sharp D/C to remove infected material; patient is subsequently at increased risk for what? = answer = amenorrhea (Asherman syndrome).
- What does progestin withdrawal test mean? → if progestin is given then withdrawn, bleeding should occur (hormonal stabilization of lining followed by allowing it to slough, akin to forcing a menstruation); if bleeding occurs, **estrogen is not deficient** and the Dx is anovulation (PCOS is just anovulation leading to 11+ cysts bilaterally + hirsutism; anovulation as independent term is same mechanism as full-blown PCOS) → if anovulation occurs, there's no corpus luteum and therefore no

progesterone released → cannot establish endogenous rise + fall of progestin, therefore no sloughing/menstruation; in contrast, if bleeding does not occur with progestin withdrawal test, either estrogen is deficient (primary ovarian failure or hypogonadotropic disorder) or the uterus is scarred (Asherman).

- 18F + no bleeding after progestin withdrawal test; Q asks, if not Tx over ten years, what is patient at risk for? → answer = osteoporosis (progestin withdrawal result means low estrogen).
- Question shows you a graph where basal body temperature increases ~0.5F mid-cycle and stays at this higher temp; why? → answer = progesterone (ovulation).
- 45F + she asks about best way to decrease risk of osteoporosis; answer = weight-bearing exercise, not calcium + vitamin D.
- 72F + already has osteoporosis + Q asks best way to most greatly decrease fracture risk; answer = going on long walks; wrong answer is swimming / pool exercises (weight-bearing component makes sense, but actually tricky considering elderly have high falls risk).
- 69F + Caucasian + nulliparous + on beta-blocker + drinks daily + compression fracture of vertebra; what is strongest predisposing risk factor (family Hx not discussed or listed)? → answer = race; white race confers higher risk of osteoporosis; wrong answers are alcohol use, beta-blocker, nulliparity, HTN.
- 42F + 8-month Hx of severe pelvic pain and heavy bleeding during menses + regular periods + two kids + does not want more kids + husband to get vasectomy soon + no other abnormalities; next best step? → answer = endometrial ablation.
- 11F + Tanner stage 3 breast and pubic hair; these findings are most predictive of what? → answer = “menarche is imminent.” USMLE wants you to know that menarche is imminent once girl is Tanner stage 3. Normal sequence is adrenarche → thelarche → pubarche → menarche.
- 13F + Tanner stage 2 + never had menstruation + brought in by mom concerned about lack of menstruation; answer = follow-up in 6 months (Tanner stage 2 so menarche is not yet imminent).
- 14F + 4x6cm mass in left breast + slightly tender + vitals normal + aunt died of breast cancer; next best step? → follow-up in 6 months → virginal breast hypertrophy is normal response to increased estrogens in adolescence (also seen in males; asked on peds and FM shelves).

- 23F + 10 weeks' gestation + nausea and vomiting for 4 weeks + lost 1.8kg; what is the most likely adverse effect on the fetus? → answer = "no significant adverse effect."
- How to Dx hyperemesis gravidarum (HG)? → answer = urinary ketones.
- When does HG present + what's the mechanism? → 8-10 weeks' gestation; an effect of beta-hCG (levels are highest at 8-10 weeks).
- Biochemical disturbance in HG? → hypokalemic, hypochloremic, metabolic alkalosis (low K, low Cl, high bicarb); yes, they ask this on Obgyn shelf.
- Tx for HG? → answer = admit to hospital and give parenteral anti-emetic therapy.
- Important drug causing hyperprolactinemia apart from antipsychotics? → metoclopramide → D2 antagonist.
- Amenorrhea in patient with anorexia; why? → decreased GnRH pulsation (hypogonadotropic) → decreased LH + FSH; Q wants "↓ FHS, ↓ estrogen" as the answer; in contrast, premature ovarian failure, Turner syndrome, and menopause have "↑ FHS, ↓ estrogen" as the answer.
- 28F + tight-fitting sports bra and/or breast trauma; Dx? → fat necrosis (can calcify).
- 36F + rubbery, mobile, painless mass in breast; Dx? → fibroadenoma → first Dx with USS only if age <30; do USS +/- mammogram if age >30; do FNA next; if confirmed, Tx = surgical excision; should be noted that guidelines vary (i.e., observe for change, etc.), but excision is definitive. Obgyn shelf will only ask you for Dx based on presentation.
- Mammogram guidelines? → start age 50 + every two years until age 75.
- 44F + painless unilateral cyst in breast that drains brown serous fluid; Dx? → answer on Surg form 6 = fibrocystic change; everyone says wtf because, yes, classic presentation is bilateral breast tenderness in woman 20s-40s that waxes and wanes with menstrual cycle; Tx is supportive (Evening Primrose oil / warm bath); histological descriptors can be: sclerosing adenosis; blue dome cysts; apocrine metaplasia.
- 25F + sharp pain in outer quadrant of right breast + exam shows 2cm tender area in right breast but no mass found; Dx? → answer = fibrocystic change.
- 47F + breast lump self-palpated + breast USS shows 3cm complex cyst + FNA performed of the cyst revealing straw-colored fluid + mass still present after aspiration; next best step? → answer = biopsy of the mass.

- 45F + unilateral rusty nipple discharge; Dx? → intraductal papilloma until proven otherwise.
- 45F + unilateral rusty nipple discharge + biopsy shows stellate morphology; Dx? → answer = invasive ductal carcinoma, not intraductal papilloma.
- 45F + mammography shows cluster of microcalcifications in upper-outer quadrant; next best step? → answer = needle-guided open biopsy (FNA wrong answer) → microcalcifications are ductal carcinoma in situ (DCIS) until proven otherwise.
- 45F + inverted nipple + greenish discharge; Dx? → mammary ductal ectasia (widening of lactiferous duct).
- 42F + recurrent miscarriage + SLE; Dx? → antiphospholipid syndrome (lupus anticoagulant) → Obgyn shelf will ask for "uteroplacental insufficiency" as the answer → Tx with aspirin or heparin; warfarin is contraindicated in pregnancy (bone abnormalities + bleeding in fetus).
- 45F + SLE + commencing third course of corticosteroids during past 18 months; Q asks what else she should be given; answer = "alendronate now" → give bisphosphonate to patients commencing steroids indefinitely, or to patients receiving steroids frequently.
- Intrauterine growth restriction (IUGR) of the fetus; which lifestyle factor most contributory; answer = smoking, not alcohol → causes decreased placental blood flow → answer = "Doppler ultrasonography of the umbilical artery."
- Which fetal parameter most reflective of IUGR? → abdominal circumference; sounds wrong, as you'd expect perhaps femur length, or biparietal diameter, etc., but answer is abdominal circumference.
- 23F + 33 weeks' gestation + FVL mutation + intrauterine female demise; Q asks which vessel the thrombosis most likely occurred in; answer = uteroplacental artery.
- Female at 24 weeks' gestation + HTN + proteinuria; most likely cause for her findings? → answer = "uteroplacental insufficiency" or "placental dysfunction"; this is the cause of preeclampsia.
- Female at 16 weeks' gestation + HTN + proteinuria + fundal height measured at the umbilicus; Dx? → answer = hydatidiform mole, not preeclampsia; preeclampsia will occur after 20 weeks' gestation; molar pregnancy presents large for gestational age → fundal height at umbilicus is normally reflective of 20 weeks' gestation.
- Uteroplacental insufficiency can cause what issue on the fetal heart tracing? → answer = late decelerations (fetal hypoxia).

- What do early, variable, and late decels mean? → early = fetal head compression; variable = cord compression; late = fetal hypoxia.
- Fetus has HR at 120bpm (NR 110-160), however there's zero variability; Dx? → answer on Obgyn NBME = fetal sleep state.
- Fetus has HR at 180bpm, however there's zero variability; Dx? → answer on Obgyn NBME = maternal fever.
- What are accelerations? → fetal well-being → rise of ~20bpm lasting ~20 seconds; 2-3 occurences every 20 minutes.
- What is a biophysical profile? → assesses fetal wellbeing; often done when non-stress test (checking for accelerations) is non-reactive; five components of biophysical profile (you do *not* need to have these memorized for the USMLE; more just be aware that if the vignette mentions qualitative non-reassurance of any aspect of the biophysical profile, then there is possibly fetal/maternal pathology):
    - **Non-stress test** shows at least two accelerations in 20 minutes.
    - **Rhythmic breathing** episode of >30 seconds in 20 minutes.
    - **Fetal movements** (at least 2 or 3 of the limbs).
    - **Fetal muscle tone** (at least one episode of flexion/extension of the trunk + limbs together).
    - **Amniotic fluid volume** (at least 2cm in vertical axis, or fluid index >5cm).
- 21F + 41 weeks' gestation + 4cm dilated + variable decels; next best step? → answer on Obygn NBME = amnioinfusion (wrong answers were external cephalic version, forceps delivery, amniocentesis, cordocentesis) → can't attempt delivery if not 10cm dilated + forceps not tried first anyway because it can cause nerve damage or sternocleidomastoid trauma (vacuum extraction / suction cup delivery first).
- What is external cephalic version? → transabdominal manipulation of a breech fetus into cephalic engagement; only performed after 36 weeks, as the fetus can spontaneously engage cephalically prior.
- What is internal podalic version? → reorienting fetus within the womb during a breech delivery; may be attempted for transverse and oblique lies when C-section not performed; also used for delivery of second twins. I've never seen this as correct answer on NBME assessment; it just shows up a lot as an incorrect answer choice, so I'm mentioning it here because students always ask, "what's that?"

- 2-day-old neonate + purplish fluctuant mound on scalp + crosses suture lines; Dx? → caput succedaneum
- Difference between caput succedaneum and cephalohematoma?
    - Caput succedaneum is poorly defined soft tissue edema on the scalp; caused by pressure of fetal scalp against cervix during parturition, leading to transient decreased blood flow and reactive edema; crosses suture lines; can be purplish in color similar to cephalohematoma (i.e., don't use color to distinguish); complications rare; disappears in hours to few days.
    - Cephalohematoma is well-defined, localized, fluctuant swelling; caused by subperiosteal hemorrhage; does not cross suture lines; may be associated with underlying skull fracture, clotting disorders, jaundice; disappears in weeks to months.
- 32F + G1P0 + third trimester + itchy hives-like eruptions within abdominal striae; Dx + Tx? → answer = pruritic urticarial papules and plaques of pregnancy (PUPPP); occurs in ~1/200 pregnancies (usually primigravid); cause is unknown, presents as pruritic hives-like eruption within striae; Tx is with topical emollients; for severe cases, topical steroids can be given; resolves spontaneously within a week of delivery.
- 25F + G1P0 + third trimester + itchy palms + soles; Dx + Tx? → answer = intrahepatic cholestasis of pregnancy (ICP); usually occurs third trimester; pruritis, particularly of palms + soles; diagnosis is achieved by ordering serum bile acids (elevated); Tx = ursodeoxycholic acid (ursodiol); important to note that **ICP is associated with increased risk of third-trimester spontaneous abortion – i.e., it is not benign;** delivery at 35-37 weeks may be considered; if bile acid levels normal, new literature suggests waiting until 39 weeks is acceptable.
- 32F + 30 weeks' gestation + 10-day Hx of nausea and generalized itching + bilirubin 2.1 mg/dL + ALT/AST/ALP all normal; Dx? → Obgyn shelf answer = intrahepatic cholestasis of pregnancy; no mention of palms + soles itching in vignette.
- 36F + G1P0 + 36 weeks' gestation + nausea/vomiting + jaundice + high bilirubin + high ALT and AST + no mention of pruritis of palms/soles; Dx? → answer = acute fatty liver of pregnancy; caused by deficiency of long-chain 3-hydroxyacyl-CoA dehydrogenase (sounds absurdly pedantic but asked on Obgyn shelf); often fatal; Tx is IV hydration + hepatology/high-risk obgyn consults + delivery.

- 29F + G1P0 + 2nd or 3rd trimester + intensily itchy eruption around umbilicus that spreads outward; Dx + Tx? → answer = herpes gestationis (gestational pemphigoid); **not** HSV, but instead an idiopathic autoimmune phenomenon; Tx = topical steroids.
- 13F + never had menstrual period + morning nausea/vomiting + suprapubic fullness; next best step? → answer = beta-hCG → can get pregnant before first menstruation; Q also on peds NBME.
- Tx for preeclampsia? → HTN Mx (labetalol, methyldopa, etc.); definitive Tx is delivery.
- Tx for eclampsia? → Mg for seizures; definitive is delivery.
- Tx for HTN emergencies in pregnancy? → just know hydralazine can be used for this purpose.
- Female at 8 weeks' gestation + cysts visualized bilaterally on pelvic USS; Dx? → theca-lutein cysts → benign finding in pregnancy + will almost always naturally regress → increased occurrence in high beta-hCG states like multiple gestation pregnancy, moles, choriocarcinoma.
- Complete vs partial mole? → complete mole = karyotype of 46; empty egg fertilized by a sperm that duplicates; bunches of grapes / snowstorm appearance on USS; chance of progression to choriocarcinoma higher than partial; partial mole = karyotype of 69; fetal parts visible on USS; lesser chance of progression to choriocarcinoma.
- Anovulation + hirsutism + BMI 27; Dx? → PCOS.
- Anovulation; mechanism USMLE wants? → **insulin resistance** → causes abnormal GnRH pulsation → high LH/FSH → LH high enough to precipitate ovulation but follicle not yet adequately primed → no ovulation (anovulation) → follicle retained as cyst.
- Why hirsutism in anovulation → higher relative LH → more androgen production by theca interna cells.
- What's LH do? → Stimulates theca interna cells (females) and Leydig cells (males) to make androgens.
- What's FSH do? → Stimulates granulosa cells (females) and Sertoli cells (males) to make aromatase; also primes follicles.
- Best Tx for PCOS? → if high BMI, weight loss first always on USMLE; if they ask for meds and/or weight loss already tried? → OCPs (if not wanting pregnancy); clomiphene (if wanting pregnancy; estrogen receptor partial agonist → leads to increased GnRH outflow).
- PCOS increases risk of what → endometrial cancer (unopposed estrogen); insulin resistance also greater risk of T2DM.

- 32F + unable to conceive for 3 years + BMI 30 + acanthosis nigricans; Dx? → answer = T2DM (PCOS or anovulation not listed as answers; wrong answer is "hypercortisolism") → Q doesn't mention any characteristic features such as purple striae, muscle wasting, or central obesity.
- 40F + vasomotor Sx; which hormone to confirm Dx? → answer = high FSH for premature ovarian failure.
- 28F + Hashimoto thyroiditis + hot flashes for 6 months + high FSH; Dx? → answer = "autoimmune ovarian failure"; this is a cause of premature ovarian failure (autoimmune diseases go together).
- Thyroid and pregnancy? → TSH normal, T3 normal, **free T4 normal, total T4 elevated** → due to increased thyroid-binding globulin due to higher estrogen.
- What do we order to evaluate thyroid function in pregnancy? → always choose **free T4** if you are asked. TSH is for screening in non-pregnant persons. Free T4 can be an answer in non-pregnant persons if they ask for most definitive marker for thyroid function.
- Levothyroxine dose in pregnancy for those with Hashimoto? → may need to be increased up to 50%.
- Thionamides in pregnancy? → methimazole is teratogenic in first trimester (causes aplasia cutis congenita); give PTU in first trimester; 2nd trimester onward switch to methimazole (PTU significantly hepatotoxic + methimazole only teratogenic early in pregnancy).
- 27F + 34 weeks' gestation + thyroid storm; Tx? → Obgyn NBME answer = PTU.
- 27F + gave birth to healthy boy 6 months ago following uncomplicated labor + no weight change or mood disturbance + on no meds + vitals WNL + dry skin + thyroid gland enlarged and non-tender + TSH high + T4 low; most likely explanation for these findings? → answer = "thyroiditis" → Dx = **postpartum thyroiditis** (a type of **silent thyroiditis**) → characterized by thyrotoxicosis followed by hypothyroidism (1/3 of women experience both phases; 1/3 experience just hyperthyroid phase; 1/3 only hypothyroid phase); affects 5-10% of women postpartum; hyperthyroid phase usually occurs 1-4 months postpartum; hypothyroid phase occurs about 4-8 months postpartum; thought to be caused by postpartum immunologic rebound (immune system normally suppressed during pregnancy); Dx w/ Hx + ordering serum TSH; increased risk of progression to Hashimoto; Tx w/ short course of propranolol if hyperthyroid; give short course of levothyroxine if hypothyroid.
- Neonate born with cretinism; what could have prevented this? → answer = "routine newborn screening"; yes, on obgyn shelf.

- 16F + anterior vaginal wall pain and dysuria for 6 months + U/A normal + vitals normal; Dx? → chronic interstitial cystitis → Tx is supportive; do **not** choose steroids.
- Important factoids about acute appendicitis in pregnancy? → can be upper right quadrant; if appendicitis, yes, perform laparascopic appendectomy.
- Beta-hCG in mole vs ectopic? → super-high in mole; low in ectopic (and slow rate of increase).
- 32F + presentation similar to stroke + beta-hCG hundreds of thousands; Dx? → choriocarcinoma (brain mets); chorio loves to metastasize to lungs.
- 24F + pregnancy visualized in the corneum of the uterus; Dx? → answer = ectopic pregnancy.
- 27F + pregnancy visualized in the parametrium of the uterus; Dx? → answer = ectopic pregnancy.
- Most common location for ectopic? → ampulla of fallopian tubes.
- Most common etiology for ectopic? → Hx of PID → scarring of fallopian tubes.
- Tx for ectopic pregnancy? → laparoscopic removal (salpingostomy / salpingectomy).
- When to give methotrexate to Tx ectopic? → all must be fulfilled: beta-hCG <6,000; < 3 cm in size; fetal HR not detectable; no evidence of fluid leakage in the cul de sac; mom stable vitals.
- Organisms causing PID + Tx? → chlamydia and/or gonorrhea; Tx = IM ceftriaxone, PLUS either oral azithromycin or oral doxycycline. If patient is septic (2+ SIRS), answer = admit to hospital and give IV antibiotic therapy (they make this distinction on Obgyn shelf).
- PID + fever does not improve after several days on Abx; next best step? → adnexal USS to look for tubo-ovarian abscess → must drain if present.
- Difference between inevitable and threatened abortions? → inevitable = bleeding + open cervix; threatened = bleeding + closed cervix; Tx for inevitable = vacuum aspiration; Tx for threatened = bed rest.
- 32F + 9 weeks' gestation + bleeding and passage of clots per vaginum + intrauterine pregnancy seen on USS; Dx? → answer = incomplete abortion (passage of clots means it's already underway).
- Difference between complete and missed abortions? → Complete = no products of conception seen on USS (abortion is literally over/complete); missed = fetal demise without passage of products of conception.
- 35F + vaginal bleeding at 6 weeks' gestation and beta-hCG 450 mIU/mL + USS shows thickened endometrial stripe and no fetal pole + one week later beta-hCG is 90 mIU/mL; next best step? →

answer = “third measurement of beta-hCG within one week” → Dx here is spontaneous abortion; must measure beta-hCG weekly until negative; same for gestational trophoblastic disease (moles).

- 43F + bleeding per vaginum + uterus is large and smooth; Q asks for which type of uterine fibroid; answer = submucosal leiomyomata.
- 43F + no bleeding per vaginum + uterus is globular; which type of fibroid? → answer = subserosal.
- 43F + beefy red mass protruding from the vagina; Dx? → answer = pedunculated submucosal leiomyomata uteri, not cervical cancer → the latter will often be described as an ulcerated, exophytic mass.
- 42F + comes in for routine exam + no complaints + large uterus on exam + USS shows various leiomyomata; next best step? → answer = observation (because asymptomatic); otherwise Tx = NSAIDs, OCPs.
- 44F + dysmenorrhea + menorrhagia + USS shows large, smooth uterus with no overt masses; Dx? → answer = adenomyosis (endometrium growing within myometrium); may present similar to submucosal fibroids, with vaginal bleeding, however uterus is diffusely enlarged and no masses seen on USS; Tx with NSAIDs, OCPs; leuprolide; definitive is hysterectomy.
- 27F + 30 weeks’ gestation + weakness of thumb abduction bilaterally; Dx? → carpal tunnel syndrome (normal in pregnancy).
- 23F + unintended pregnancy + fever of 104F + vaginal discharge + abdo pain + laceration visualized on cervix; Dx? → septic abortion → she tried to self-abort using, e.g., a hanger.
- 32F + rupture of membranes (ROM) >18 hours + abdo pain + fever; Dx + Tx? → chorioamnionitis; Tx = ampicillin + gentamicin + clindamycin (amp + gent alone seen as answer on one Obgyn shelf Q).
- 32F + C-section 12 hours ago + abdo pain + fever; Dx + Tx? → postpartum endometritis; Tx = ampicillin + gentamicin + clindamycin.
- Organism(s) causing chorioamnionitis + endometritis? → answer = polymicrobial.
- 25F + postpartum endometritis + low BP; Dx? → answer = puerperal sepsis; gynecologic infection starting 1-10 days after parturition leading to sepsis.
- Lump seen at 4 or 8 o’clock position on vulva; Dx + Tx? → Bartholin gland cyst/abscess; Tx = warm compresses for cyst; drain if abscess.
- Organism(s) causing Bartholin gland abscess? → answer = polymicrobial.

- 37F + Bartholin gland abscess + Q asks “most serious complication of this condition?” → answer = necrotizing fasciitis; wrong answer = “gram positive sepsis” (polymicrobial; need not be gram +).
- Grey/whitish patchy/rough area on the vulva or perineum; Dx + Tx? → lichen sclerosus → must do punch biopsy first to rule out SCC; if confirmed LS, do topical steroids; if SCC, surgically excise.
- SCC of perineum in diabetic; biggest risk factor in this patient? → answer = HPV, not dysglycemia.
- 24F + sharp adnexal pain + no adnexal mass mentioned in vignette + 10-15 mL of serosanguinous fluid aspirated from the cul de sac; Dx? → ruptured cyst (usually corpus luteal); Tx = supportive.
- 24F + Hx of ovarian cyst + colicky pelvic pain past few weeks + pain has become constant past couple days + 6x8cm palpable adnexal mass; Dx? → ovarian/adnexal torsion (cyst is a risk factor).
- 24F + Hx of ovarian cyst + intermittent pelvic pain for four hours that has become constant past two hours + 8x10cm palpable adnexal mass; Dx? → ovarian/adnexal torsion (pain may be weeks or hours).
- 24F + increasingly severe pelvic pain the past couple days + 6x8cm mass palpable in the adnexa; Dx? → torsion.
- 25F + normal periods + LMP 20 days ago + 5cm mobile mass in right adnexa on examination + slightly tender to palpation; Dx? → answer = hemorrhagic corpus luteum cyst; wrong answer is endometrioma (chocolate cyst seen in endometriosis).
- 18F + tampon use + diffuse rash + BP 90/60; Dx? → toxic shock syndrome (*S.aureus*).
- 24F + 30 weeks’ gestation + spotting on underwear 12 hours after sexual intercourse + bleeding gradually increasing since + USS normal; Dx? → answer = cervical trauma.
- 36F + 26 weeks’ gestation + severe flank pain + feels faint when attempting to urinate; Dx? → urolithiasis (progesterone slows ureteral peristalsis).
- Mechanism for increased cholesterol gallstones in pregnancy? → progesterone slows biliary peristalsis + estrogen increased activity of HMG-CoA reductase (compensatory for lowering serum levels of cholesterol).
- 26F + three first-trimester miscarriages + has single kidney; Q asks most likely reason for recurrent miscarriage; answer = congenital uterine abnormalities.
- Tx for torsion? → laparoscopic detorsion.

- 32F + dull right-sided pelvic pain + beta-hCG negative + USS shows simple 5cm cyst; Tx? → answer = "oral contraceptive therapy and a second pelvic examination in 6 weeks"; the wrong answer is "reassurance and schedule follow-up examination in 1 year."
- 23F + extremely painful periods + needs to miss grad school classes sometimes because of the pain + examination shows no abnormalities; Dx? → answer = primary dysmenorrhea = "prostaglandin production" = PGF2alpha hypersecretion.
- Above 23F; next best step in Mx? → answer = NSAIDs; pregnancy test is wrong answer.
- 23F + extremely painful periods + needs to miss grad school classes sometimes because of the pain + examination shows nodularity of the uterosacral ligaments; Dx? → answer = endometriosis. Obgyn shelf will often omit details such as pain with defecation or dyspareunia because they're too easy.
- How to Dx endometriosis? → answer = diagnostic laparoscopy.
- 26F + dull pelvic pain + USS shows cystic mass with calcification; Dx? → answer = dermoid cyst (mature cystic teratoma); details such as "hair, skin, teeth" are too easy for Obgyn shelf.
- 65F + multiple masses "caked" on the omentum; Dx? → ovarian cancer.
- 31F with epilepsy + 10 weeks' gestation + has seizure + phenytoin serum level below therapeutic range; next best step? → answer = increase dose of phenytoin (yes, during pregnancy) → seizure leads to fetal hypoxia, which is worse case scenario, so must prevent at all costs.
- 31F on valproic acid wanting to get pregnant; what do we do? → stop valproic acid (contraindicated in pregnancy due to high chance of neural tube defects) → can use other anti-epileptics during pregnancy instead.
- Hx of many pregnancies + downward movement of vesicourethral junction → stress incontinence → answer on one Obgyn NBME Q is "decreased external urethral tone."
- Tx of stress incontinence → pelvic floor exercises (Kegel); if insufficient → mid-urethral sling.
- Hyperactive detrusor or detrusor instability → urge incontinence.
- Need to run to bathroom when sticking key in a door → urge incontinence.
- Incontinence in multiple sclerosis patient or perimenopausal → urge incontinence.
- 52F + hot flashes + urge incontinence; Q asks mechanism; answer = "estrogen deficiency."
- Tx of urge incontinence → oxybutynin (muscarinic cholinergic antagonist) or mirabegron (beta-3 agonist).

- Incontinence + high post-void volume (usually 3-400 in question; normal is <50 mL) → overflow incontinence.
- Incontinence in diabetes → overflow incontinence due to neurogenic bladder.
- Tx for overflow incontinence in diabetes → bethanechol (muscarinic cholinergic agonist).
- Incontinence in BPH → overflow incontinence due to outlet obstruction → eventual neurogenic bladder.
- What is the only approved indication for hormone-replacement therapy (HRT)? → severe vasomotor Sx (hot flushes, urge incontinence); HRT is not used for preserving bone density; increases risk of thromboembolic and cerebrovascular events; estrogen increases fibrinogen and factor VIII levels.
- 57F + blood stains on underwear for 6 months + painful sexual intercourse + atrophic, friable vaginal mucosa on exam + cervix and bimanual exams normal; Dx + Tx? → atrophic vaginitis → answer = "hypoestrogenic state" → Tx = lubricants; if insufficient, topical estrogen may be used.
- 25F + currently breastfeeding + menstruation not yet resumed + dyspareunia + erythematous vagina with no discharge; next best step in Mx? → answer = "recommendation for use of a lubricant" → high prolactin levels during breastfeeding → hypoestrogenic state → Sx similar to atrophic vaginitis in menopause.
- HRT increases the risk of what kind of cancer? → answer= breast, not endometrial; greater absolute amount of estrogen over female's life increases breast cancer risk; HRT does not increase endometrial cancer risk; latter is unopposed estrogen as risk factor, which is why HRT is estrogen + progesterone; only time HRT is given as estrogen only is for women with Hx of hysterectomy.
- 53F + taking HRT past six months + stopped taking progesterone component because she didn't like how it affected her moods + vaginal bleeding; next best step? → answer on Obgyn shelf = endometrial biopsy.
- 53F + started HRT three months ago + normal mammogram when started HRT + now has cyst seen on ultrasound after self-palpation; next best step? → answer = FNA biopsy of the cyst.
- How do combined oral contraceptive pills affect cancer risk: **↓↓ ovarian (~50% ↓ risk),** ↓ endometrial, ↔ breast; ↑ cervical (from decreased barrier protection → ↑ HPV infections; not from pill itself). Some studies have suggested possible increased risk for breast, but no significance.

- 16F + aunt died of ovarian cancer + asks GP how to screen for ovarian cancer; what is your response? → answer = no screening, but offer her information about oral contraceptive pills.
- 25F + BRCA mutation confirmed + three first-degree family members with gynecologic cancers; next best step? → answer = total abdominal hysterectomy and bilateral salpingo-oophorectomy.
- 47F + total abdominal hysterectomy and bilateral salpingo-oophorectomy performed for leiomyomata uteri; Q asks what we do re Pap smears; answer = "no longer indicated."
- 22F + T1DM + 33 weeks' gestation + fundal height 38cm; Dx? → polyhydramnios (fundal height in cm should approximately = # of weeks pregnant).
- Neonatal girl with karyotype 46XX + has phallus and scrotum; Q asks mechanism; answer = "ACTH hypersecretion" → in congenital adrenal hyperplasia caused by 21- and 11-hydroxylase deficiency, cortisol is low, so ACTH goes up to compensate, leading to cortical hyperplasia; in addition, precursors are shunted to DHEA-S and androstenedione, leading virilization of newborn.
- 33F + prenatal USS shows two fetuses with thick dividing membrane; what kind of twin pregnancy is this? → answer = dichorionic diamniotic; thick dividing membrane = two chorions; # of placentae = # of chorions.
- 33F + prenatal USS shows one fetus much larger than the other; what kind of twin pregnancy is this? → most likely to be monochorionic monoamnionic in the setting of twin-twin transfusion syndrome, where one fetus "steals"/siphons nutrients and blood flow from his or her twin.
- 43F + receiving beta-hCG as part of IVF protocol + develops severe abdo pain + ascites; Dx? → answer = ovarian hyperstimulation syndrome → almost always due to iatrogenic beta-hCG administration; causes vascular hyperpermeability.
- 21F + requests OCPs + Pap smear is normal; Q asks what else needs to be done; answer = check for chlamydia → should be noted that whilst Pap smears always start at 21, STI checks are done from age of sexual onset.
- 33F + regular periods + Hx of multiple sexual partners + unable to conceive with husband for 3 years + husband has normal semen sample; next best step? → answer = hysterosalpingogram (assess tubal patency and uterine architecture; possible Hx of PID leading to tubal occlusion (despite no Hx of ectopic in the patient).

- 35F + hysterosalpingogram shows spillage of dye into the peritoneal cavity; Dx? → normal finding (fallopian tubes are normally open at both ends).
- What is uterine didelphys? → uterus develops as paired organ (double uterus) + double cervix +/- double vagina.
- 52F + presents for routine screening for first time in 4 years; Q asks "in addition to cholesterol screening, Pap smear, and mammography; what does she need? → answer = colonoscopy. Similar answers might be influenza vaccine if fall/winter (every year).
- How often are Pap smears indicated, and when are they started and stopped? → commenced at age 21, then every 3 years; starting age 30, can become every 5 years if co-test for HPV; performed until age 65 (past ten years must be normal findings + no Hx of moderate or severe dysplasia).
- Pap smears in HIV? → at time of diagnosis, then every year.
- Mx of Pap smear result: atypical squamous cells of undetermined significance (ASC-US) → repeat cytology in a year, OR test for HPV; if positive, do colposcopy + biopsy; if negative, repeat co-testing in three years.
- Mx of LSIL on Pap smear? → if negative HPV testing, repeat co-testing in one year; if (+) HPV testing or no testing, do colposcopy + biopsy.
- Mx of high-grade squamous intraepithelial neoplasia (HSIL) on Pap smear? → regardless of HPV status: immediate loop electrosurgical excision procedure (LEEP), OR colposcopy + biopsy.
- Mx of cervical intraepithelial neoplasia (CIN) I seen on biopsy → immediate LEEP, OR colposcopy + cytology every 6 months.
- Mx of CIN II/III seen on biopsy → immediate LEEP demonstrating clear margins, then do Pap + HPV contesting 1 and 2 years postoperatively.
- 57F + vaginal hysterectomy performed for CIN III; next best step? → Obgyn shelf answer = "Pap smear annually."
- 32F + colposcopy is performed for LSIL + entire squamocolumnar junction cannot be visualized; next best step? → answer on Obgyn NBME = cone biopsy.
- 47F + Pap smear shows atypical glandular cells + colposcopy normal + endocervical curettage shows benign cells; next best step? → Obgyn NBME answer = endometrial biopsy.

- 35F + two minutes after separation of placenta has shortness of breath + tachycardia + bleeding from venipuncture sites; Dx? → amniotic fluid embolism; can cause DIC; supportive care.
- 35F + two days after C-section + gets up to go to the bathroom + SoB + tachycardia; Dx? → pulmonary embolism → heparin followed by spiral CT (if not pregnant) or V/Q scan (if pregnant).
- 39F + pregnant + Sx of pulmonary embolism + V/Q scan performed showing segmental defects; next best step in Dx? → answer = spiral CT; student says "wait but I thought we don't do CT in pregnancy." Right, we don't. But if they ask for next best step after V/Q scan, that's still the answer they want.
- 27F + two days after C-section + temp 100.8F + breath sounds decreased at both lung bases + urinary catheter specimen is negative + remainder of exam unremarkable; Dx? → answer = atelectasis (most common cause of fever within 24 hours of surgery (but shelf has two days after C-section for one Q).
- 27F + triad of third-trimester painless bleeding + ROM + fetal bradycardia; Dx? → answer = vasa previa (fetal vessels overlying the internal cervical os); associated with velamentous cord insertion (vessels not protected by Wharton jelly).
- 22F + uncomplicated delivery of newborn + heavy vaginal bleeding + placenta shows large, non-tapering vessel extending to margin of membranes; Dx? → answer = succenturiate placental lobe; students says wtf? → just need to know sometimes placenta can have auxiliary lobe with connecting vessels; this is a cause of vasa previa, in addition to velamentous cord insertion.
- 35F + C-section 6 weeks ago + required 3 units of transfused RBCs + 9kg weight loss + has cold intolerance + could not breastfeed; Dx? → Sheehan syndrome (arrow Q on shelf; answer is ↓ for prolactin, ACTH, GH, FSH, TSH); should be noted tangentially that on newer NBME for Step 1, Q with Sheehan syndrome has ↑ for aldosterone (not hyperaldosteronism, but higher baseline to compensate for lower cortisol).
- 15F + never had menstrual period + one-wk Hx of constant, severe pelvic pain + 10-month Hx of intermittent pelvic pain + BP of 90/50 + bluish bulge in upper vagina; Dx? → hematometra → imperforate hymen with blood collection in the uterus → vagal response causes low BP → Tx = cruciate incision of the hymen.
- 15F + never had menstrual period + one-wk Hx of constant, severe pelvic pain + 10-month Hx of intermittent pelvic pain + BP normal + bluish bulge in upper vagina; Dx? → hematocolpos → blood

collection in the vaginal canal, but not backed up to the uterus like hematometra → Tx = cruciate incision of the hymen.

- 27F + delivered newborn 5 days ago + pain in calf with dorsiflexion of foot; next best step in Dx? → answer = duplex ultrasonography of the calf; positive Homan sign for DVT in hypercoagulable state.
- Down syndrome important testing?
    - First trimester screen (11-13 weeks): ↓ pregnancy-associated plasma protein A (PAPPA), ↑ beta-hCG, ↑ nuchal translucency, hypoplastic nasal bone.
    - Second trimester screen (16-18 weeks): ↓ AFP, ↑ beta-hCG, ↓ estriol, ↑ inhibin-A; in Edward syndrome, all decreased; Patau is variable.
    - Cell-free DNA (as early as 10 weeks).
- Most common cause of abnormal AFP measurement? → answer = dating error.
- 32F + AFP measurement comes back 2.6x upper limit of normal; next best step? → answer = re-ultrasound; wrong answer = perform AFP measurement again → need to simply do ultrasound to reapproximate dates.
- Important locations for the "celes":
    - Cystocele: anterior superior vaginal wall.
    - Urethrocele: anterior inferior vaginal wall.
    - Enterocele: posterior superior vaginal wall (Q on shelf says "high on posterior vaginal wall; another Q says the patient can feel movement within her vagina → weird, but presumably gut peristalsis).
    - Rectocele: posterior inferior vaginal wall.
- 32F + protrusion of distal urethra through urethral meatus; Dx? → urethral prolapse; sounds reasonable, but don't confuse with stress incontinence; the latter will sometimes be described as "downward mobility of vesicourethral junction with Valsalva" (not urethral prolapse).
- 22F + 24 weeks' gestation + fundal height 20cm + no cervix palpated + examination shows fetus in breech position in vagina; Dx? → cervical incompetence; Tx w/ cervical cerclage; notable risk factor is prior conization.
- 30F + 37 weeks' gestation + fetus in breech position; during labor, risk of which complication is greatest? → answer = cord prolapse.

- 32F + 14 weeks' gestation + Hx of two LEEP + cervix flush against upper vagina and measures 2cm in diameter + pelvic USS shows funneled lower uterine segment; Dx? → cervical incompetence → "funnel" means cervical incompetence ("cervical funneling" / "funneled lower uterine segment").
- 87F + partial prolapse of uterine cervix through the introitus + uterus can easily be pushed back into the uterus; next best step? → answer = vaginal pessary.
- Stages of labor:
    - Stage 1 latent: 0-6cm cervical dilation **(old guidelines: 0-4cm)**
    - Stage 1 active: 6-10cm (complete) cervical dilation. **(old guidelines: 4-10cm)**
    - Stage 2: 10cm (complete) cervical dilation to delivery of fetus.
    - Stage 3: delivery of fetus to delivery of placenta.
    - Obgyn NBME has Q where 32F has been at 5cm dilation for past 4 hours; answer = "arrest of active phase"; the wrong answer is "protracted latent phase."
- What is definition of protracted latent phase? → dilating <1-2cm per hour, which reflects the 95%tile in contemporary women. Women <6cm are in latent phase; regardless of parity, may take 6-7 hours to progress from 4-5cm, and 3-4 hours to progress from 5-6cm.
- What does "arrest of active phase" mean? → no cervical change in >4 hours despite adequate contractions (>200 Montevideo unites [MVU]), or >6 hours if contractions inadequate.
- 28F + 38 weeks' gestation + cervix completely dilated + strong contractions + fetal station remains unchanged over next hour; Dx? → answer = cephalopelvic disproportion (baby too big for pelvis).
- 5F + foul-smelling yellow vaginal discharge + blood spotting on underpants + no dysuria + mild vulvar erythema seen on exam; Dx? → answer = vaginal foreign body, not sexual abuse; presumably sexual abuse there would be lacerations or trauma seen on physical exam.
- 82F + Alzheimer + brought in by daughter for blood on underwear + 3cm vaginal laceration + erythematous, edematous perineal body; Dx? → answer = sexual assault.
- 23F + dysuria + bacteriuria + pyuria; Q asks how to decrease future episodes; answer = "voiding immediately after coitus."
- 23F + three UTIs over past year + Hx of UTIs being Tx successfully with TMP-SMX; Q asks for most appropriate med for daily UTI prophylaxis; answer = TMP-SMX; slightly unusual question, but it's on the Obygn NBME.

- 37F + dysuria + urinalysis shows 20-50 WBCs/hpf + one week of TMP-SMX does not improve Sx; next best step? → answer = urethral culture for chlamydia → if patient doesn't improve with Tx of UTI, check for STIs.
- 20F + 40 weeks' gestation + epidural catheter placed + lidocaine and epinephrine injected + develops metallic taste in mouth; Dx? → answer = "intravascular injection of anesthetic."
- 25F + 5 weeks post-delivery + insomnia + irritable + finds baby's cry annoying and leaves him in crib crying for long periods of time; next best step? → answer = "arrange for immediate psychiatric evaluation" → post-partum depression; Tx = sertraline (SSRI) and CBT; if mania, delusions, or hallucinations → post-partum psychosis; if more mild + within 7-10 days of delivery → post-partum blues.
- 25F + 42 weeks' gestation + oligohydramnios + cervix long, closed, and posterior; next best step? → answer = "administer a prostaglandin"; wrong answer is amnioinfusion (do for variable decelerations with ROM).
- 34F + pregnant + low serum iron and ferritin + microcytic anemia + proceeds to take iron for three weeks + three weeks later, iron and ferritin are normal but still has microcytic anemia; next best step in Mx? → answer = "hemoglobin electrophoresis"; Dx is thalassemia (alpha trait usually, as this is asymptomatic + picked up in pregnancy) → **microcytic anemia non-responsive to iron supplementation;** Hb electrophoresis will show presence of HbA2.
- 28F + 7 weeks' gestation + started taking prenatal vitamin 3 weeks ago + microcytic anemia; next best step? → answer = hemoglobin electrophoresis; same as above, **the implication is that the supplement contains iron + she is possibly non-responsive to it → thalassemia.**
- 28F + African American + 7 weeks' gestation + microcytic anemia + Hb electrophoresis shows 95% HbA1; Dx? → answer on Obgyn shelf = iron deficiency anemia; thalassemia would show HbA2.
- 28F + pregnant + MCV 87 + Hb 10.5 g/dL; Dx? → answer = physiologic dilution of pregnancy → Hb drop to 10.5 g/dL is normal finding.
- "What about platelets in pregnancy?" → reduction normal; gestational thrombocytopenia is the Dx when level drops to <150,000 per uL.

- 24F + immune thrombocytopenic purpura (ITP); Q asks the potential effect on the fetus → answer = "fetal platelet destruction"; maternal IgG against her own platelet GpIIb/IIIa can cross placenta, attacking the fetal platelets. This is on new Obgyn form.
- 20F + 42 weeks' gestation + shoulder dystocia + neonate born with arm pronated, adducted, and internally rotated; Dx? → "injury to the 5^{th} and 6^{th} cervical nerve roots" (Erb-Duchenne palsy).
- Most common cause of postpartum bleeding? → uterine atony (hypocontractile uterus).
- Tx for uterine atony? → uterine massage first, followed by oxytocin, then ergonovine.
- 33F + postpartum bleeding despite uterine massage and oxytocin; next best step? → answer = ergonovine therapy (do not give in HTN).
- Diabetic mom giving birth + shoulder dystocia + McRoberts maneuver implemented; what is notable risk to the fetus here? → answer = clavicular fracture (anterior shoulder caught behind pubic symphysis → McRoberts maneuver is flexing mom's hips + applying suprapubic pressure → clavicular fracture not uncommon).
- Diabetic mom giving birth + shoulder dystocia + McRoberts maneuver implemented + postpartum bleeding + uterus is firm on palpation; most likely cause of bleeding? → answer on Obgyn shelf = vaginal laceration, not uterine atony.
- 34F + delivers term neonate + placenta delivers after gentle cord traction + now has moderate vaginal bleeding + HR 60 + BP 60/40 + IV saline doesn't help + uterus cannot be palpated on physical exam; Dx? → answer = uterine inversion.
- Episiotomy performed posterior in the midline; what does the obstetrician cut into if he cuts too far? → answer = external anal sphincter.
- 37F + 40 weeks' gestation + Hx of C-section + constant, sharp abdominal pain + maternal vitals all normal + fetal late decels + "Leopold maneuvers show fetal small parts above the fundus"; Dx? → answer = uterine rupture.
- 37F + 40 weeks' gestation + oxytocin administered + robust contractions occurring every two minutes + abdo pain + hypotension + fetal head palpated in RUQ; Dx? → uterine rupture.
- What are tachysystole and uterine hypertonus? → tachysystole is >5 contractions every ten minutes; uterine hypertonus is a sustained contraction >2 minutes.

- What are Leopold maneuvers? → abdominal palpatory maneuvers used to determine the position and lie of the fetus.
- 62F + ovarian mass + bleeding per vaginum + endometrial biopsy shows atypical complex hyperplasia; Q asks for which ovarian cancer is the Dx? → answer = granulosa cell tumor → unopposed estrogen → endometrial hyperplasia → endometrial cancer risk.
- 47F + 9-month Hx of irregular periods where they occur at 2-3-month intervals + endometrial biopsy shows proliferative endometrium; next best step? → answer on shelf = "cyclic progestin therapy" → control irregular menses and prevent endometrial hyperplasia.
- 32F + menometrorrhagia + LMP 2 weeks ago + periods 28-30-day intervals + just started taking OCPs for Tx; what is the most likely explanation for improvement in patient's bleeding? → answer = "synchronization of endometrium."
- 27F + G3P2 + Rh negative + received RhoGAM both prior pregnancies + arrives now at first prenatal visit for third pregnancy; next best step? → Obgyn shelf answer = "indirect antiglobulin (Coombs) test" → must see if she's developed antibodies to Rh antigen.
- 29F + G1P0 + O+ blood type + fetus is A or B blood + goes on to develops pathologic jaundice postpartum; Dx? → hemolytic disease of the newborn (ABO type) → mothers with O blood type will have fractional IgG (instead of IgM) against A and B antigens → cross placenta → fetal hemolysis → severity highly variable; Obgyn shelf will always give first pregnancy and an O+ mom so that student can't accidentally get lucky with the Dx if he/she only knows about Rh type hemolytic disease of the newborn.
- 29F + G2P1 + Rh negative + fetus experiences hydrops; Dx? → hemolytic disease of the newborn (Rh type) → presumably mother made antibodies against fetal Rh antigen from prior pregnancy following mixing of circulations.
- When to give RhoGAM? → normally at 28 weeks' gestation + again at parturition; also give for spontaneous or instrumental abortions + procedures (e.g., amniocentesis) + trauma/insults (e.g., abruptio placentae).
- 34F + G3P2 + Rh negative + all pregnancies with same male partner + indirect Coombs test positive for anti-Kell antigens at titer of 1:256; next best step? → answer = "Kell typing of the father's blood"; implication is mom is Kell negative but prior fetus(es) Kell positive; fetal blood must have entered

maternal blood during prior pregnancy, however mom has no titers against Rh, just Kell, because RhoGAM was presumably given.

- Painful third-trimester bleeding following MVA or cocaine use; Dx? → abruptio placentae.
- Painless third-trimester bleeding; Dx until proven otherwise? → placenta previa → placental implantation site can spontaneously move off the internal os before 36 weeks, so don't plan for Caesar before then.
- Postpartum hemorrhage due to placental issue; Dx? → placenta accreta/increta/percreta.
- 21F + recently took Abx + red vaginal introitus and itching + cervical and vaginal discharge are normal + KOH prep and wet mount show no abnormalities; Dx? → answer on Obgyn NBME = vaginal candidiasis (thick white discharge is otherwise classic). Tx = topical nystatin or oral fluconazole.
- 67F + T2DM + vaginal candidiasis Tx with topical miconazole + doesn't respond to Tx; Q asks why; answer = T2DM.
- 21F + mucopurulent discharge + no organisms grow; Dx? → chlamydia → oral azithromycin or doxycycline. Azithromycin is ideal because it's one-off stat oral dose; doxy is BID for a week.
- 21F + mucopurulent discharge + gram negative diplococci; Dx? → gonorrhea → cotreat for chlamydia → IM ceftriaxone + oral azithro, OR IM cefixime + oral azithro.
- 21F + erythematous cervix + yellow/green discharge + wet mount confirms Dx; Dx? → trichomoniasis (flagellated protozoa) → Tx = topical metronidazole *for patient and partner.*
- 21F + erythematous vaginal canal + thin, watery discharge + wet mount confirms Dx; Dx? → bacterial vaginosis (Gardnerella vaginalis) → met mount shows clue cells (squamous cells covered in bacteria) → Tx = topical metronidazole.
- 21F + thin, grey discharge + KOH prep Whiff test is performed yielding fishy odor; Dx? → bacterial vaginosis.
- 21F + VDRL positive at titer of 1:4 + physical exam shows no abnormalities + complains of no Sx + chlamydia and gonorrhea testing negative; next best step? → answer = Obgyn shelf answer = fluorescent treponema antibody (syphilis).
- 19F + painless vulvar ulcer + rapid plasmin reagin negative + all other tests negative; next best step? → Obgyn NBME answer = repeat rapid plasma reagin (slightly unusual answer, but can sometimes be negative early in primary syphilis).

- 21F + one-week Hx of 0.25-cm crusty, painless papule on the posterior fourchette; Dx? → condylomata acuminata → HPV6+11.
- 22F + soft pink papillary lesions on labia minora and posterior fourchette; Tx? → answer on obgyn NBME = podophyllum resin; student says wtf? → used to treat warts.
- Gardasil HPV vaccine protects against which types? → 6, 11, 16, 18 (6+11 warts; 16+18 SCC).
- 24F + recently went backpacking in Asia + painful vulvar crater + gram (-) rods cultured; Dx + Tx? → answer = chancroid (haemophilus ducreyi); Tx with azithromycin.
- 35F + G1P0 + exposed to child with chickenpox + never been vaccinated against VZV; next best step? → administer VZV IVIG within 96 hours (to be most effective, but still advised up to 10 days post-exposure).
- When is VZV IVIG advised for neonates? → maternal active lesions between 5 days prior to and 2 days post-delivery.
- Neonate born with patent ductus arteriosus; what Sx did the mom have while pregnant? → answer = arthritis, not rash; Dx is congenital rubella syndrome in the neonate (causes PDA).
- 25F + 22 weeks' gestation + develops low-grade fever and rash + fetus develops hydrops; Dx? → maternal infection with parovirus B19.
- 21F + painful vesicles on vulva; do we give oral or topical acyclovir? → answer = HSV → always oral if asked.
- Herpes and pregnancy? → acyclovir indicated to reduce chance of active lesions at time of labor; if active lesions **or prodromal Sx** present at parturition, C-section is indicated; acyclovir is safe during pregnancy.
- HIV and pregnancy? → most important USMLE point is **HAART therapy during pregnancy is more important than not breastfeeding** in terms of decreasing vertical transmission; sounds strange, as the virus is literally in breastmilk, but the answer is HAART therapy to decrease viral load is most important to prevent vertical transmission; in addition, administer zidovudine to mom prior to C-section, then zidovudine within 12 hours to neonate post-delivery (latter Q on peds NBME).

- Hepatitis B and pregnancy? → if mom HepB +, give both HBIG + vaccine within 12 hours of birth; if mom HepB negative, give just vaccine within 12 hours of birth; if mom status unknown, give vaccine within 12 hours of birth, and give HBIG within 7 days if mom's test comes back + or remains unknown.
- 27F + 14 weeks' gestation + not immune to HepB; next best step? → answer = vaccinate to HepB now.
- Influenza and pregnancy? → safe to give IM killed vaccine during pregnancy (in fall or winter).
- MMR vaccine and pregnancy? → vaccinate *before* pregnancy; do not give during pregnancy.
- TB and pregnancy? → Tx for latent and active TB, yes; for active, Tx with RIPE for 2 months, followed by RI for 7 more months (9 months total); if not pregnant, RI is only given for 4 more months.
- Breastfeeding and OCPs? → Obgyn shelf wants you to know that estrogen-containing contraception **decreases protein content of breastmilk;** also linked to lower milk supply + shorter duration of breastfeeding; contraindicated < 6 weeks postpartum; if hormonal contraception used, progestin-only recommended.
- How to differentiate between androgen insensitivity syndrome and Mullerian (paramesonephric duct) agenesis? → both phenotypically female teenagers with normal Tanner stage development; both have vagina that ends in blind pouch; the clinical difference is that in androgen insensitivity syndrome, they will say **absent or sparse pubic and axillary hair;** in Mullerian agenesis, the hair pattern will be normal, or they'll even explicitly say "coarse" pubic and axillary hair. If androgen insensitivity syndrome suspected, next best step = karyotyping (46XY); Mullerian agenesis is 46XX.
- 16F + never had menstrual period + 5'9" + sparse pubic and axillary hair; Dx? → AIS → pointing out that the Q will say "a 16-year-old girl comes in," but karyotypically the patient is still a male.
- 12F + 1-year Hx of progressive hair growth and acne + 2-cm vaginal canal + significant clitoromegaly + posterior labioscrotal fusion + no cervix or palpable uterus; Dx? → 5-alpha-reductase deficiency → "phallus at age 12" (i.e., penis at age 12, since surge of testosterone at puberty yields significant DHT production despite deficient enzyme); Obgyn shelf will merely ask for the karyotype here; answer = 46XY (i.e., male, even though stem will say "12-year-old girl").
- 17F + never had menstrual period + high FSH + absent breast development + scant pubic hair; next best step? → answer = karyotyping (Turner syndrome).
- 15F + Tanner stage 2 + 4'11" + bone age is equal to chronologic age; answer = karyotyping (Turner).

- 37F + C-section two days ago + incision site erythematous + abdomen tender + vitals normal + two palpable lymph nodes in groin; Dx? → answer = "normal postoperative course."
- 37F + vaginal bleeding + hydroureter; Q asks for what kind of cancer; answer = cervical SCC (impingement on the ureter).
- When are OCPs contraindicated? → **smokers over 35; migraine with aura; HTN (>160/100);** current or past venous thromboemboli; thrombotic disorder (i.e., prothrombin mutation, FVL); cerebrovascular event; ischemic heart disease; current breast cancer; liver tumor; among others; Obgyn shelf will ask which is contraindicated, and the answer is **"triphasic oral contraceptives"** (same thing as OCP).
- 18F + menstrual cycles with 14-40-day intervals + beta-hCG negative; next best step? → answer = "cyclic progesterone therapy" → means OCPs, but this is shelf wording.
- What is most effective form of emergency contraception? → answer = copper IUD; second-best is ulipristal (selective progesterone-receptor modulator; SPRM).
- 31F + copper IUD in place + pelvic exam shows enlarged uterus + USS shows 4cm fibroid; next best step? → answer = "leave the IUD in place but inform the patient that the leiomyoma may cause heavier menses."
- Important points about Depo vs Implanon? → Depo is progestin injection that is effective for three months; it can cause decreased bone density; Implanon is a progestin implant contraceptive that is effective for three years; it is associated with erratic periods.
- Type of cancer patient is at increased risk for if commencing Depo? → answer on Obgyn shelf = breast.
- Important contraindication to IUD? → active STI/PID **or Hx of infection within past 3 months**; current pregnancy (obvious); Hx of gynecologic malignancy.
- 42F + HTN managed with meds + often forgets to take meds + wants contraception; what is most appropriate recommendation? → answer = levonorgestrel IUD (for patients with poor pharmacologic adherence).
- 27F + Hx of difficulty remembering to take daily meds + wants contraception + Tx for chlamydia three months ago; Q asks most appropriate form of contraception; answer = "Depo medroxyprogesterone"; IUD not ideal because of Hx of infection past three months.

- 68F + Hx of breast cancer + paresthesias bilaterally in legs; next best step? → steroids first for possible spinal mets (decrease inflammation); then do MRI of spine.
- 28F + G2P1 + 10 weeks' gestation + prior pregnancy resulted in neonate of 4540 grams; Q asks what she's at increased risk for during current pregnancy; answer = gestational diabetes.
- When to screen for gestational diabetes (GD) for normal risk women? → 24-28 weeks' gestation.
- 28F diabetic + 37-weeks' gestation + delivers neonate with neonatal respiratory distress syndrome (NRDS) + macrosomia (>4000 grams); Q asks which hormone in the serum of the fetus is responsible; answer = insulin → inhibits surfactant production; should be noted that insulin does not cross the placenta; fetus produces more endogenous insulin with maternal diabetes.
- Mechanism for NRDS? → decreased surfactant production → decreased lecithin/sphingomyelin ratio; lecithin is aka dipalmitoyl phosphatidylcholine.
- 37F + 33 weeks' gestation + C-section scheduled in 12 hours + bolus of steroids given 12 hours ago; next best step? → answer = give bolus of steroids; *two boluses of steroids* must be given within 24 hours of delivery <34 weeks.
- When to give steroids and magnesium prior to delivery? → steroids before 34 weeks (*two boluses*); if 34 0/7 – 36 6/7 weeks, give one bolus of steroids; add magnesium if before 32 weeks.
- When are tocolytics used? → <34 weeks' gestation if delivery would result in premature birth (i.e., do not use after 34 weeks); only able to delay birth up to a few days; terbutaline (beta-1/-2 agonist), ritodrine (beta-2 agonist), and nifedipine frequently used; notably effective in helping expectant mother to receive two boluses of corticosteroids in the 24-hour period prior to <34-week delivery; various contraindications, including infection, IUGR, and cervical dilation >4cm.
- What are Braxton-Hicks contractions → irregular, spontaneous contractions sometimes felt in third trimester; they are normal and benign; in contrast, labor presents are regular and increasingly sustained contractions.
- When to give GBS prophylaxis?
    - Hx of prior pregnancy with early-onset GBS disease in neonate (i.e., pneumonia, meningitis, sepsis); **do not give** if prior pregnancy demonstrated mere colonization of GBS.
    - GBS bacteriuria at any point during current pregnancy (e.g., first trimester), *even if treated successfully.*

  - Positive rectovaginal swab at 36 weeks.
  - If maternal status is unknown, give if one or more of the following:
    - Maternal fever >38C.
    - ROM >18 hours.
    - Preterm delivery (<37 weeks).
  - Successful GBS prophylaxis is IV penicillin or ampicillin within 4 hours of delivery of fetus; oral amoxicillin/clavulanate (Augmentin) is the wrong answer.

- "Can you explain that annoying Bishop score stuff real quick?"
  - 5 criteria summing to 13 points; higher is better; >8 indicates likely successful vaginal delivery; <6 suggests cervical ripening may be required.
  - **USMLE will not make you calculate, don't worry. But students sometimes ask about this.**
  - Cervical position: Posterior – 0 points; Middle – 1 point; Anterior – 2 points.
    - Becomes more anterior as labor nears.
  - Cervical consistency: Firm – 0 points; Medium – 1 point; Soft – 2 points.
    - More rigid and resistant to stretch in primigravid women.
  - Cervical effacement: 0-30% – 0 points; 30-50% – 1 point; 50-70% – 2 points; >70% – 3 points.
    - How "thin" the cervix is; normally cervix is 3cm long; becomes "paper-thin" when fully effaced.
  - Cervical dilation: Closed 0 points; 1-2cm – 1 point; 2-4cm – 2 points; >4cm – 3 points.
    - Most important indicator of progression through first stage of labor.
  - Fetal station: -3 – 0 points; -2 – 1 point; -1, 0 – 2 points; +1, +2 – 3 points.
    - Fetal head position relative to ischial spines (usually 3-4cm intravaginal and non-palpable); - numbers mean the fetal head is above the ischial spines; + numbers mean head has descended below the ischial spines for impending delivery.
- "Oh yeah can you quickly explain the fetal fibronectin test?" → fetal fibronectin (fFN) is the "glue" found between the chorion and decidua; if a woman is 22-35 weeks' gestation and having symptoms of preterm labor, fFN test predicts whether preterm labor is likely; if negative, <5% chance of delivery within next two weeks; if positive, preterm labor likely.

- 28F + 33 weeks' gestation + clear fluid leaking from vagina past two days + no contractions or bleeding; next best step? → answer = sterile speculum exam; likely preterm premature rupture of membranes (PPROM); wrong answers are fetal fibronectin test (only if premature labor / contractions).

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